Implements

Overview of all implements for the Bruut OpenAgbot: the 49 that are worked out in FreeCAD, with every variant on its own page, followed by the 24 ideas that are not worked out yet.

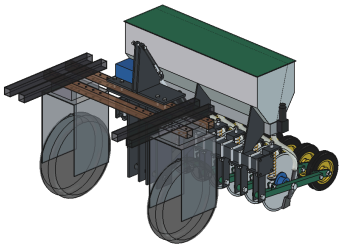
Contents

1. **Overseeder**
   Sowing & Planting
2. **Liquid fertilizer applicator**
   Spraying, Fertilizing & Irrigation
3. **Feed pusher auger**
   Transport & General
4. **Disc brushes**
   Transport & General
5. **Dockweed drill**
   Soil Cultivation & Weed Control
6. **Trencher**
   Soil Cultivation & Weed Control
7. **Field marker**
   Measuring & Sensors
8. **Mobile refiller**
   Spraying, Fertilizing & Irrigation
9. **Around-pole fence mower**
   Mowing
10. **Weed picker**
    Mowing
11. **Sickle bar mower**
    Mowing
12. **Irrigation**
    Spraying, Fertilizing & Irrigation
13. **Spot sprayer**
    Soil Cultivation & Weed Control
14. **In-row weeder**
    Soil Cultivation & Weed Control
15. **Belt rake**
    Soil Cultivation & Weed Control
16. **Weeds burner (flame weeder)**
    Soil Cultivation & Weed Control
17. **Push blade**
    Soil Cultivation & Weed Control
18. **Fertilizer spreader**
    Spraying, Fertilizing & Irrigation
19. **Grassland harrow**
    Crop & Soil
20. **Soil sampler**
    Crop & Soil
21. **Inter-row hoe**
    Soil Cultivation & Weed Control
22. **Grassland aerator**
    Soil Cultivation & Weed Control
26. **Battery pack**
    Power & Drive
27. **Solar canopy**
    Power & Drive
28. **Compost spreader**
    Spreading & Bedding
29. **Stone picker**
    Crop & Soil
30. **Mulch layer**
    Crop & Soil
31. **Grass meter**
    Measuring & Sensors
32. **Soil compaction meter**
    Measuring & Sensors
33. **Charging dock**
    Power & Drive
34. **UV-C mildew light**
    Weed & Crop Protection
35. **Sawdust spreader**
    Spreading & Bedding
36. **Bedding rake**
    Livestock & Farm
37. **Dumper barrow**
    Transport & General
38. **Biocontrol spreader**
    Weed & Crop Protection
39. **Bag trailer**
    Transport & General
40. **Broom / sweeper**
    Transport & General
41. **Lime spreader**
    Spraying, Fertilizing & Irrigation
42. **Cover crop undersower**
    Crop & Soil
43. **Straw spreader**
    Spreading & Bedding
44. **Electric weeder**
    Weed & Crop Protection
45. **Hydraulic power pack**
    Transport & General
46. **Irrigation pump**
    Spraying, Fertilizing & Irrigation
47. **Cambridge roll and overseeder**
    Sowing & Planting

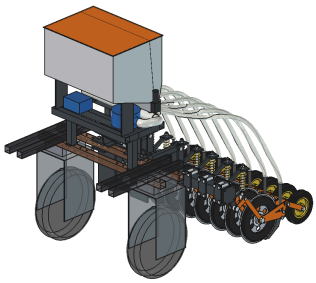
IMPLEMENT · SOWING & PLANTING

Overseeder

Grassland slit seeder. 1 m modules with 8 rows at 125 mm and coupling flanges. A thin disc cuts a 15 mm slot, the seed is placed about 12 mm deep and a sprung press wheel closes the slot. Every row follows the ground on its own arm; two gas springs push the floating lift frame down with robot weight.



Simple

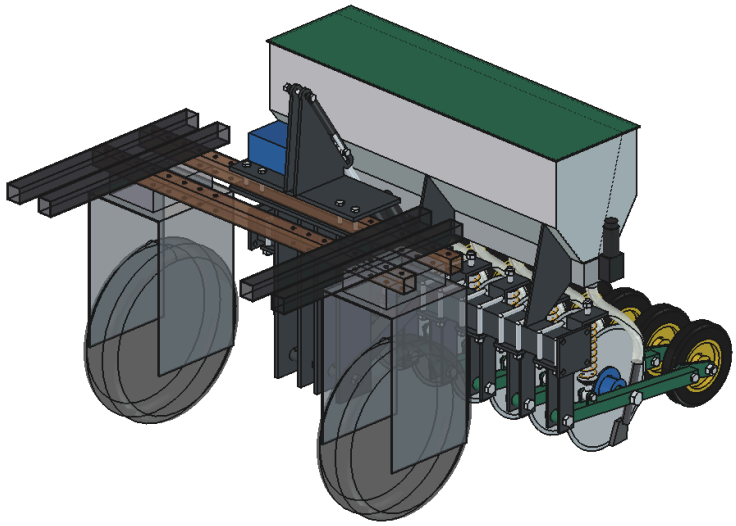


Advanced

2 variants: Simple · Advanced
Folder: `overseeder/`

OVERSEEDER · VARIANT 1 OF 2

Simple



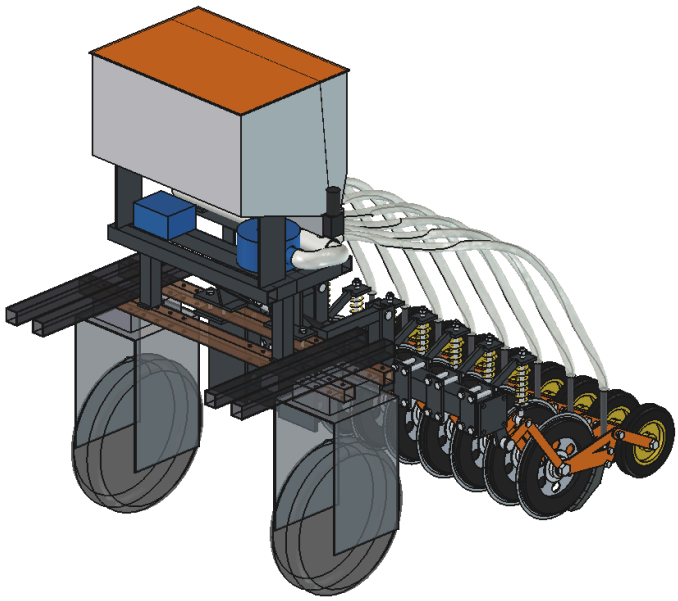
Angled single disc (7°, Ø300 × 3) with a seed boot in its lee and a depth ring. Gravity-fed seed hopper with a compartment for grass (41 l) and one for fine seed (18 l), each with its own cam roller and a motor that follows the driving speed. Built on the headstock and lift frame of the simple applicator v2.

- 8 rows at 125 mm = 1 m per module
- Draft force approx. 335 N (normal)
- Depth within ±3 mm: 94 %
- About €1,675 in parts

Folder: `overseeder/simple/`

OVERSEEDER · VARIANT 2 OF 2

Advanced



Parallel linkage so the toolbar stays parallel, straight Ø300 × 3 disc with depth rings on both sides, a curved seed coulter directly behind the disc and a press wheel in a fork. Air seeder (62 + 17 l hopper, 12 V fan) on a frame above the rear axle blows the seed through 8 hoses to the coulters.

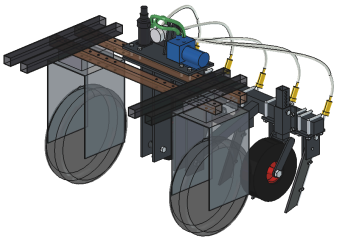
- 8 rows at 125 mm = 1 m per module
- 2 gas springs in the slotted hole
- Depth within ±3 mm: 98 %
- About €2,280 in parts

Folder: `overseeder/advanced/`

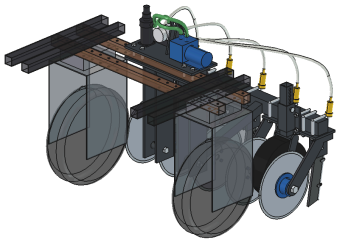
IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Liquid fertilizer applicator

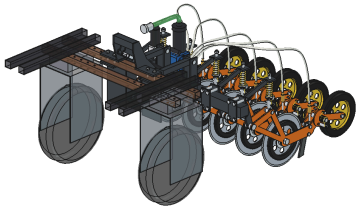
Injects liquid fertilizer (such as RENURE) just under the sward, without mist or drift. Concept versions (simple, simple v2, advanced) and the precision applicator, the first production version with a roller pump.



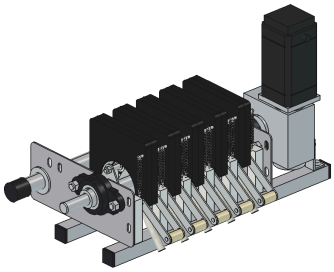
Simple



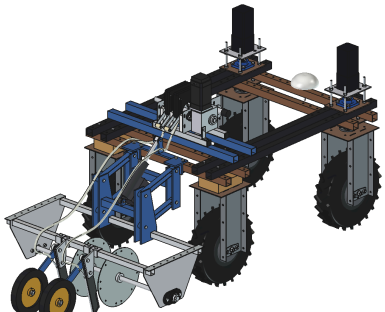
Simple v2



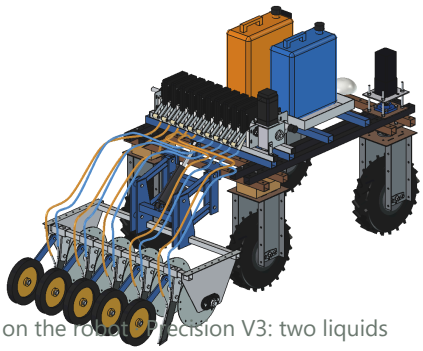
Advanced



Precision: roller pump



Precision V2 on the robot

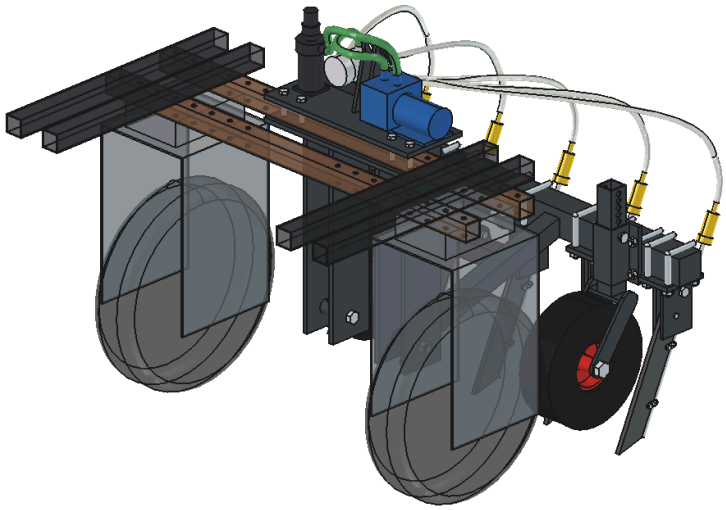


Precision V3: two liquids

6 variants: Simple · Simple v2 · Advanced · Precision: roller pump · Precision V2 on the robot · Precision V3: two liquids
Folder: `liquid fertilizer applicator/`

LIQUID FERTILIZER APPLICATOR · VARIANT 1 OF 6

Simple



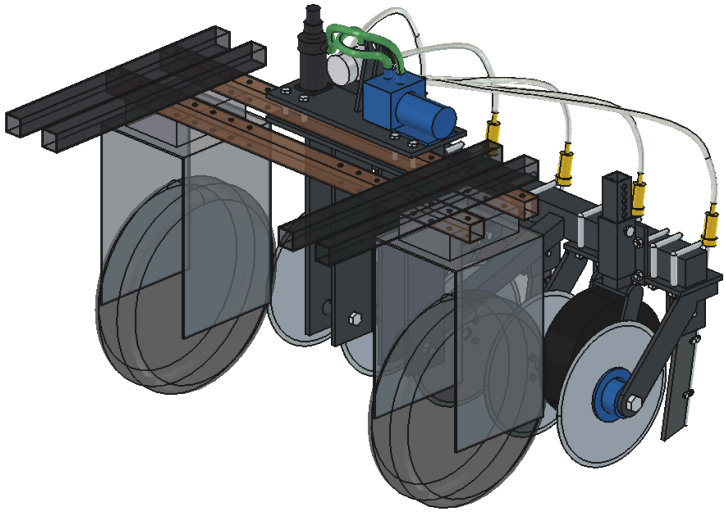
Cheap and robust: 5 fixed injection knives with a stainless steel tube, two wheelbarrow wheels as depth wheels, a lift frame with a single low pivot and one cheap linear actuator, a 12 V diaphragm pump with fixed regulator and orifice plates. Only flat bar, tube and standard parts.

- 5 knives at 200 mm
- 12 V diaphragm pump, dose depends on speed
- About €600 in materials

Folder: `liquid fertilizer applicator/concept versions/simple/`

LIQUID FERTILIZER APPLICATOR · VARIANT 2 OF 6

Simple v2

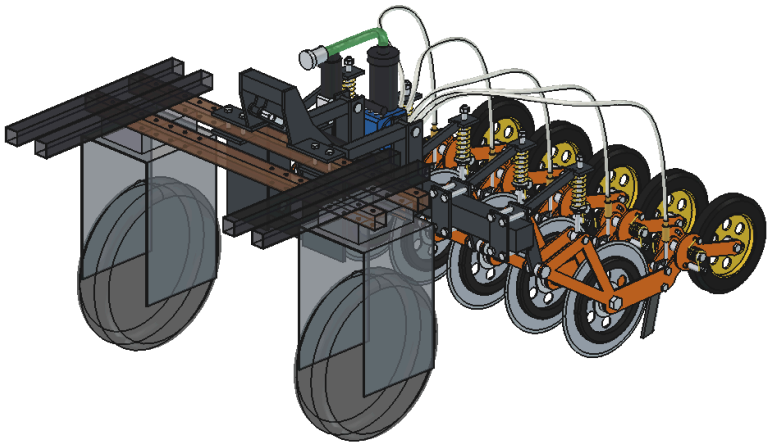


Simple version plus a flat coulter disc Ø300 in front of each knife, cutting 45 mm deep; the knife runs 14 mm behind it in the same plane. Costs 26 kg extra on the lift frame (70 kg), a 1500 N actuator and up to 50 kg of ballast in hard sward.

- Disc + knife per row
- Lift frame 70 kg
- About €915 in materials

Folder: `liquid fertilizer applicator/concept versions/simple_v2/`

Advanced



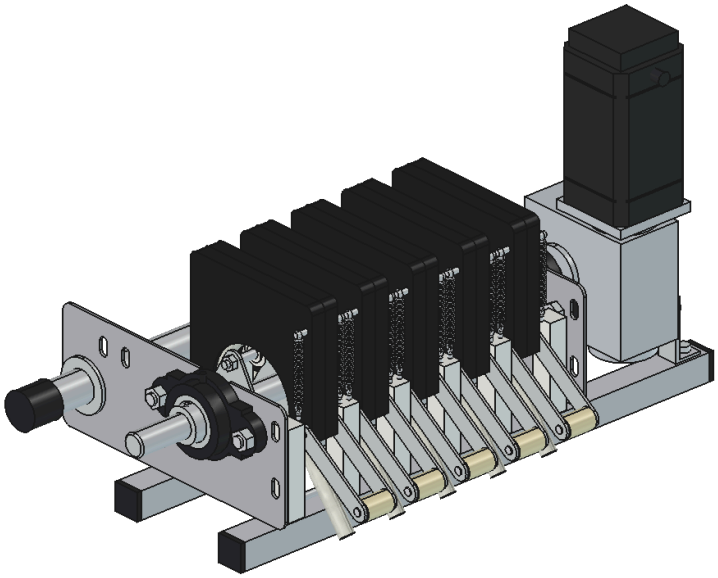
Disc, knife, tube and press wheel per element on spring arms; a 5-channel peristaltic pump driven by a worm-gear motor whose speed follows the robot's driving speed, and an electrically lifted parallel linkage that floats while working (slotted hole). No ground wheel.

- 5 elements at 200 mm
- 89–1,316 l/ha, 18 settings
- Includes ground-following animation

Folder: `liquid fertilizer applicator/concept versions/advanced/`

LIQUID FERTILIZER APPLICATOR · VARIANT 4 OF 6

Precision: roller pump

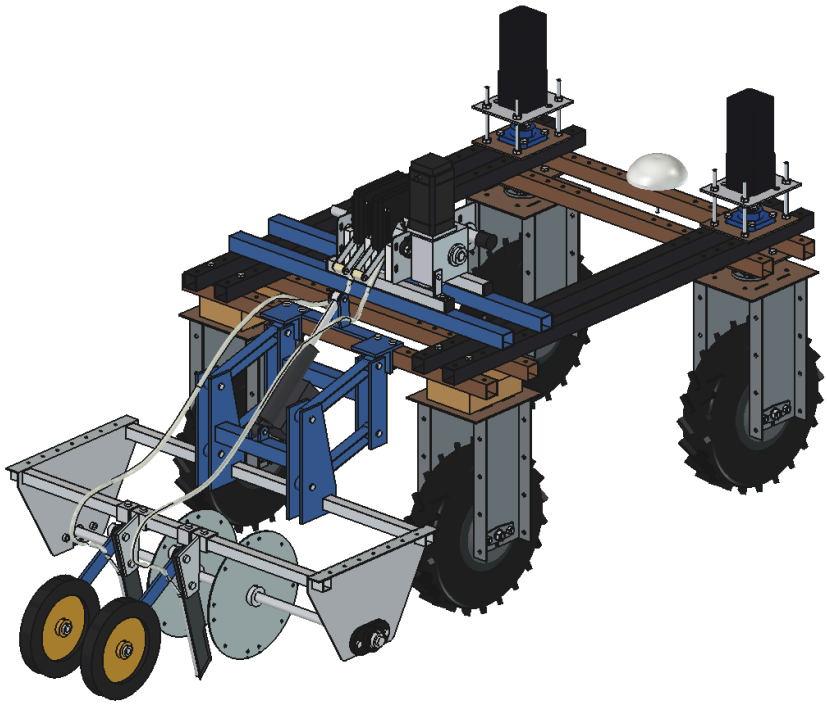


The first production version, built with Herre Bartlema from measured parts. A modular peristaltic roller pump on the robot, stepper-driven so the dose follows the driving speed; 5 or 10 sections.

- 10 ml per revolution per tube
- 5 or 10 sections
- Includes a running animation

Folder: `liquid fertilizer applicator/precision_applicator/roller_pump/`

Precision V2 on the robot

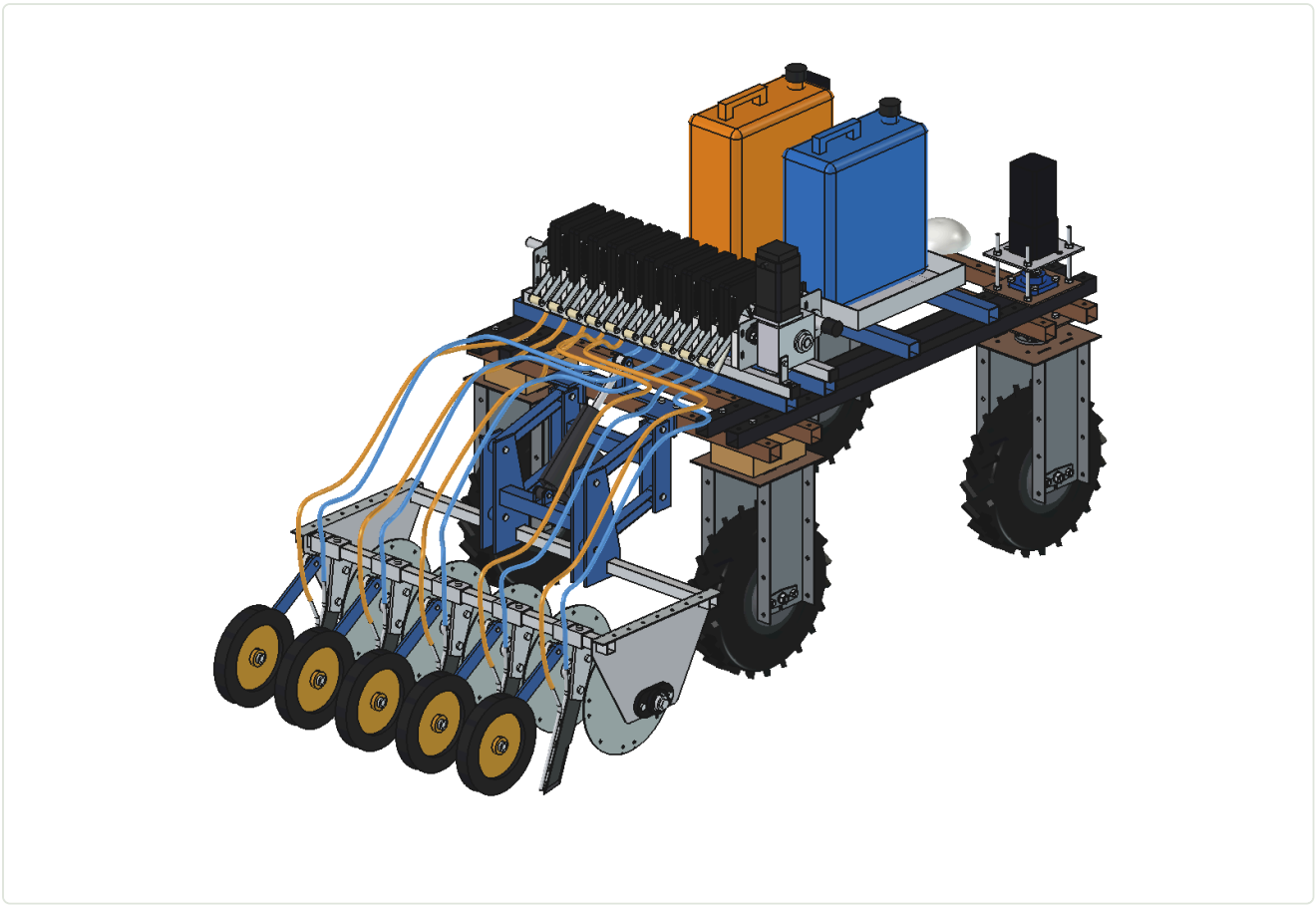


Applicator V2 with the roller pump on the robot: Ø 325 discs with injector knives behind them on a parallel linkage with an electric actuator, and a press wheel behind every knife that closes the slot. 5 rows, or 2 rows in the middle (applicator V1, combined).

- Disc, knife and press wheel per row
- Electric actuator lift
- Includes hoses and an animation

Folder: `liquid fertilizer applicator/precision_applicator/combined_v2/`

Precision V3: two liquids



Applicator V3 with 5 rows of trailing double knives (both outlets 50 mm deep) and press wheels, fed by the 10-section roller pump: 5 sections per liquid, so two products go in at once.

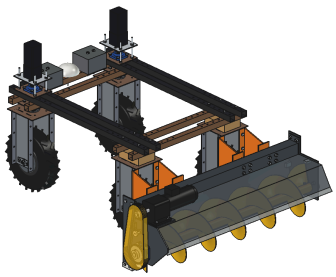
- 5 rows, double knives
- 10-section roller pump, 2 liquids
- Includes an animation

Folder: `liquid fertilizer applicator/precision_applicator/combined_v3/`

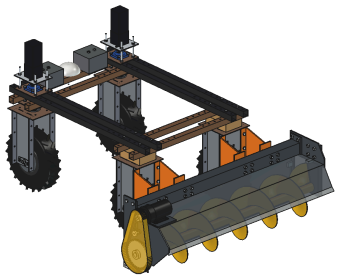
IMPLEMENT · TRANSPORT & GENERAL

Feed pusher auger

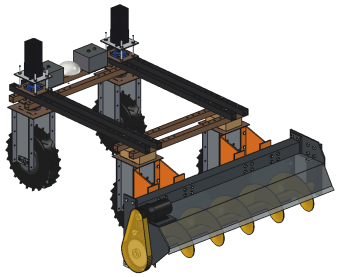
Pushes feed back to the feed fence. Sits at the back of the robot; the robot drives with the auger first. The auger picks up the pushed-away feed and carries it sideways to the fence.



V1



V2

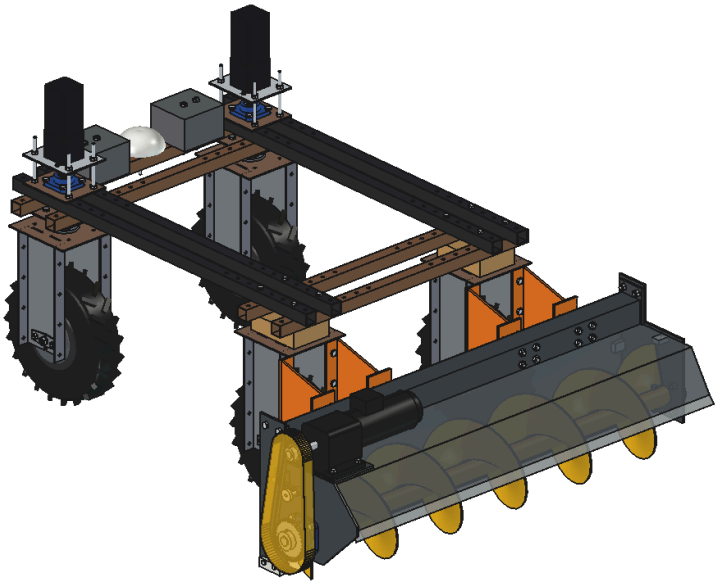


V3

3 variants: V1 · V2 · V3
Folder: `feed pusher auger/`

FEED PUSHER AUGER · VARIANT 1 OF 3

V1



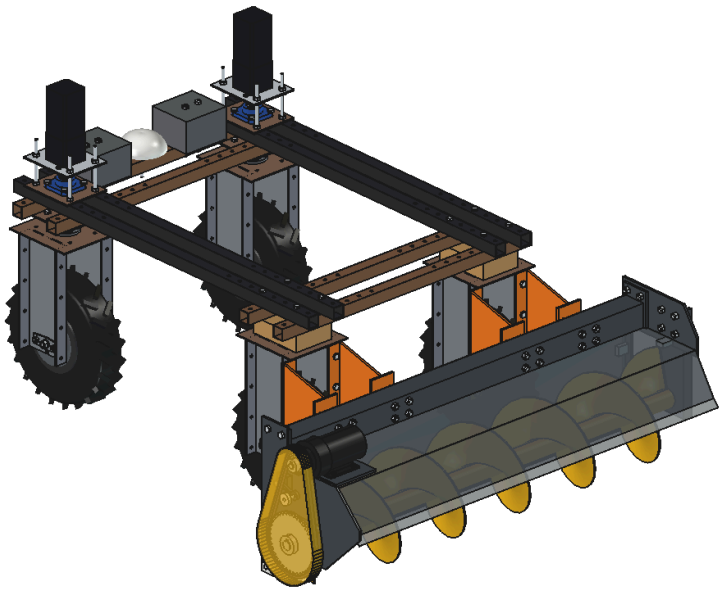
Ø320 auger under a 2 mm folded hood with 4 bends. Foot-mounted gearmotor on top of the hood, 08B-1 chain with tensioner along the left side plate. Mounted with 4 arms on the rear flange holes of the rear wheel modules. Open discharge end. Includes a feed-transport model and a force animation.

- 24 V gearmotor with chain drive
- Open discharge end
- Feed-transport model and force animation

Folder: `feed pusher auger/V1/`

FEED PUSHER AUGER · VARIANT 2 OF 3

V2



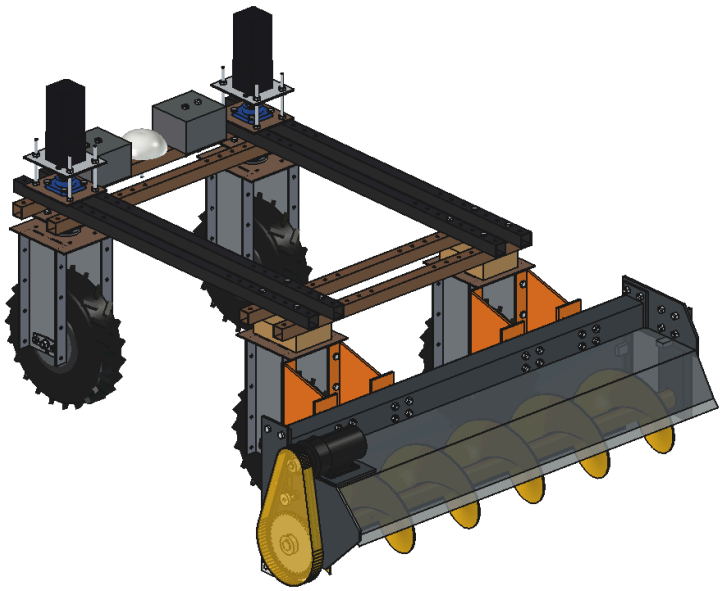
V1 without a through shaft: the free end hangs on a strap with a Ø25 pin and bearings inside the auger, the drive end has a stepped drive bush with an M6 shear bolt. The auger slides out through a cutout in the end plate.

- Auger removable without opening the frame
- Frame, hood and drive as V1
- M6 shear bolt breaks at 252 Nm

Folder: `feed pusher auger/V2/`

FEED PUSHER AUGER · VARIANT 3 OF 3

V3



V2 with the existing, measured auger: Ø300 flight on a Ø60 x 3 core tube, M12 shear bolt between welded rings, and a wear-resistant plastic ring on a fixed Ø20 pin at the free end instead of bearings.

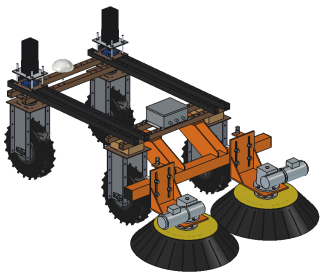
- Measured existing auger
- No bearings at the free end
- Frame, hood and drive as V2

Folder: `feed pusher auger/V3/`

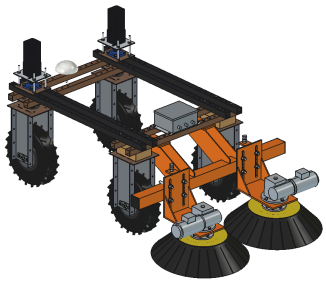
IMPLEMENT · TRANSPORT & GENERAL

Disc brushes

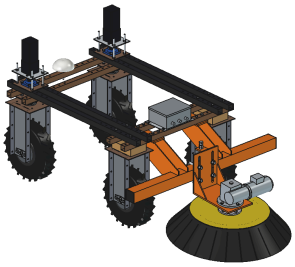
Sweeper disc brushes for loose material on concrete. Each brush on its own 48 V worm gearmotor (NMRV063, 550 W); reversing the motors makes a windrow in the middle or spreads material wide. Mounted on an 80 × 80 toolbar under the rear beams with U-bolts. Includes an animation with a particle model.



Double 750



Double 650

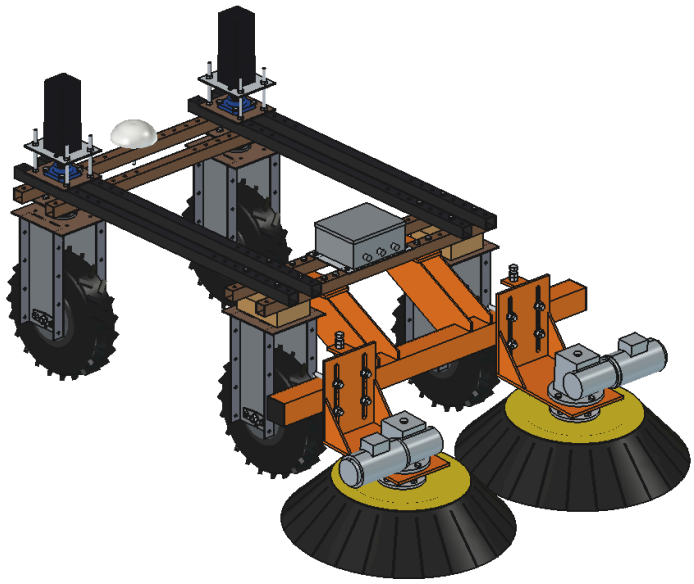


Single 850

3 variants: Double 750 · Double 650 · Single 850
Folder: `brushes/`

DISC BRUSHES · VARIANT 1 OF 3

Double 750



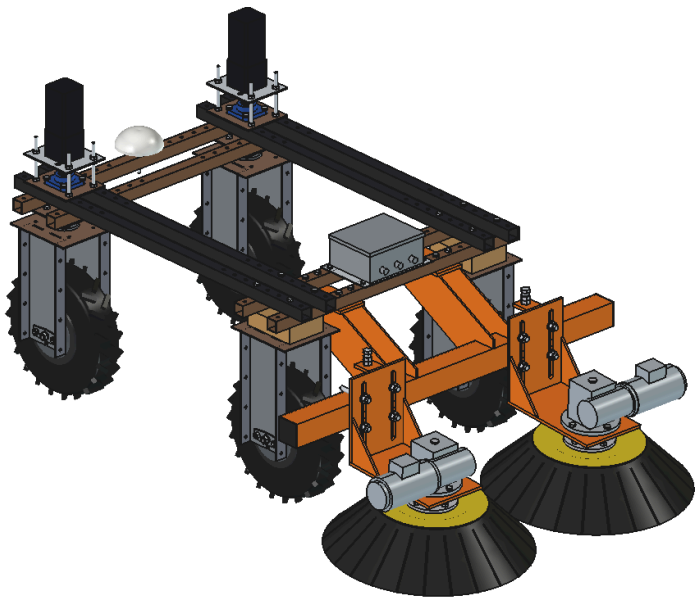
Two Ø750 brushes (standard). Working width 1.50 m with brushes touching to 1.70 m with 200 mm between them. Spread wide or make a windrow.

2 × Ø750	1.50–1.70 m

Folder: `brushes/simple/`

DISC BRUSHES · VARIANT 2 OF 3

Double 650



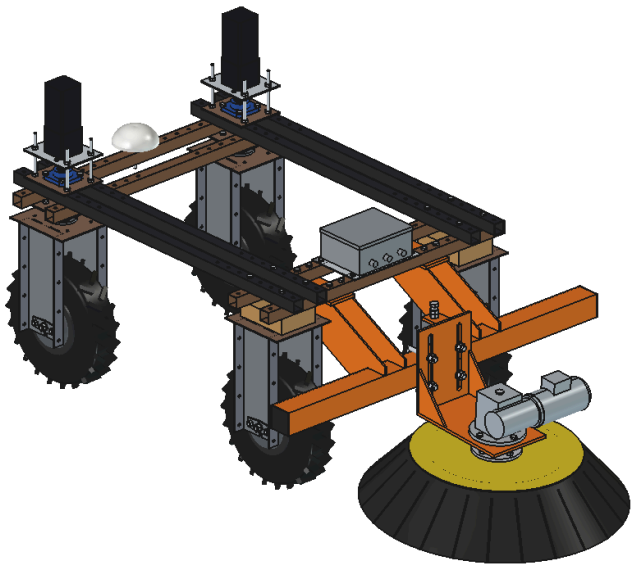
Two Ø650 brushes: the same job, narrower and lighter.

2 × Ø650	1.30–1.50 m

Folder: `brushes/simple/`

DISC BRUSHES · VARIANT 3 OF 3

Single 850



One Ø850 brush covering the robot's own width, sweeping the strip in front of the robot to one side.

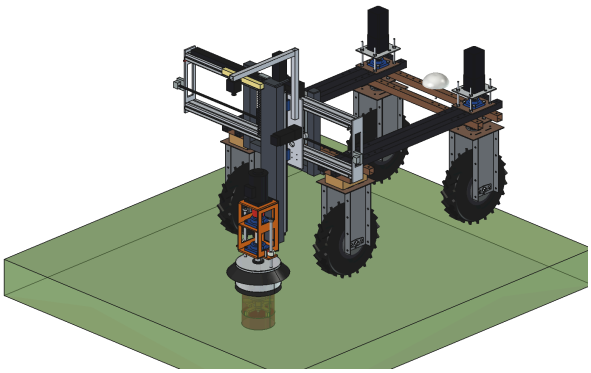
1 × Ø850	0.85 m

Folder: `brushes/simple/`

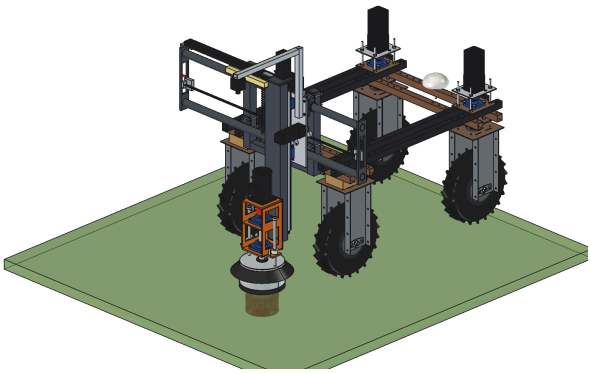
IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Dockweed drill

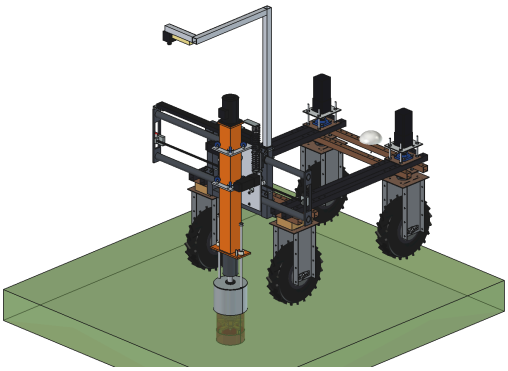
Finds a dock plant (broad-leaved dock) and mills out its root on a CNC gantry behind the robot, chemical-free. Six numbered versions that each change one part of the machine.



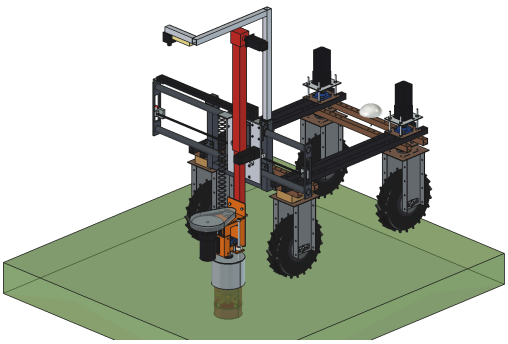
v1



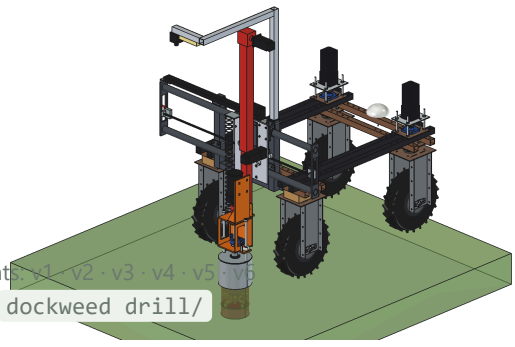
v2



v3

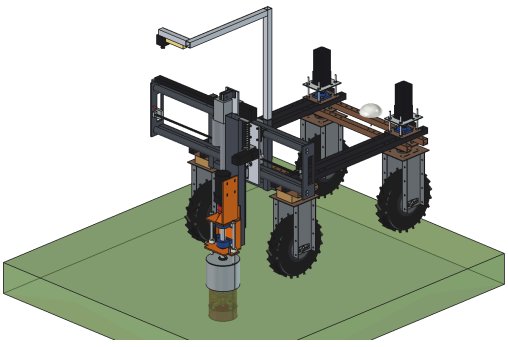


v4



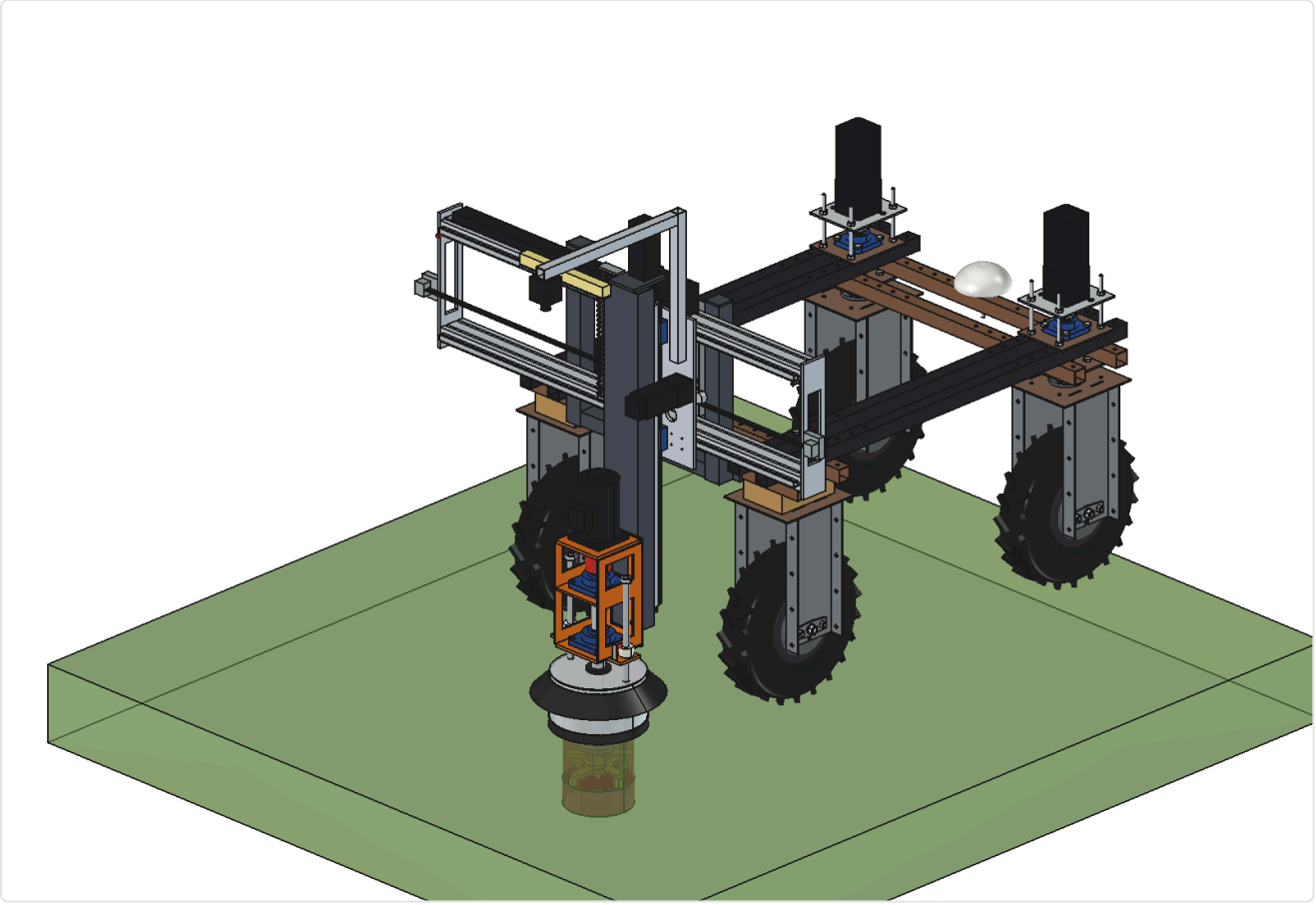
6 variants: v1 · v2 · v3 · v4 · v5 · v6
Folder: `dockweed drill/`

v5



v6

v1

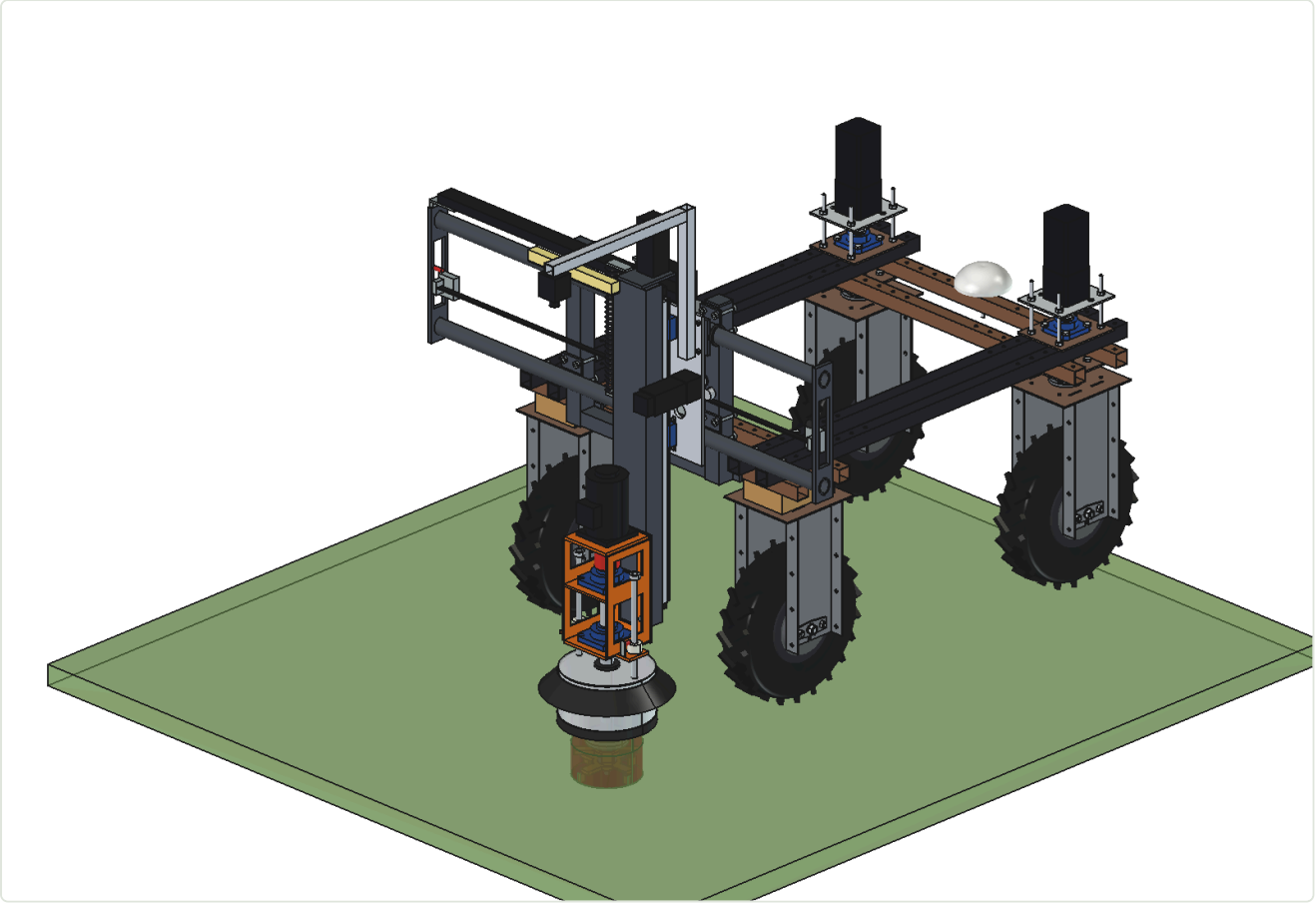


X slide on 2 aluminium profiles 40 x 80 with HGR15 rails and 4 carriages; cutter Ø180 at 1500 rpm in a pot Ø250 and a mowing disc as second tool.

- X stroke 1000 mm
- HGR15 rails
- Includes an animation

Folder: `dockweed drill/v1/`

v2

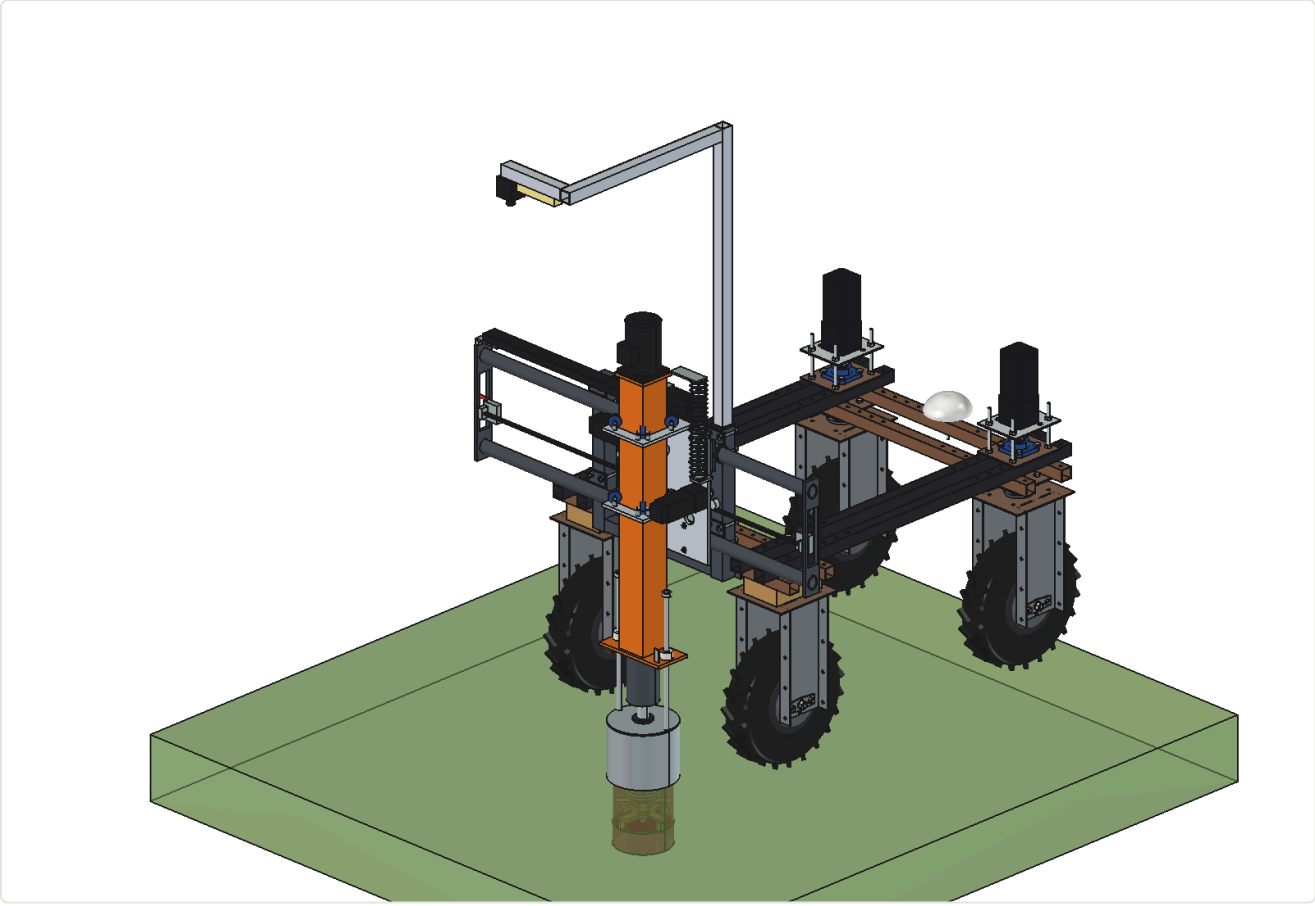


X slide on PTO-shaft tubes. The robot is the Y axis and stops with the plant under the spindle; the cutter goes 150 mm deep and grinds up the root crown and upper taproot.

- X slide on PTO tubes
- Cutter 150 mm deep
- Animations: three plants, nettle mowing

Folder: `dockweed drill/v2/`

v3

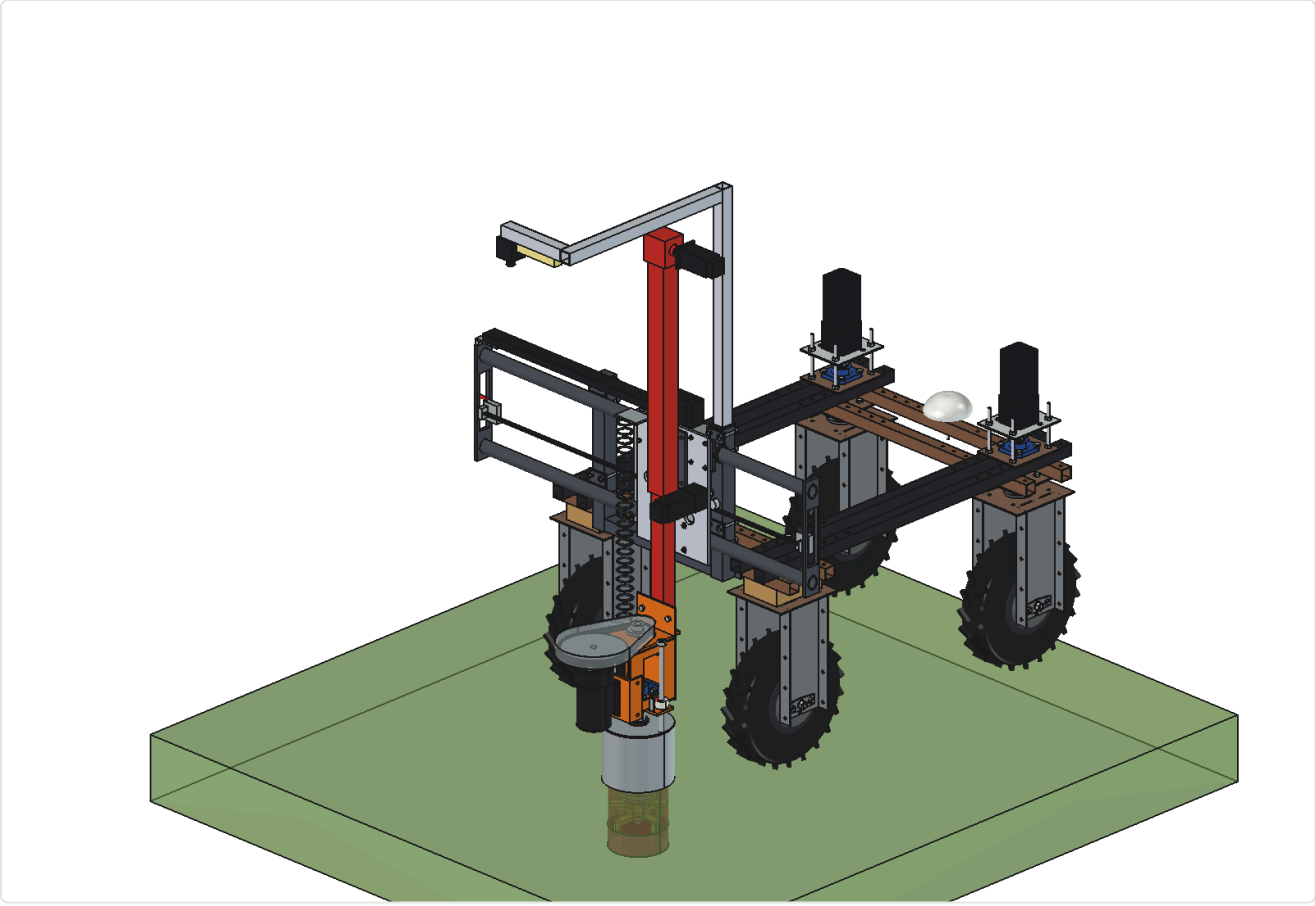


v2 with a cheaper Z axis: a 100 x 100 mast with the spindle inside, rolling on 8 V-groove corner rollers, and a steel can like Robot Ruud's.

- Z mast on corner rollers
- Ruud-style can
- Includes an animation

Folder: `dockweed drill/v3/`

v4

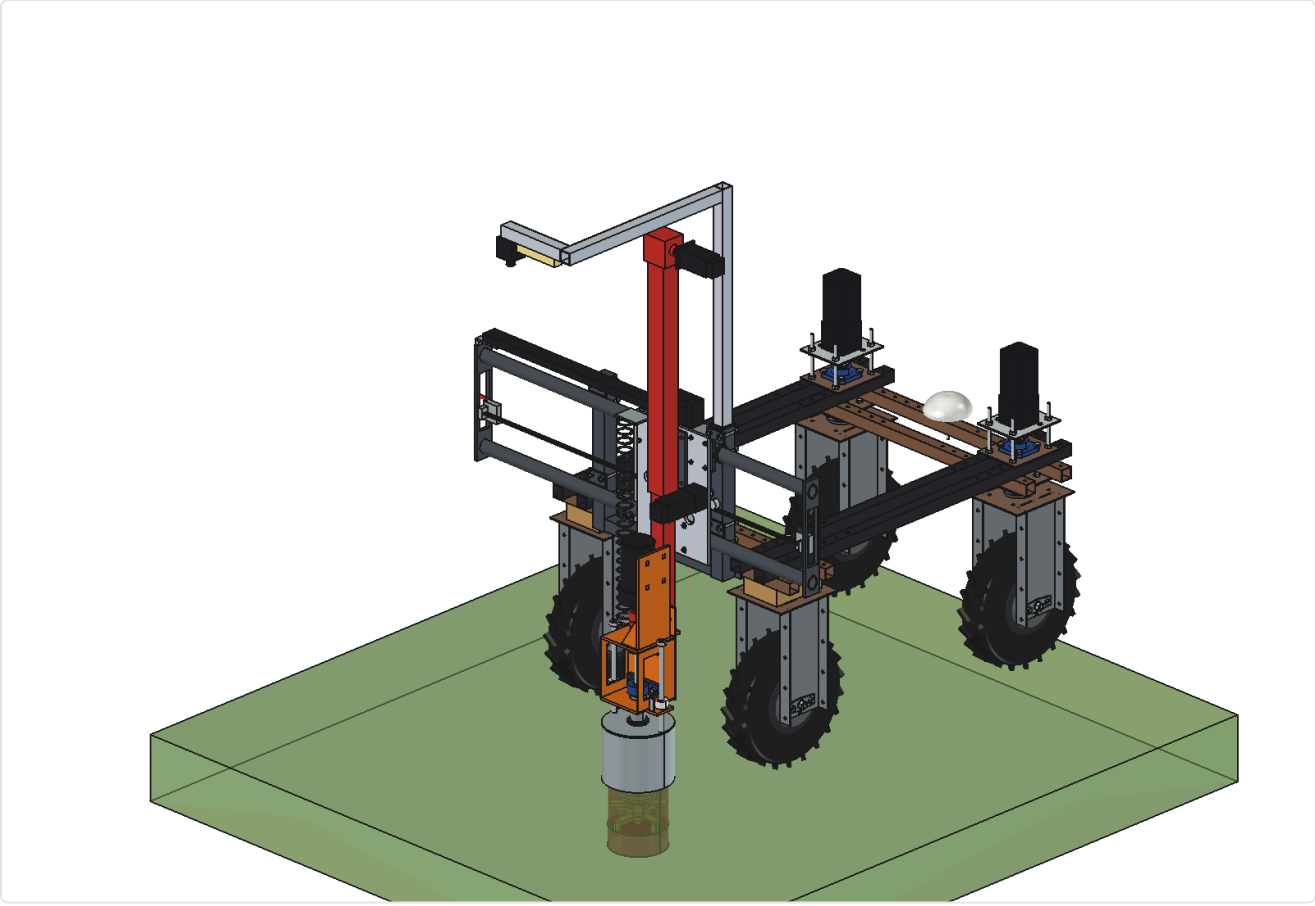


v3 with a drop-leg trailer jack as the Z axis (stroke 380 mm, NEMA 23) and the 800 W gearmotor with chain on 2 x UCP 204.

- Trailer jack as Z
- 800 W gearmotor + chain
- Includes an animation

Folder: `dockweed drill/v4/`

v5

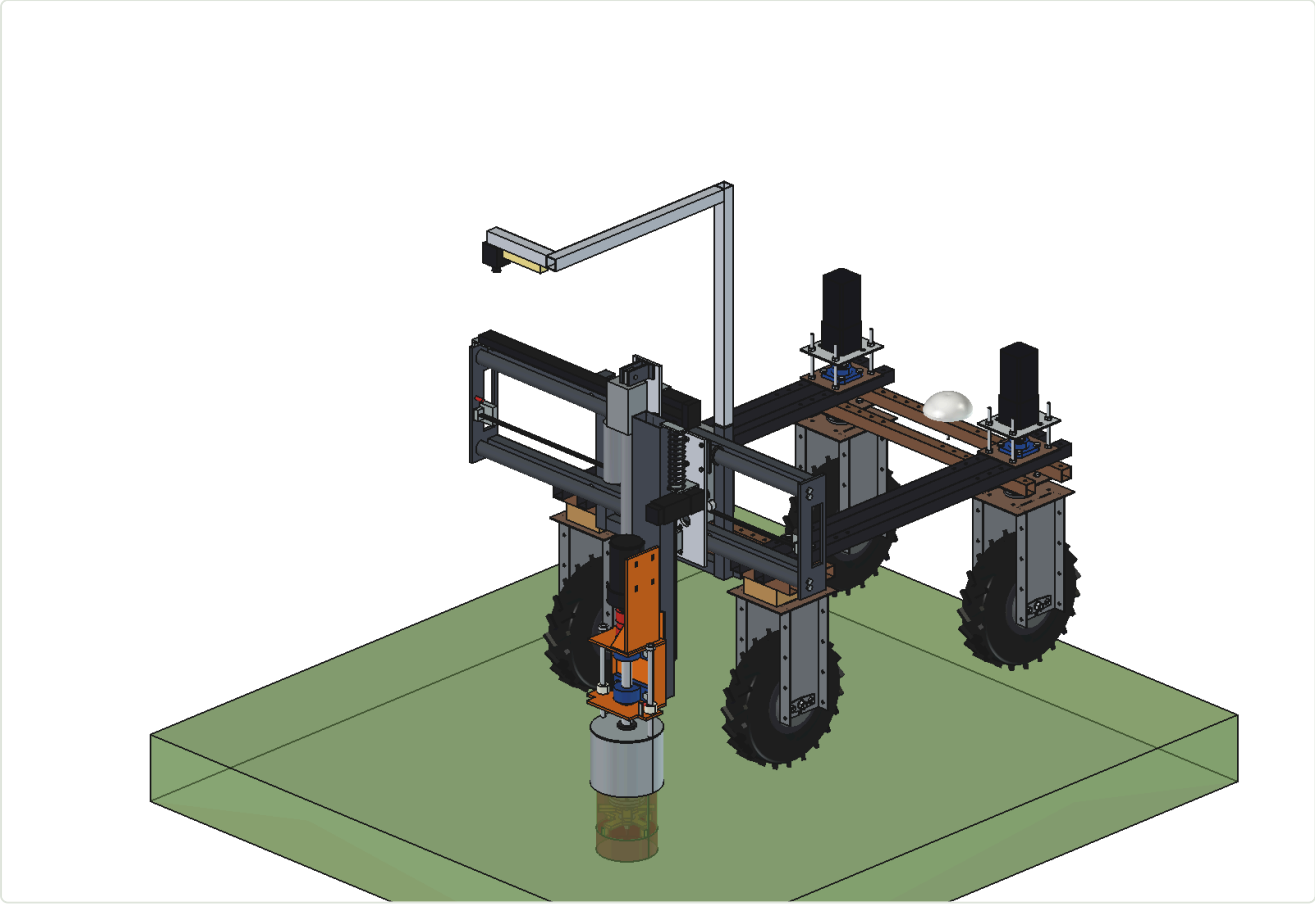


v4 with the 800 W gearmotor directly above the spindle through a flexible jaw coupling: no chain, no sprockets.

- Direct drive, jaw coupling
- No chain guard
- No animation

Folder: `dockweed drill/v5/`

v6



v5 with a 40 x 40 frame, PTO tubes bolted between 12 mm side plates, a Z tube on HGR15 rails driven by a 24 V actuator, and a Ø30 spindle in 2 x UCP 206.

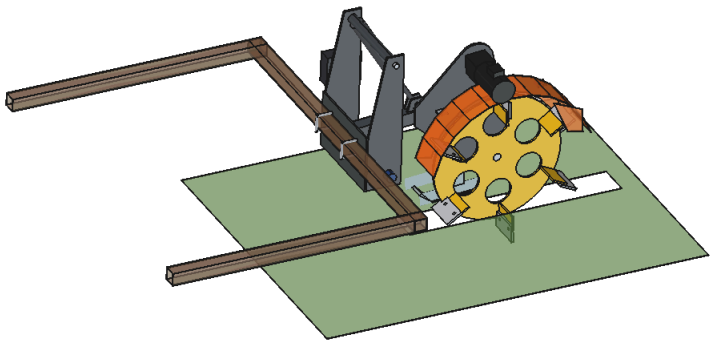
- Bolted PTO tubes
- 24 V actuator Z axis
- Ø30 spindle in UCP 206

Folder: `dockweed drill/v6/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Trencher

Wheel trencher for cutting narrow trenches.

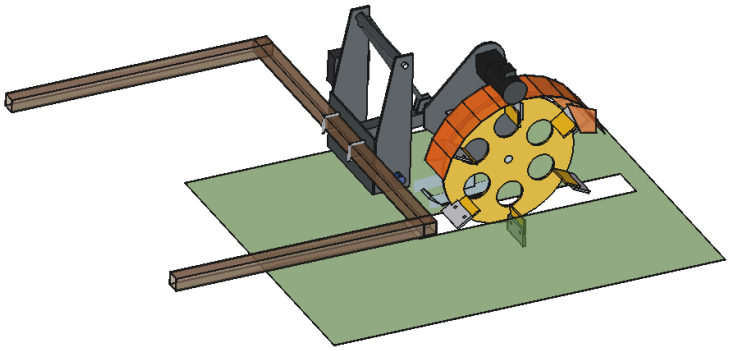


One design

1 variant: One design
Folder: `trencher/`

TRENCHER · VARIANT 1 OF 1

One design



Wheel trencher mounted on the Bruut. Renders and STEP export. An older single-script design (trencher_design.py).

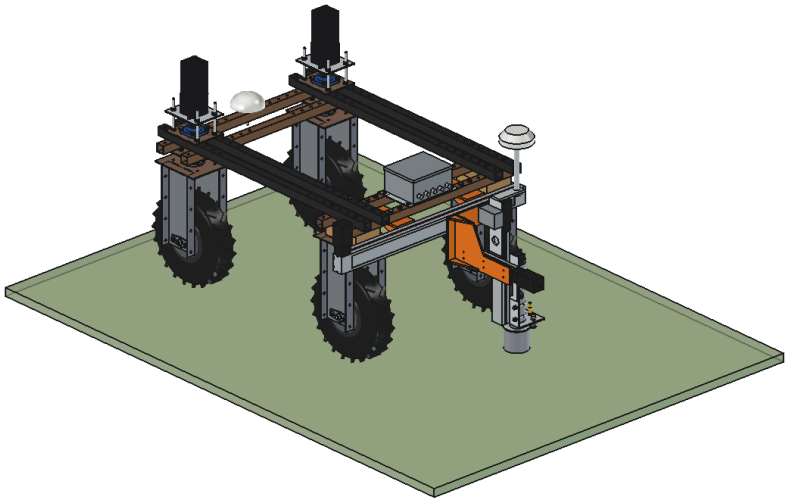
> Renders and STEP export

Folder: `trencher/`

IMPLEMENT · MEASURING & SENSORS

Field marker

Stakes out points from a coordinate list (piles, posts, planting positions, plot corners). The robot stops roughly behind the point; an XYZ stage puts a spray-paint head exactly on it.

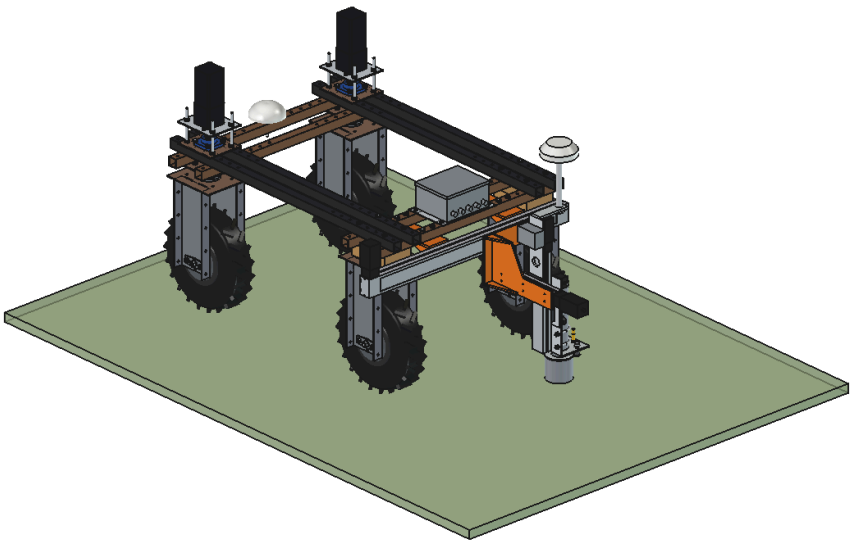


Simple

1 variant: Simple
Folder: `marker/`

FIELD MARKER · VARIANT 1 OF 1

Simple



Three bought linear modules: X 800 mm belt module, Y 200 mm and Z 300 mm ball screws. Paint head: upside-down marking-spray can in a PVC drain pipe pushed by a servo; sprung PVC 110 shroud with an inductive sensor finds the ground. Second GNSS antenna on a mast straight above the nozzle.

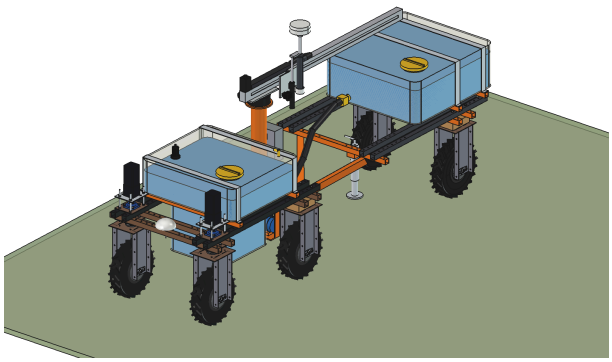
- Robot may stop ±100 mm off
- NEMA 23 closed-loop motors
- About €1,450 in parts
- Concept lists flag, peg, punch and soil-probe heads

Folder: `marker/simple/`

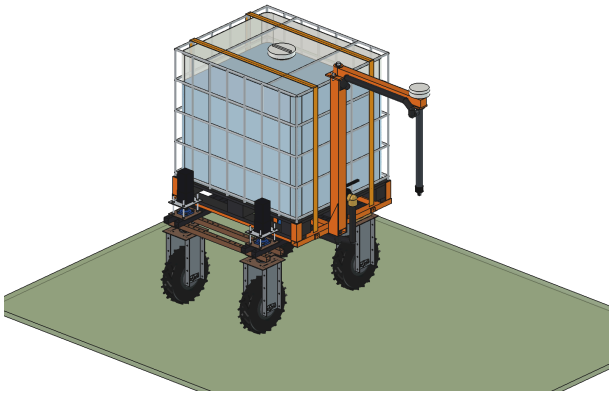
IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Mobile refiller

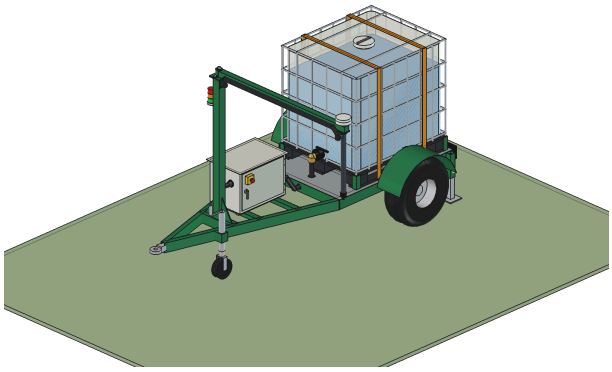
Support robot that brings liquid (and in concept seed, granular fertilizer, water, batteries) to the working robots, so they never leave the field.



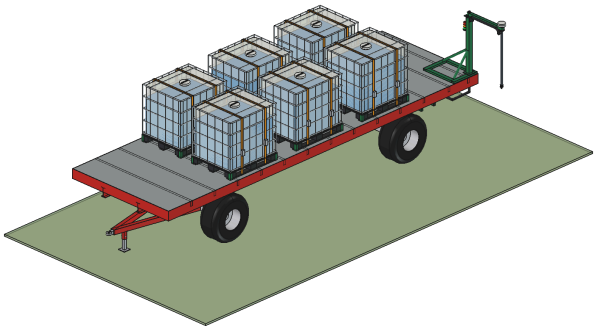
Liquid



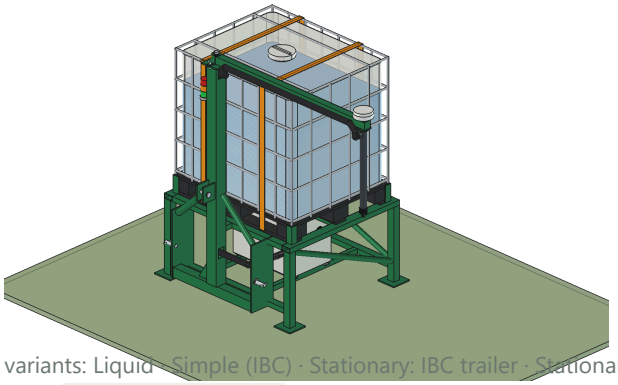
Simple (IBC)



Stationary: IBC trailer



Stationary: flat wagon



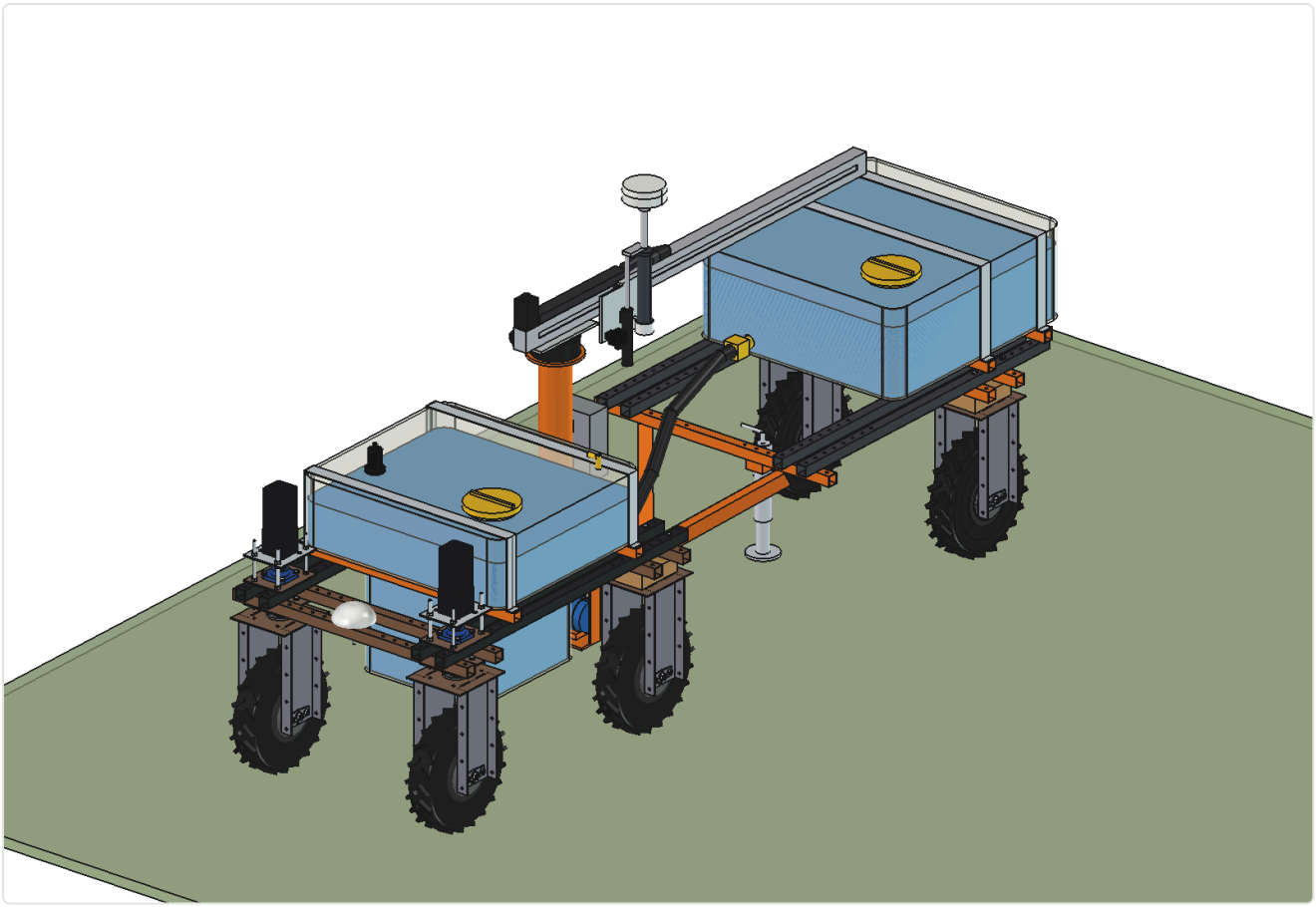
Stationary: three-point frame

5 variants: Liquid · Simple (IBC) · Stationary: IBC trailer · Stationary: flat wagon · Stationary: three-point frame

Folder: `mobile refiller/`

MOBILE REFILLER · VARIANT 1 OF 5

Liquid



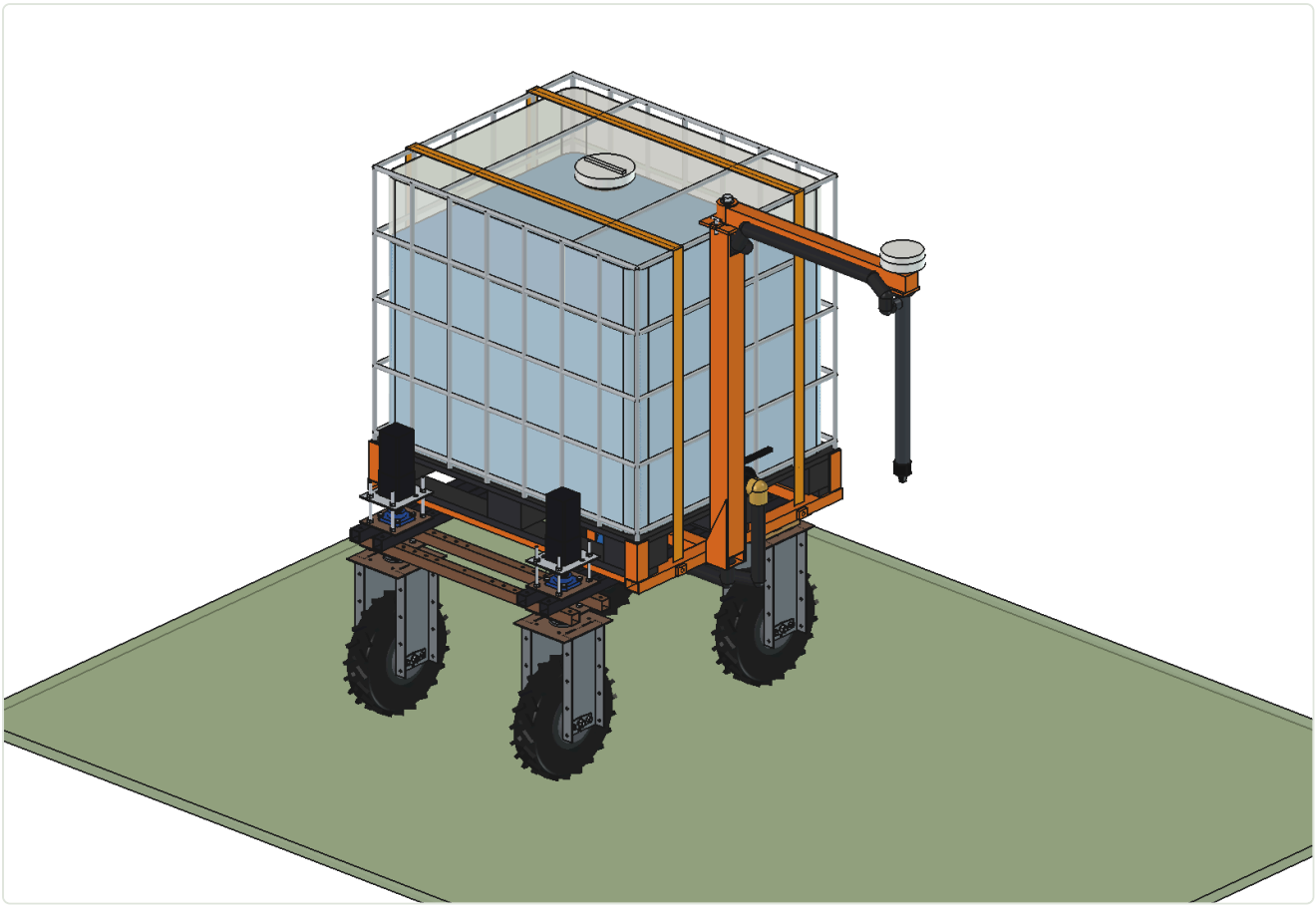
268 l T-shaped tank with a pump below the liquid level, a slewing boom with a trolley and a spout, an RTK antenna on the spout and a camera for confirmation, and a powered trailer of 246 l (6 driven wheels). Tops up the working robot in the straight reverse of its headland turn while both drive.

- About 25 l per turn, ≈ 3 s standstill
- Pump 150 l/min
- About €6,000 in parts

Folder: `mobile refiller/liquid/`

MOBILE REFILLER · VARIANT 2 OF 5

Simple (IBC)



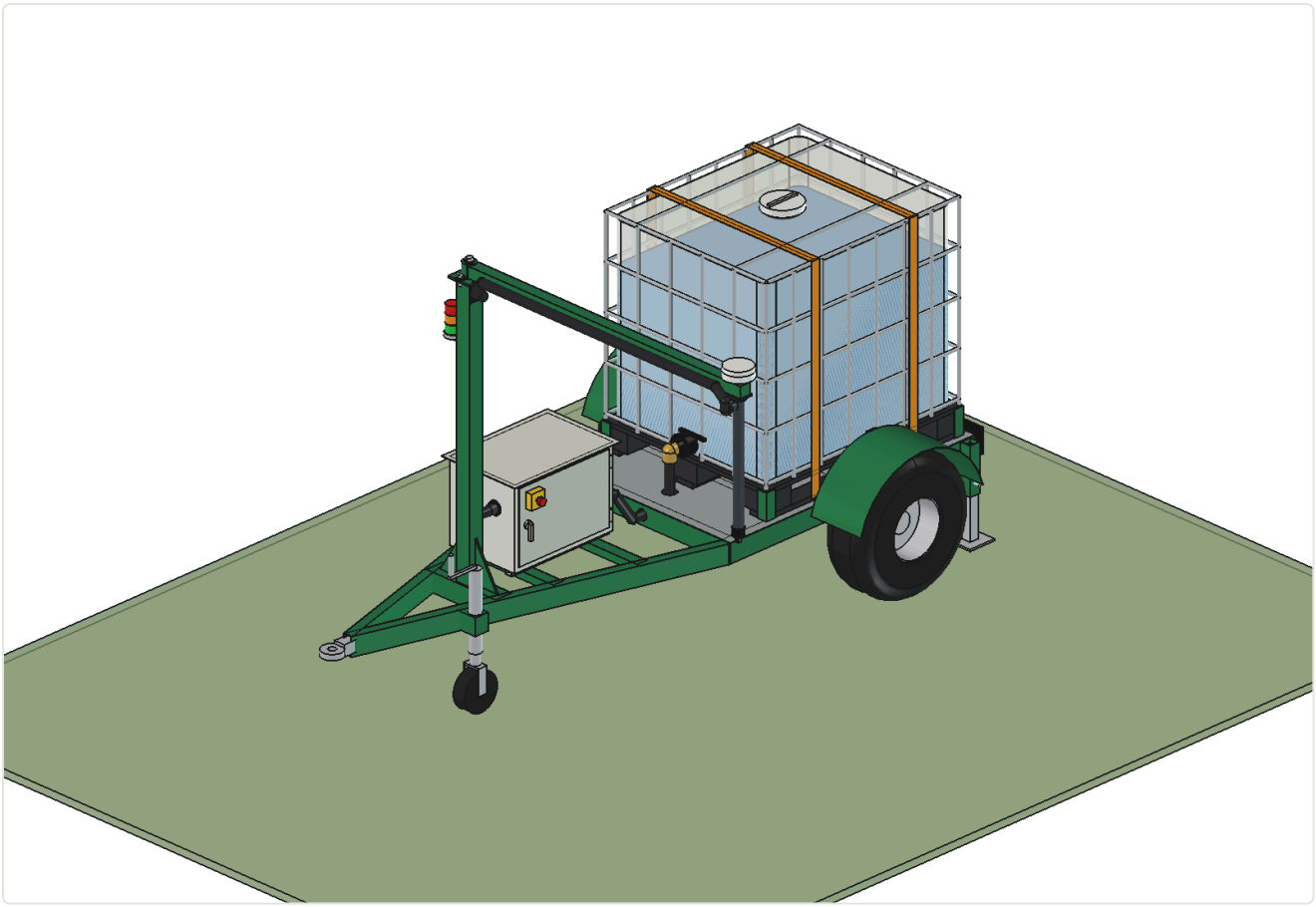
A standard 1000 l IBC set on with pallet forks, pump under the IBC, a fixed mast with a hanging hose and the RTK antenna above the nozzle. The working robot skips passes so every turn is a U-turn; the refiller stands inside the turn and the robot stops ≈ 13 s under the nozzle; 1 m further per refill.

- Standard 1000 l IBC
- No boom, trailer or camera
- About €1,650 in parts

Folder: `mobile refiller/simple/`

MOBILE REFILLER · VARIANT 3 OF 5

Stationary: IBC trailer



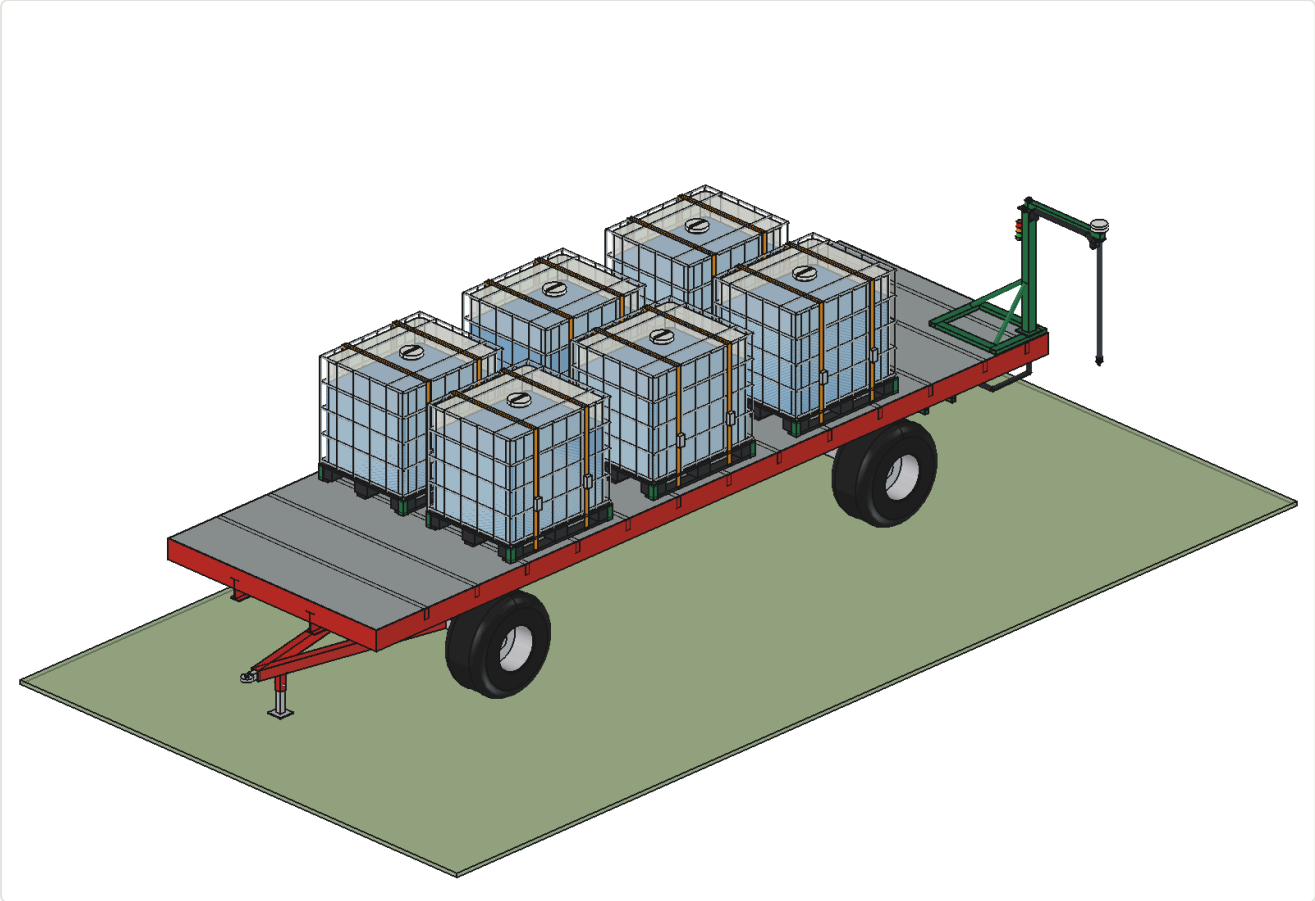
Single-axle trailer for one or two 1000 l IBCs, parked on the headland by a tractor. Control box (pump, battery, controller, RTK receiver, radio) on the A-frame drawbar and a mast with swinging arm, drop pipe, drip-stop nozzle and RTK antenna straight above the nozzle.

- 1 or 2 IBCs
- 120 l per visit
- The working robot drives to it

Folder: `mobile refiller/stationary/ibc_trailer/`

MOBILE REFILLER · VARIANT 4 OF 5

Stationary: flat wagon



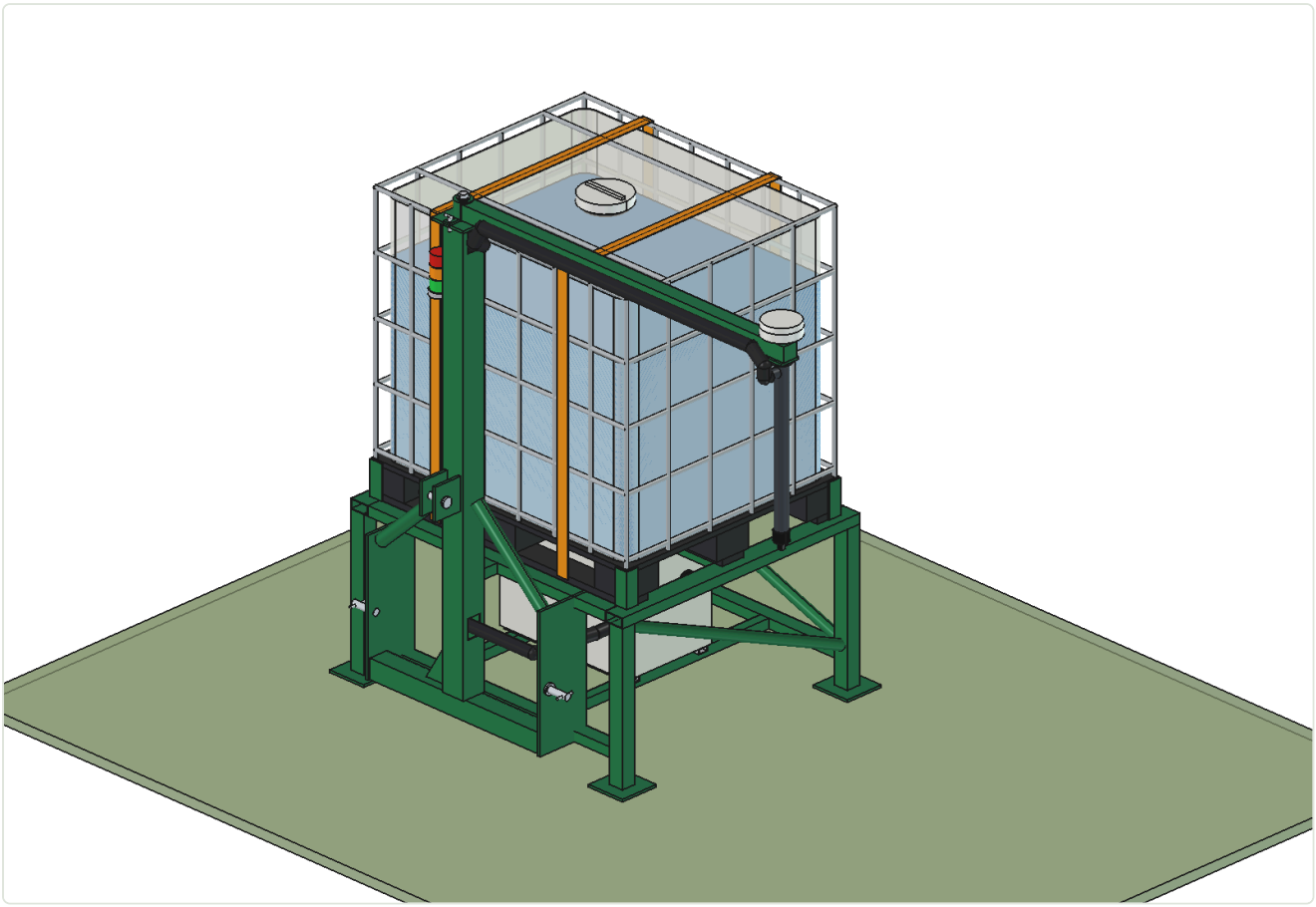
Standard two-axle flat bale wagon (8 × 2.48 m) with 6 IBCs, two across. Only two holes in the deck and two box hangers; suction header along the deck, control box under the deck, mast on a skid at the rear left corner.

- 6 IBCs
- Pump 0.8 m below the IBC bottoms
- Wagon itself unchanged

Folder: `mobile refiller/stationary/flat_wagon/`

MOBILE REFILLER · VARIANT 5 OF 5

Stationary: three-point frame



Welded frame on four legs for one 1000 l IBC, picked up on a category 2 three-point linkage and set down on the headland. The mast doubles as the hitch tower; control box under the platform.

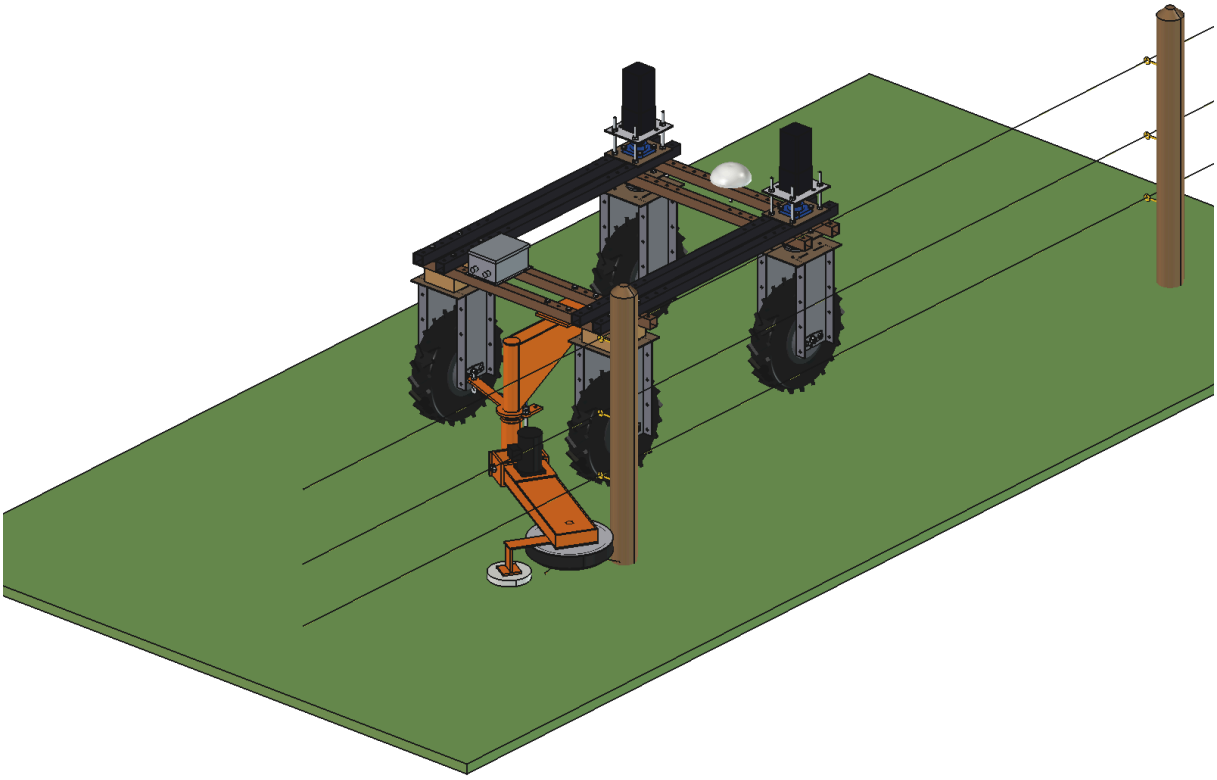
- 1 IBC
- Category 2 three-point
- Pump 535 mm below the IBC bottom

Folder: `mobile refiller/stationary/three_point/`

IMPLEMENT · MOWING

Around-pole fence mower

Mows the strip under an electric fence and goes around the posts by itself.

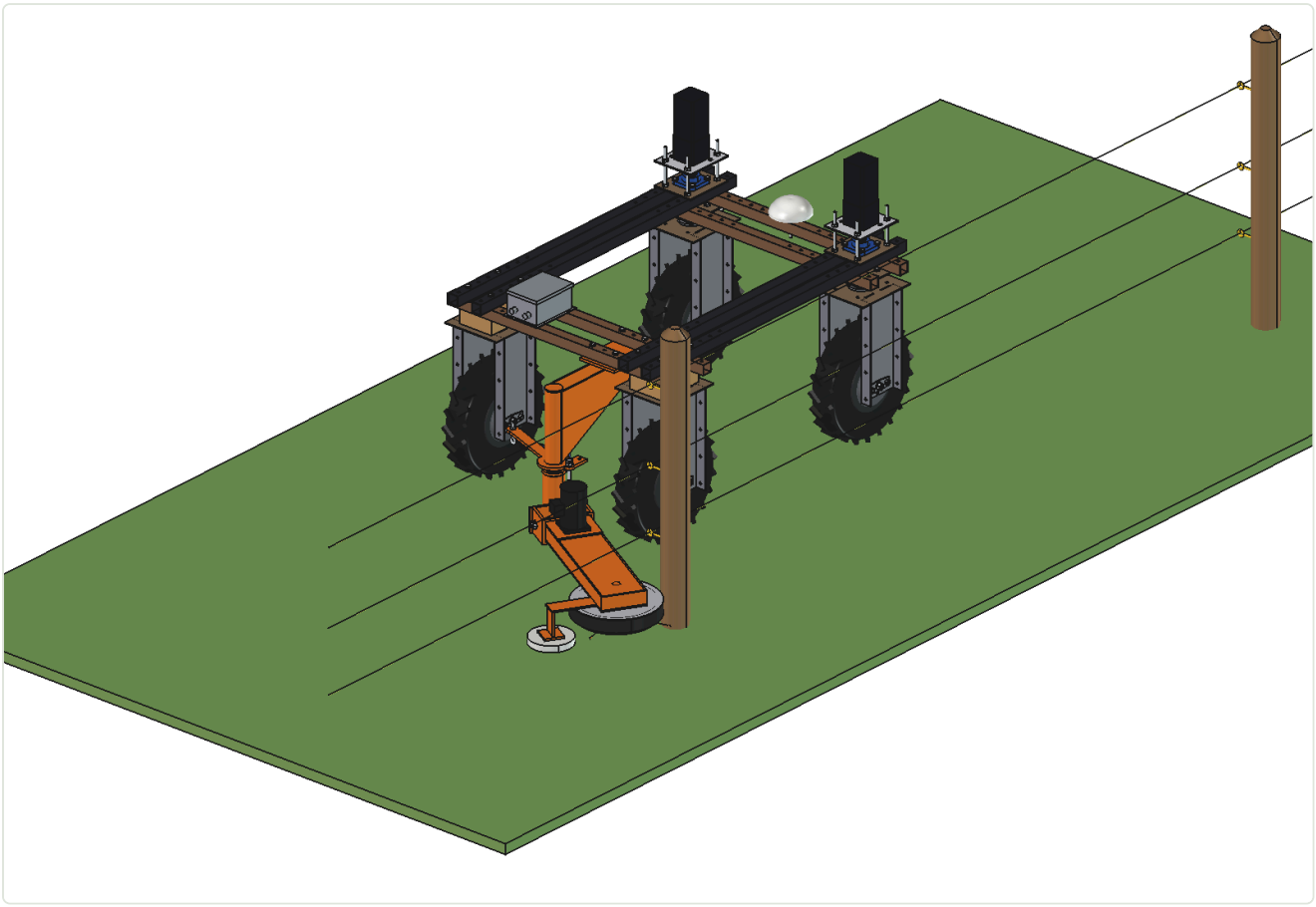


Simple

1 variant: Simple
Folder: `around_pole_mower/`

AROUND-POLE FENCE MOWER · VARIANT 1 OF 1

Simple



Trailing swing arm on a post behind the right rear wheel, held on a rest stop by a spring. A Ø400 nylon-string head under a PE-pipe guard ring slides around each post. 48 V BLDC motor at the root of the arm, V-belt inside the arm to a bought lawn-tractor deck spindle; head 182 mm high on an HDPE skid. No sensor, no actuator, no software.

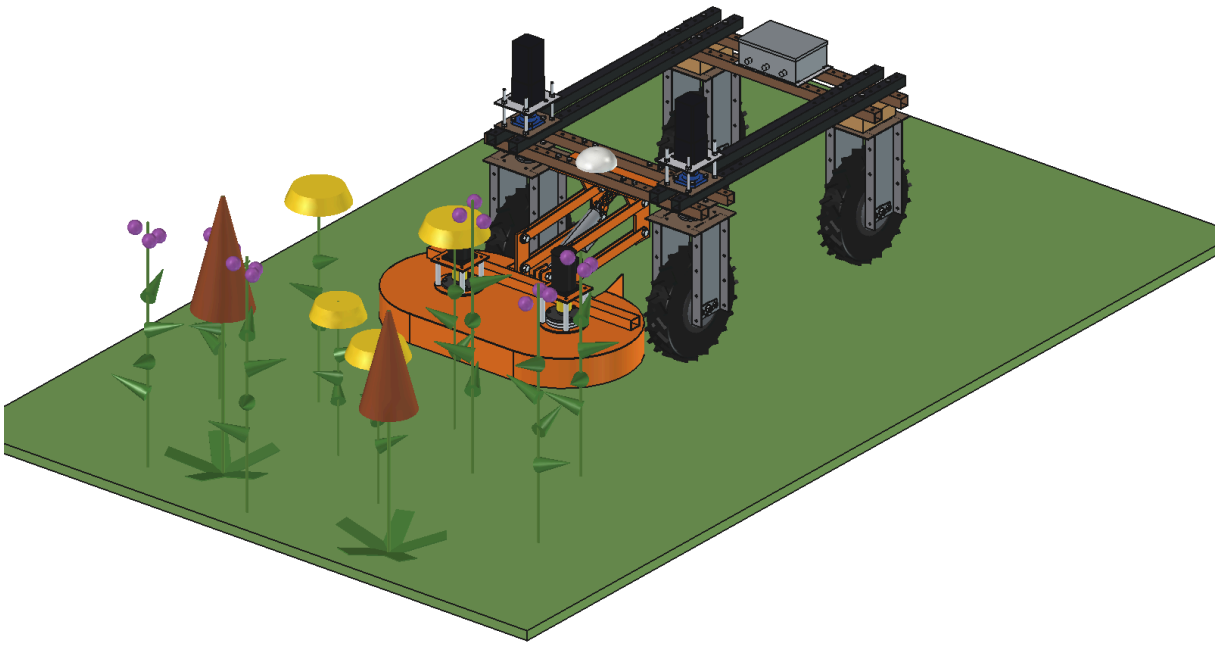
- ≈ 615 W at 0.5 m/s
- 1.8 km of fence per hour
- About €575 in parts
- Arm dynamics and animation with grass

Folder: `around_pole_mower/simple/`

IMPLEMENT · MOWING

Weed picker

Tops thistles, ragwort and dock above the grass and gathers them in one windrow, no bin. Ragwort in particular has to leave the field.

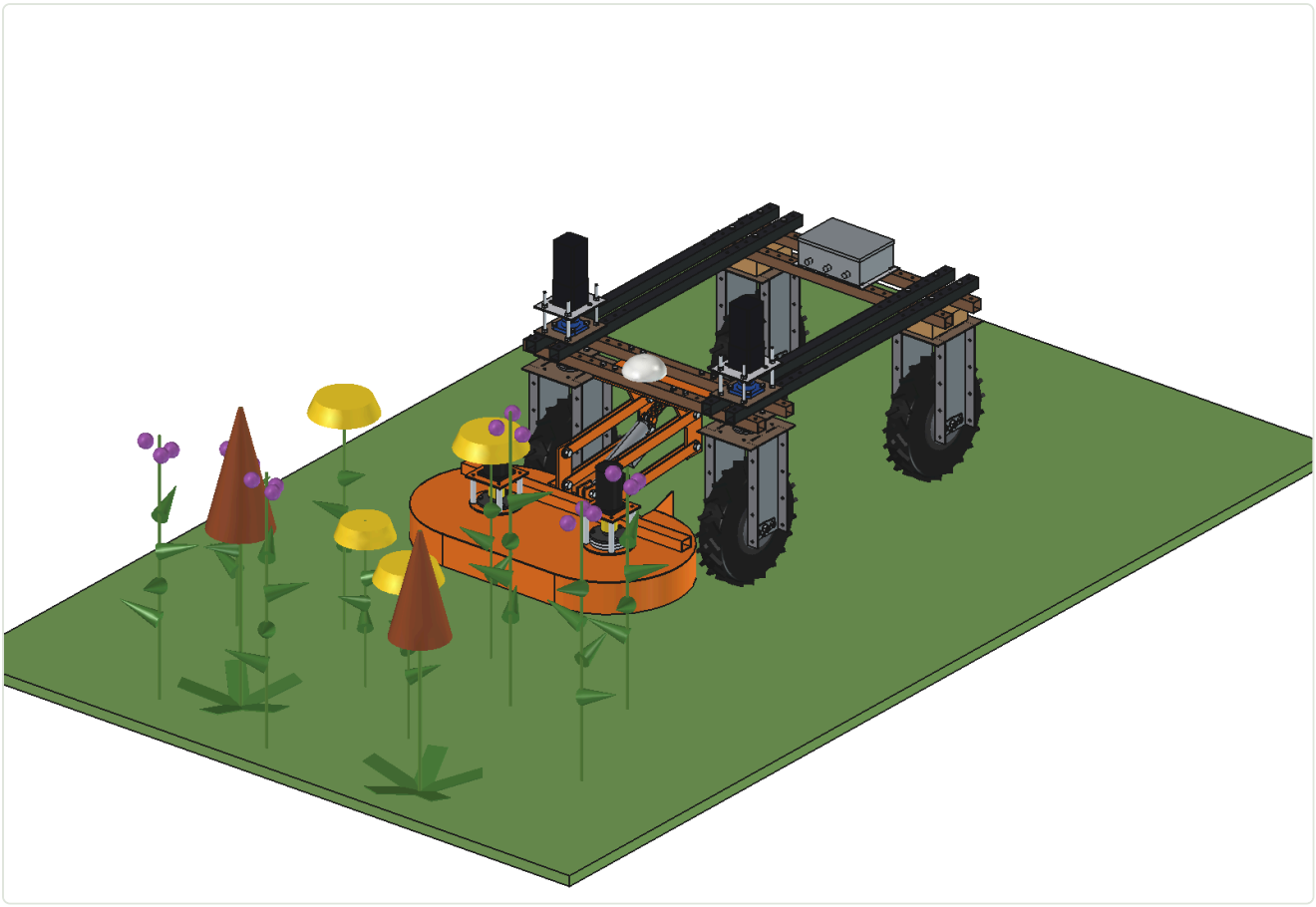


Mowing deck

1 variant: Mowing deck
Folder: `weed picker/`

WEED PICKER · VARIANT 1 OF 1

Mowing deck



Twin-rotor mowing deck in front of the robot: two rotors with free-swinging blades (Ø510, 990 mm cut) turn inwards and throw the chopped weed out of a centre opening as a windrow ≈ 320 mm wide. Each rotor: 48 V BLDC motor on a lawn-tractor deck spindle; the left rotor runs 24 mm higher. Cut height 100–400 mm from a 4000 N actuator on a parallel linkage, with a float link for humps.

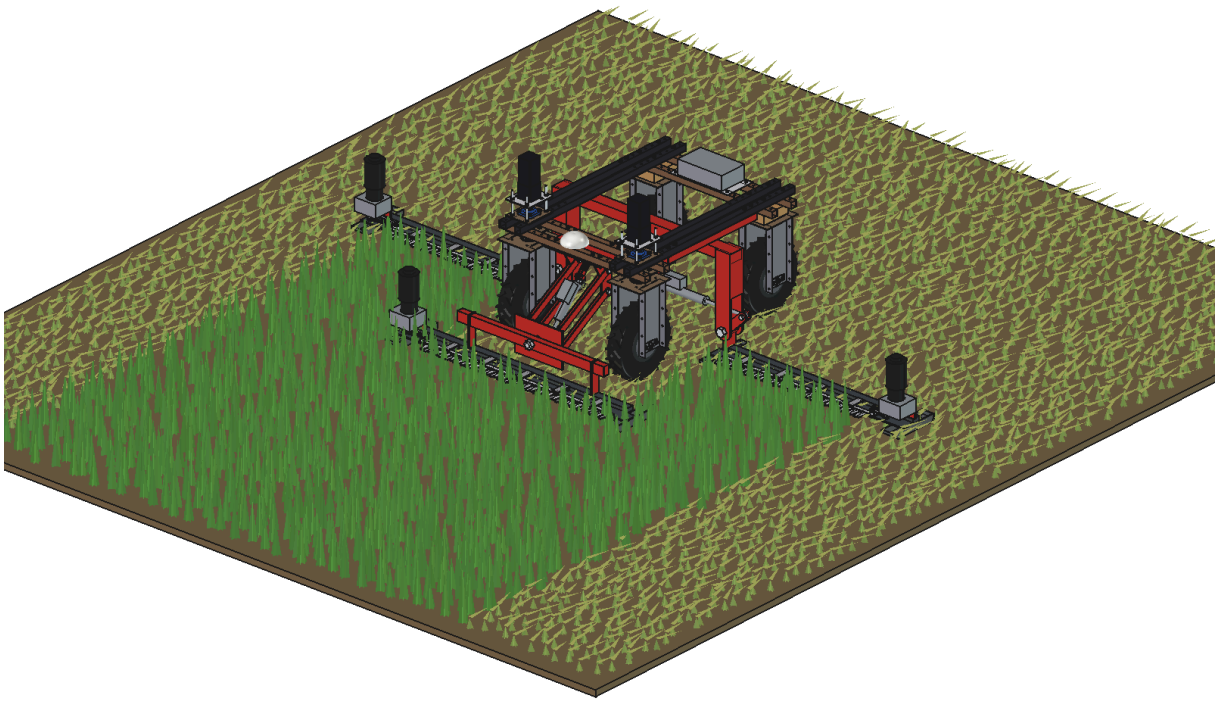
- ≈ 1 kW at 0.5 m/s
- 82 kg (rear counterweight advised)
- About €1,040 in parts
- Animation in a weed patch

Folder: `weed picker/mowing_deck/`

IMPLEMENT · MOWING

Sickle bar mower

Mows grass with double-knife cutter bars instead of rotating blades: a clean cut at low power, the grass left lying in a swath to dry. The bars follow the ground on their shoes.

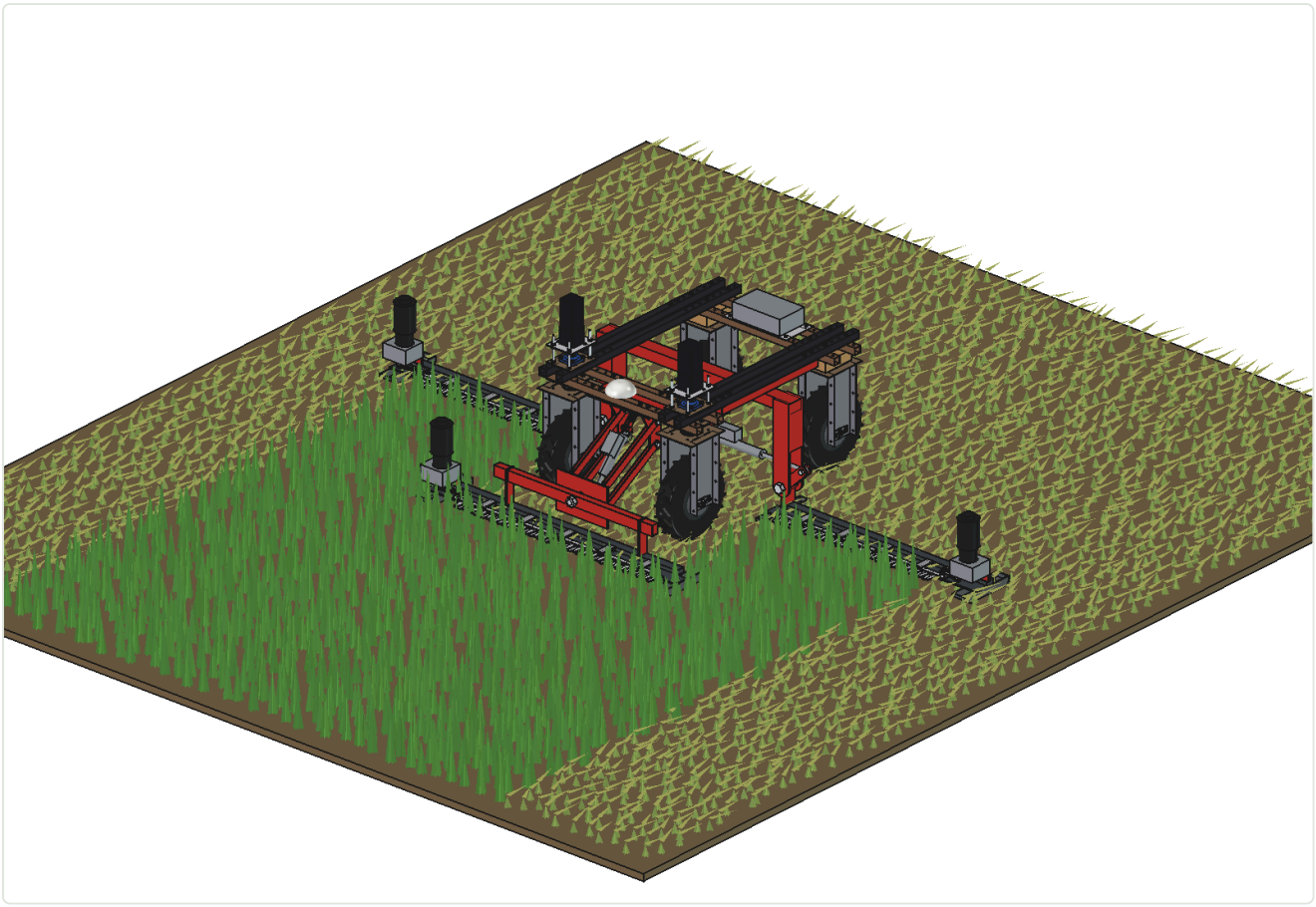


Triple

1 variant: Triple
Folder: `sickle bar mower/`

SICKLE BAR MOWER · VARIANT 1 OF 1

Triple



Three double-knife bars: 1.25 m in front of the robot on a parallel linkage with a float link and a roll pin, and two 1.0 m wings in the middle of the frame on a mid frame under the top beams. Each bar: upper and lower knife (50 mm pitch, stroke 50, 600 rpm), a crank gearbox with two cranks 180° apart so the knife forces cancel, and a 48 V 1000 W BLDC motor. The wings fold up with a 4000 N actuator each; the anchor slides in a slot so they float with the ground.

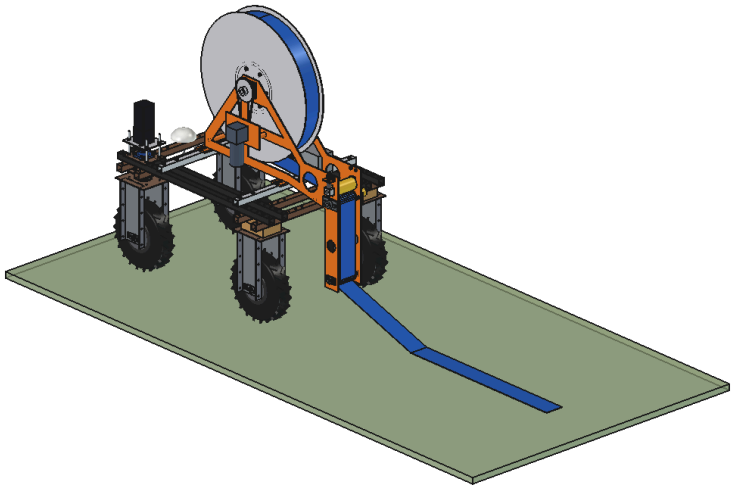
- 3.25 m working width
- ≈ 2.2 kW mowing at 1 m/s, 1.0 ha/h
- 210 kg, 163 kg on the shoes while mowing
- About €3,200 in parts

Folder: `sickle bar mower/triple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Irrigation

Irrigates grassland by flooding at low pressure from a pump at the ditch: little energy, no wind drift, no evaporation in the air.

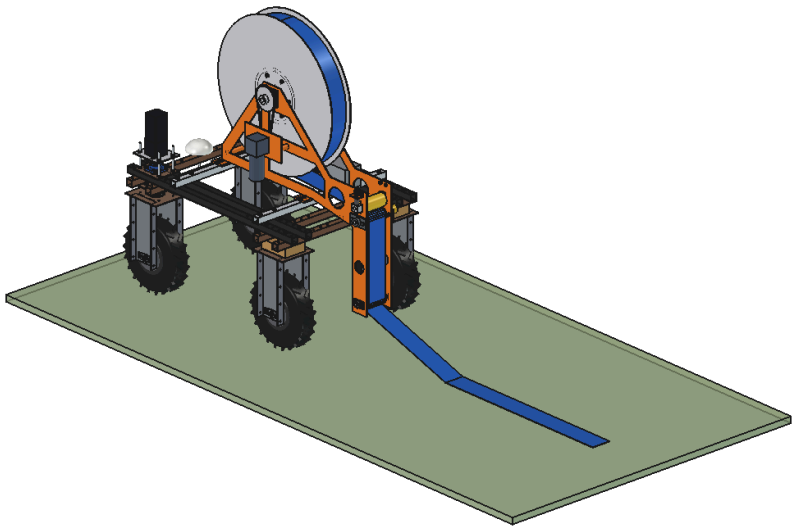


Stationary pump + hose reel

1 variant: Stationary pump + hose reel
Folder: `irrigation/`

IRRIGATION · VARIANT 1 OF 1

Stationary pump + hose reel



The robot carries a reel with 150 m of 3" lay-flat hose, lays it out empty, the petrol pump at the ditch floods a 4 m strip (outlets every 2 m), the hose runs empty and the robot winds it up again. Pinch rollers keep the reel dry and squeeze the last water out.

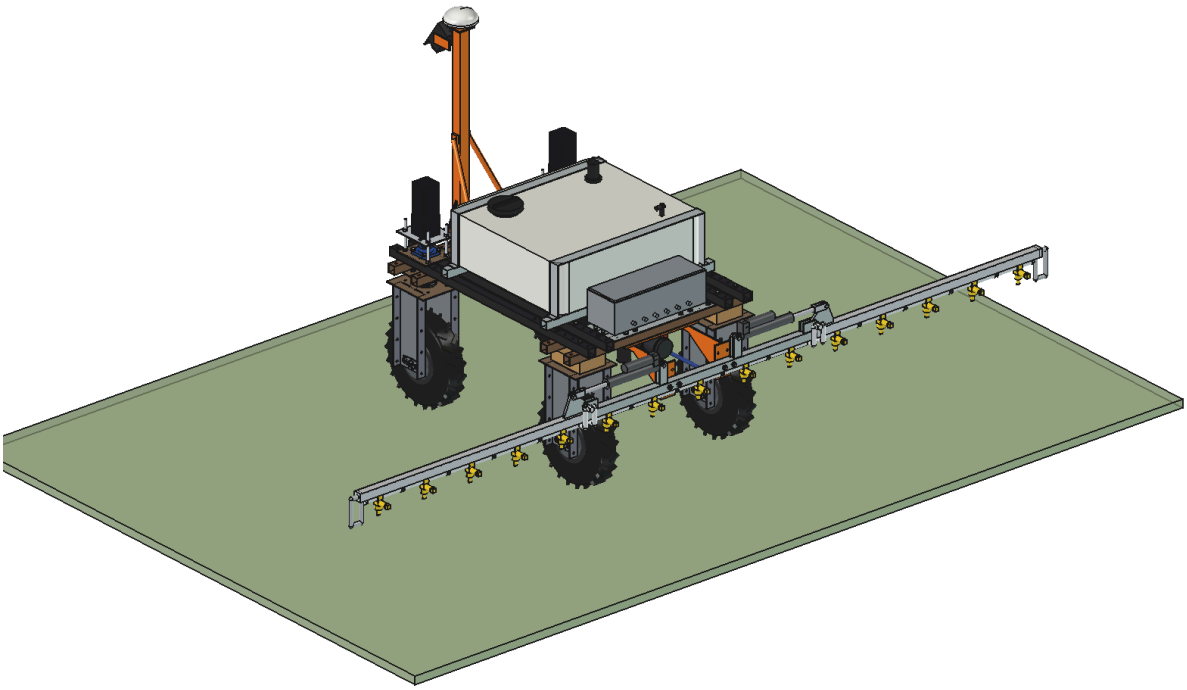
- 7.4 m^3/h at 0.17 bar, CU ≈ 99 %
- 122 min per 500 m^2 strip at 25 mm
- 146 kg, about €1,900 in parts

Folder: `irrigation/stationary + roller/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Spot sprayer

Sprays weeds plant by plant instead of the whole field. A camera in front finds the weeds; the boom behind opens only the nozzles above a plant.

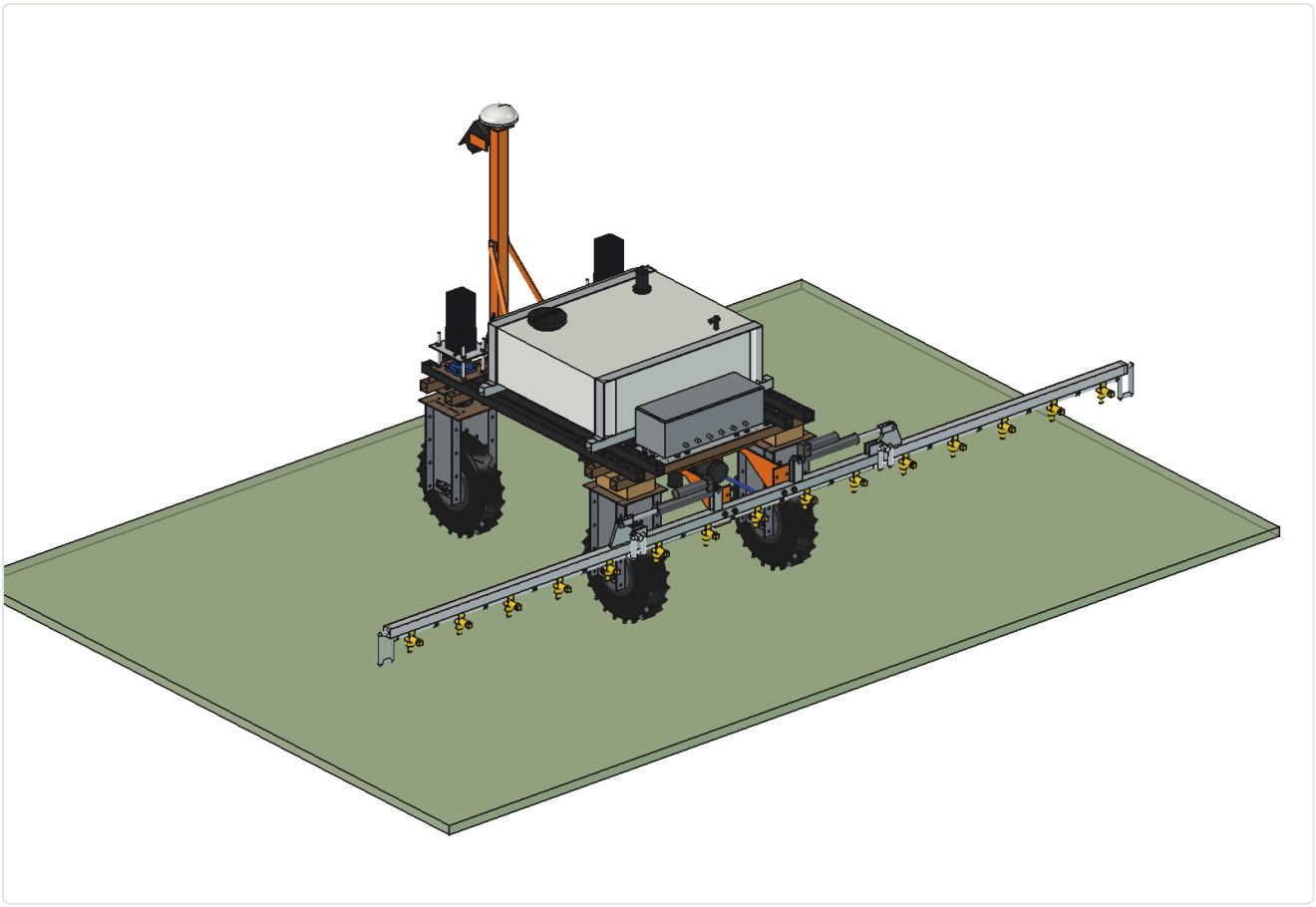


Folding 3.75 m boom

1 variant: Folding 3.75 m boom
Folder: `spotspray unit/`

SPOT SPRAYER · VARIANT 1 OF 1

Folding 3.75 m boom



Wide-angle global-shutter camera on a mast (1.48 m, 55° down) sees the whole strip from 1.8 m ahead; 2.65 s later 15 PWM nozzles at 250 mm on a stainless pipe open above the plant. Centre section 1.25 m and two 1.25 m wings that fold up in 3.2 s with a 24 V actuator each, also in the field. 120 l tank and diaphragm pump on the robot.

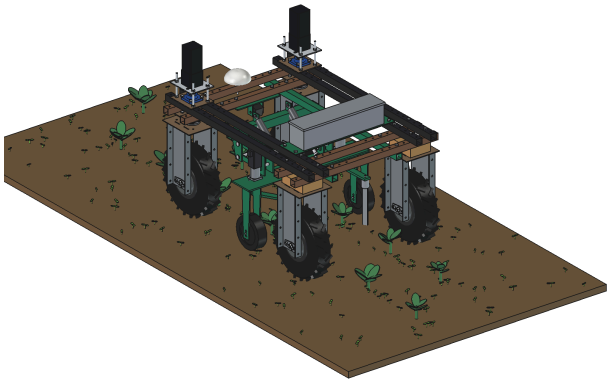
≈ 20 l/ha at 5 % weed cover, 1.1 ha/h	91 kg
About €3,580 in parts	Animation with docks and a fence post

Folder: `spotspray unit/boom/`

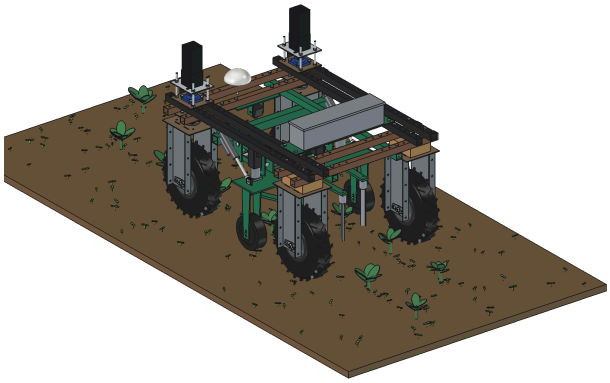
IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

In-row weeder

Hoes inside two rows of broccoli (375 mm apart, wheels in the inter-rows). Camera under the front beams finds the plants 0.84 s ahead; knives swing out around each plant.



Finger, 2 rows

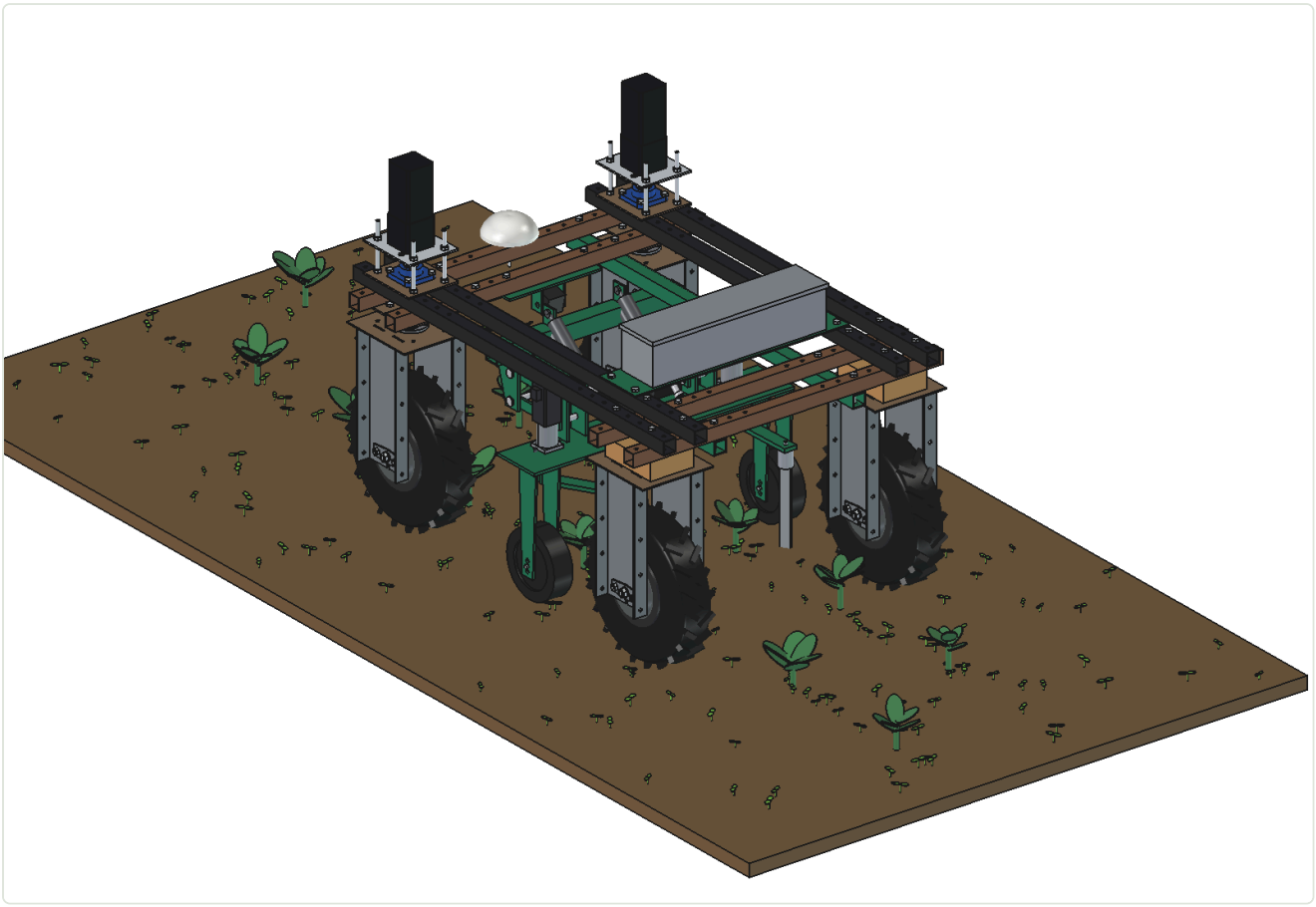


Double arms

2 variants: Finger, 2 rows · Double arms
Folder: `in row weeder/`

IN-ROW WEEDER · VARIANT 1 OF 2

Finger, 2 rows



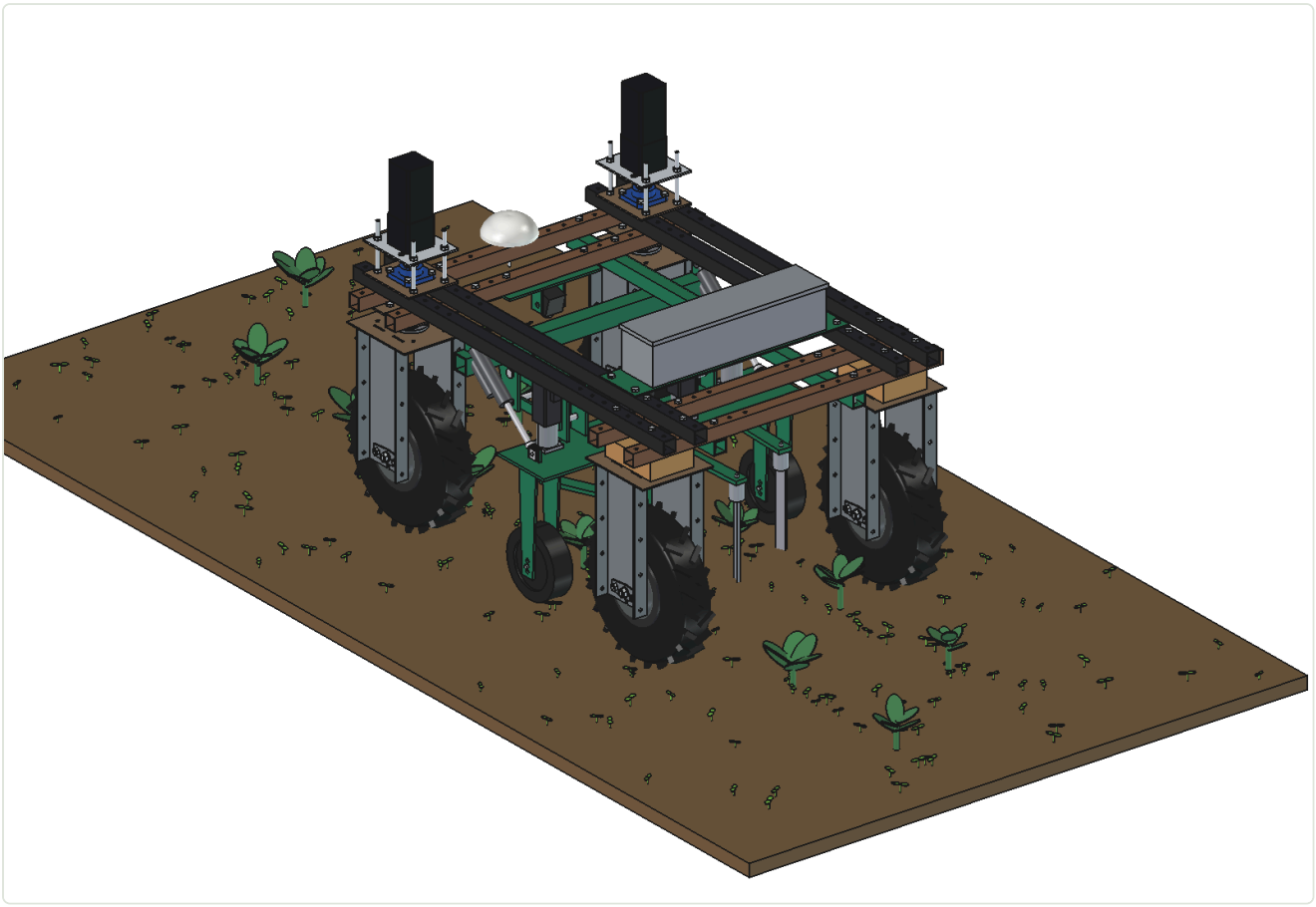
Two row units under the belly on parallelograms with gauge wheels in the robot's wheel track. A NEMA 23 closed-loop stepper with a 10:1 gearbox swings a 220 mm arm with a spring-steel finger and a flat knife 20 mm deep. Out in 98 ms, back in 148 ms; never closer than 50 mm to a stem.

- 0.5 m/s, 6,400 plants/h
- 55 kg
- About €2,240 in parts

Folder: `in row weeder/finger/`

IN-ROW WEEDER · VARIANT 2 OF 2

Double arms



Two knives per row (4 steppers) that open to both sides of each plant and close behind it, timed by the real distance to the stem. Each knife moves only 78 mm: out in 89 ms, back in 118 ms.

> 90 % of the row band outside the safe zones hoed

> 68 kg

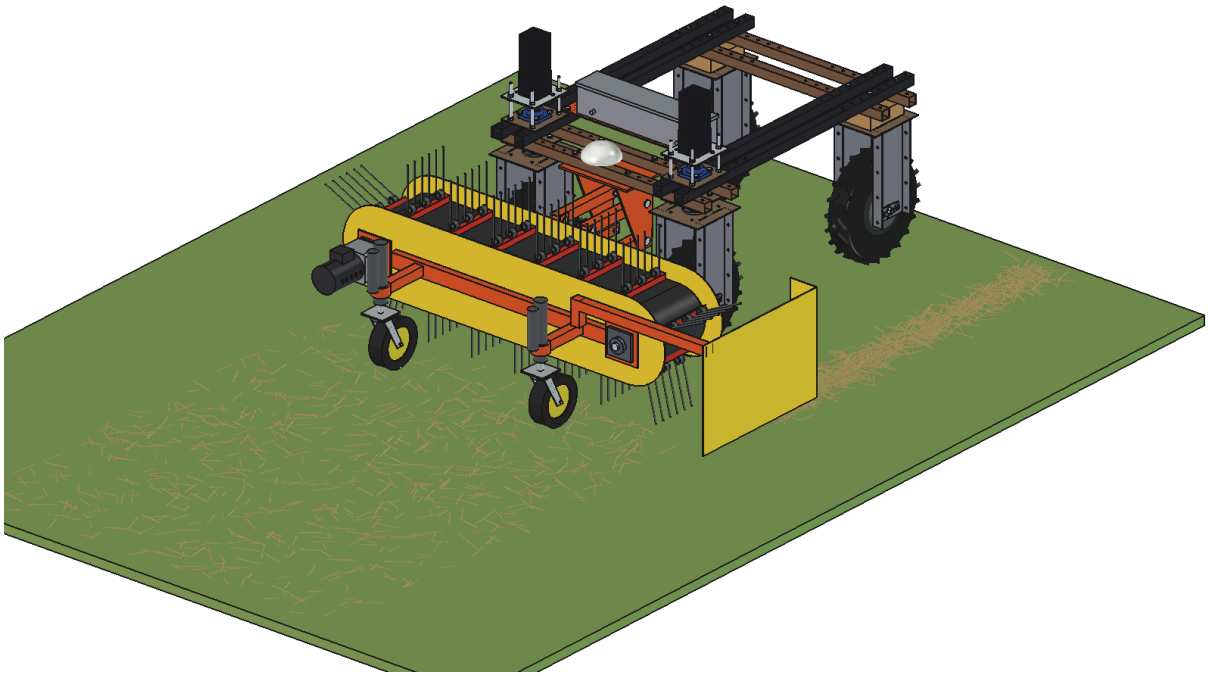
> About €2,820 in parts

Folder: `in row weeder/double_arm/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Belt rake

Band rake in front of the robot: a rubber belt with 15 tine bars around two drums (1.52 m working width). Combs weed seedlings out of nursery beds between young trees or rakes moss and thatch out of grassland into one windrow.

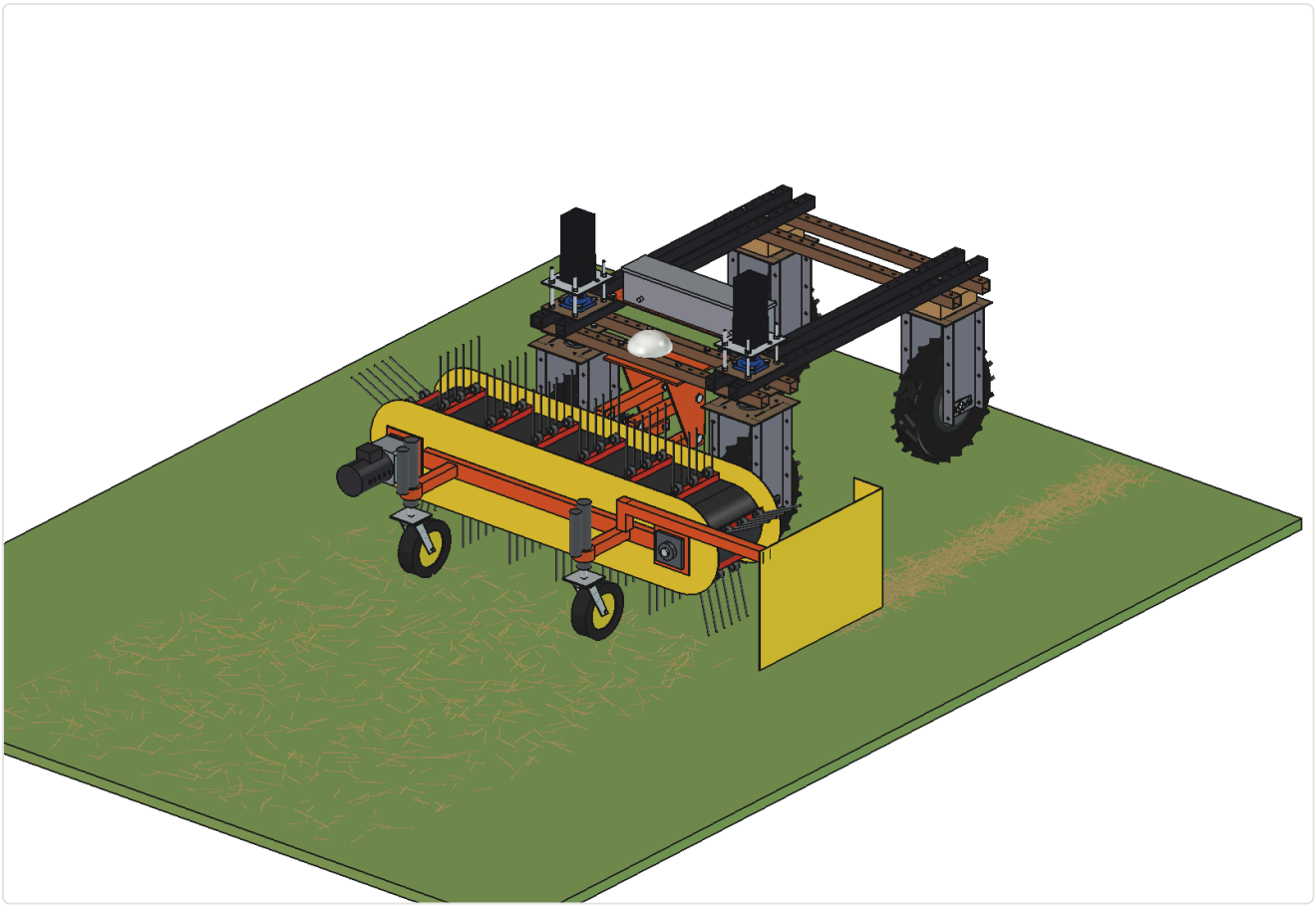


Electric

1 variant: Electric
Folder: `bandrake/`

BELT RAKE · VARIANT 1 OF 1

Electric



48 V 750 W gearmotor with a 20:1 planetary gearbox on the right drum. Two swivel castors on 24 V actuators carry the 110 kg rake, set the depth and lift it 100 mm for the headland; a free parallelogram pushes it.

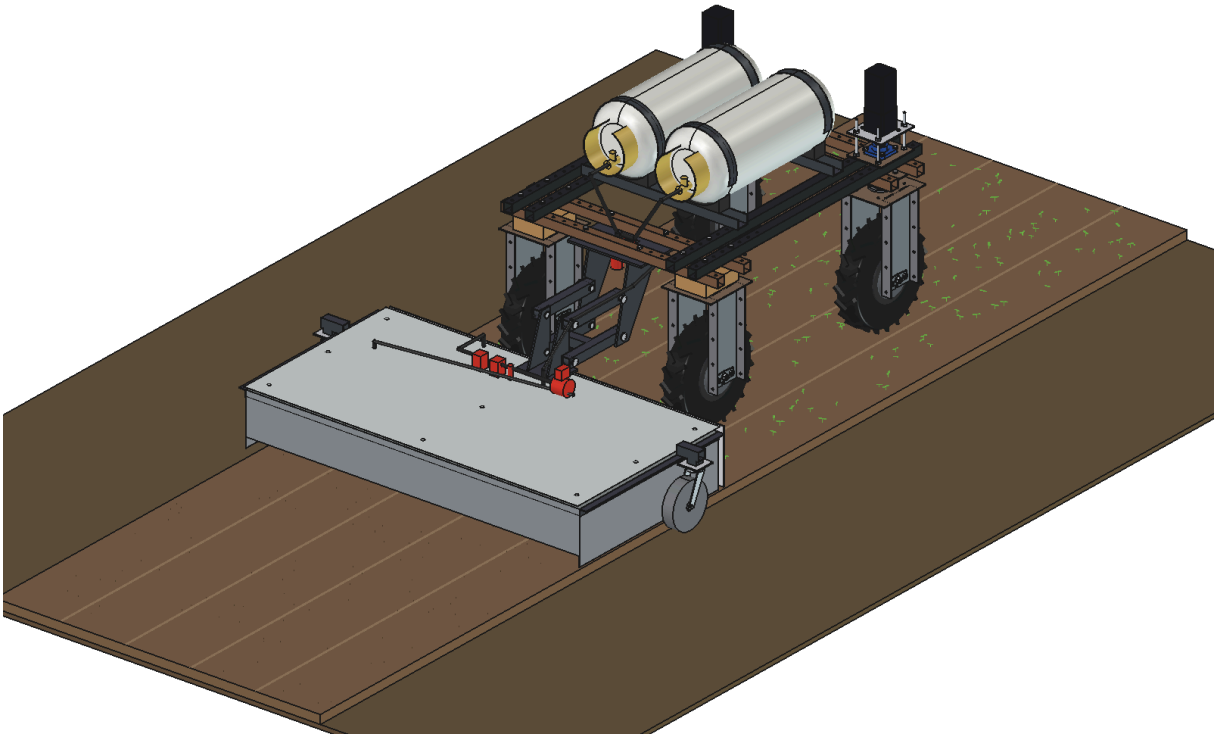
- Comb mode 0.6 m/s, rake mode 1.2 m/s, 0.49 ha/h
- About €2,380 in parts
- Tine forces and animation

Folder: `bandrake/electric/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Weeds burner (flame weeder)

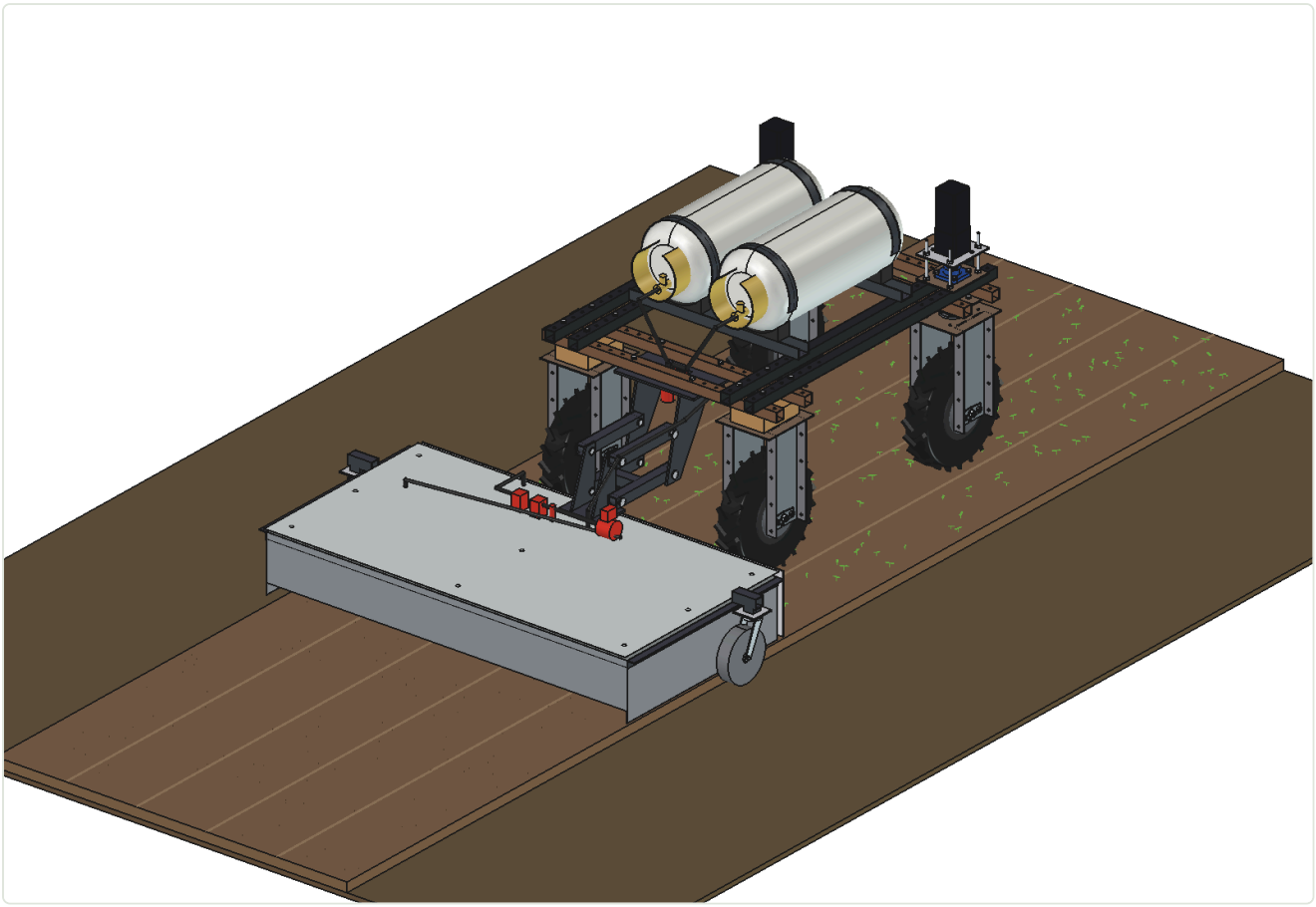
Pre-emergence flame weeder for freshly sown beds (carrot, onion, beetroot, parsnip): kills the emerged seedlings without moving the soil.



Propane hood

1 variant: Propane hood
Folder: `weeds burner/`

Propane hood



Insulated 1.5 × 0.8 m hood behind the robot with 6 flat-flame burners (13 kg/h, 167 kW). Two 18 kg forklift cylinders with liquid withdrawal on the robot, a vaporiser coil in the hood, two main solenoid valves in series, a motorised regulator, pilots with igniters and flame sensors; the valves close at once when the robot stops.

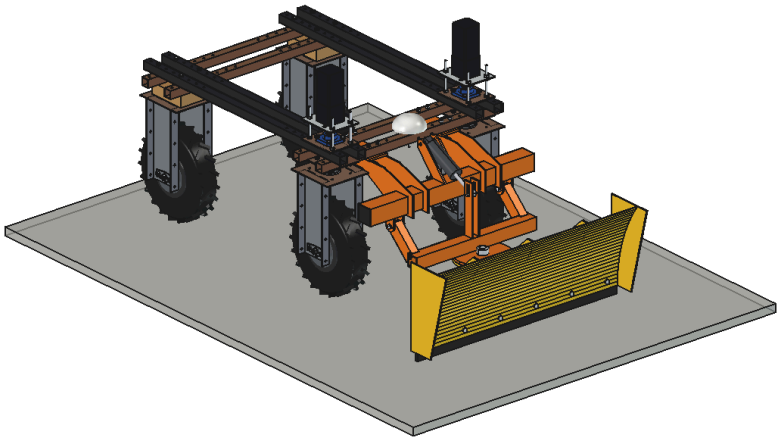
- 34 kg/ha at 0.7 m/s, 0.84 ha per filling
- Gas ≈ €55/ha
- 131 kg empty, about €3,400 in parts

Folder: `weeds burner/propane_hood/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Push blade

Front-mounted dozer / grader blade: pushes feed back to the feed fence, scrapes slurry and snow, grades soil and paths. Covers the manure scraper idea as well.

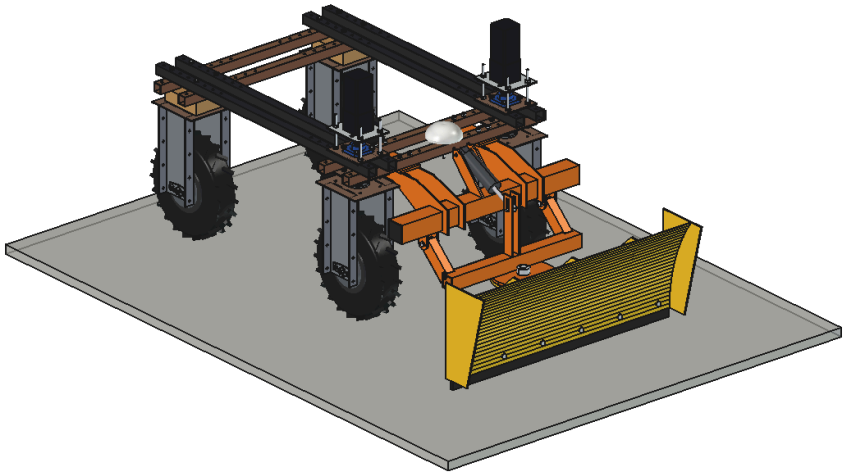


Simple

1 variant: Simple
Folder: `push blade/`

PUSH BLADE · VARIANT 1 OF 1

Simple



1100 x 450 moldboard on R700 with a bolted rubber edge (steel edge for soil), angle +-25 degrees on an M30 king pin with a lock pin, two lift arms on a front toolbar under the front lower beams, 24 V 1500 N actuator with a float slot.

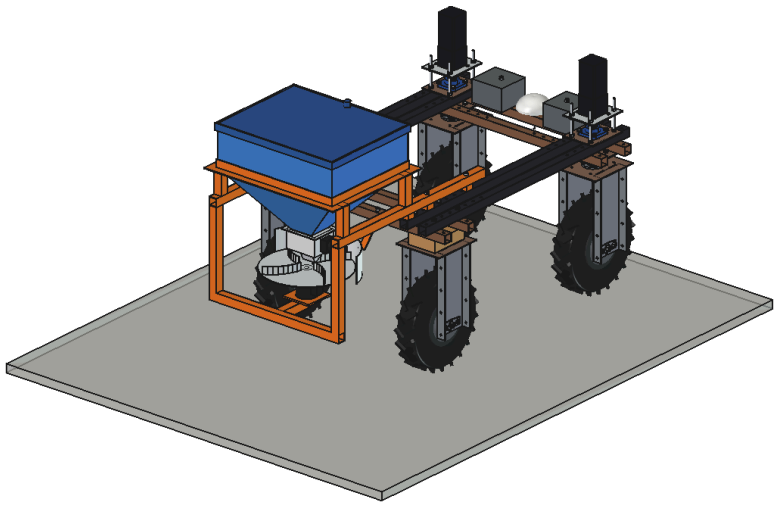
- Net push 801 N on dry concrete, 348 N with manure
- Lift 159 mm, clears the steered front tyres by 148 mm
- 81 kg

Folder: `push blade/simple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Fertilizer spreader

Spinning-disc spreader for granular fertilizer (and granular lime with a coarser roller). Dose follows the driving speed and a task map.

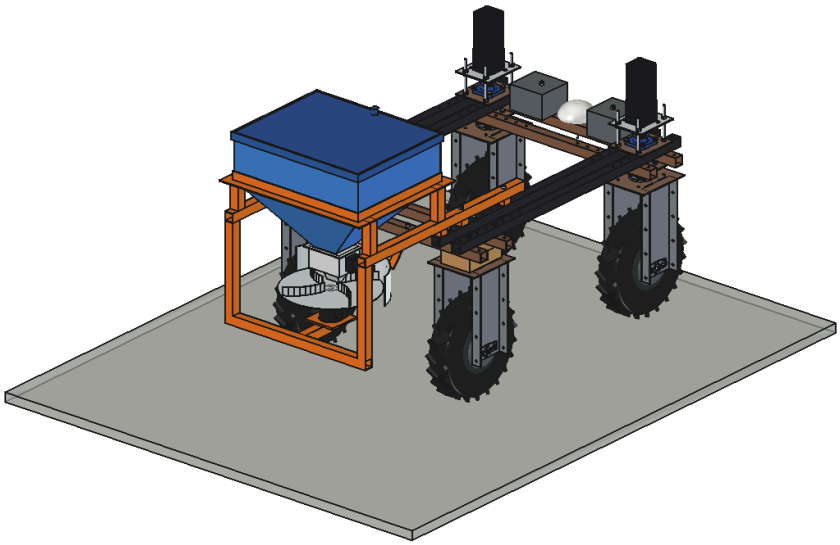


Simple

1 variant: Simple
Folder: `fertilizer spreader/`

FERTILIZER SPREADER · VARIANT 1 OF 1

Simple



88 l PE hopper over and behind the rear axle on rails on the upper beams, fluted metering roller O80 on a 12 V gearmotor, vaned disc O400 on a 48 V BLDC motor behind the robot with a guard towards the robot. Optional 2 x 20 kg front counterweight.

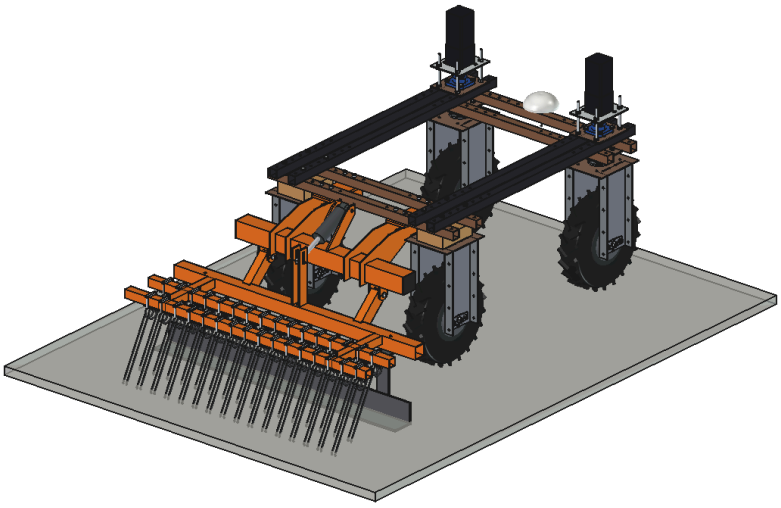
- Driving width 4.1..6.7 m (500..900 rpm)
- 100..541 kg/ha at 1 m/s
- ~1.6 ha/h, 130 kg full

Folder: `fertilizer spreader/simple/`

IMPLEMENT · CROP & SOIL

Grassland harrow

Pasture harrow behind the robot: a levelling board spreads molehills and dung pats, two rows of spring tines rake out moss and work in seed. Covers the molehill spreader idea.

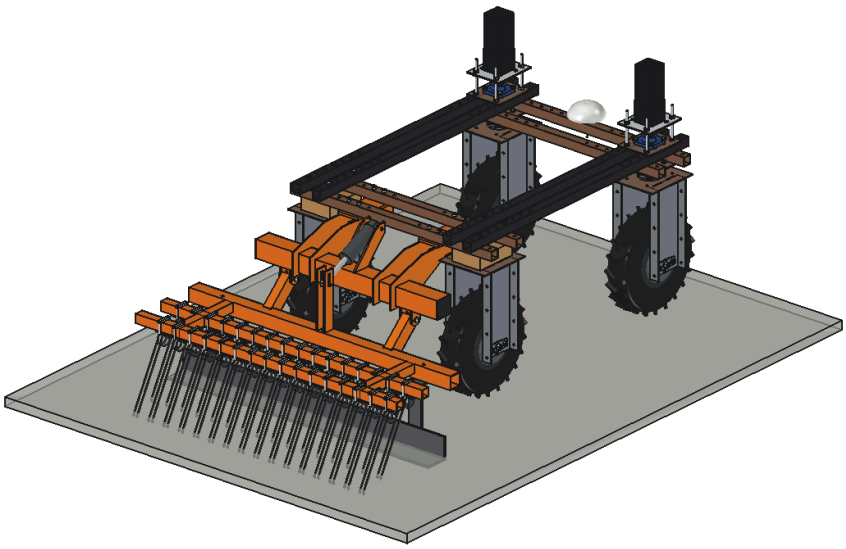


Simple

1 variant: Simple
Folder: `grassland harrow/`

GRASSLAND HARROW · VARIANT 1 OF 1

Simple



1300 mm harrow on two lift arms from a rear toolbar (the push blade geometry, mirrored): angle-iron levelling board, 32 double spring tines O8 (a track every 37.5 mm), floats on the sward, lifted by a 24 V actuator.

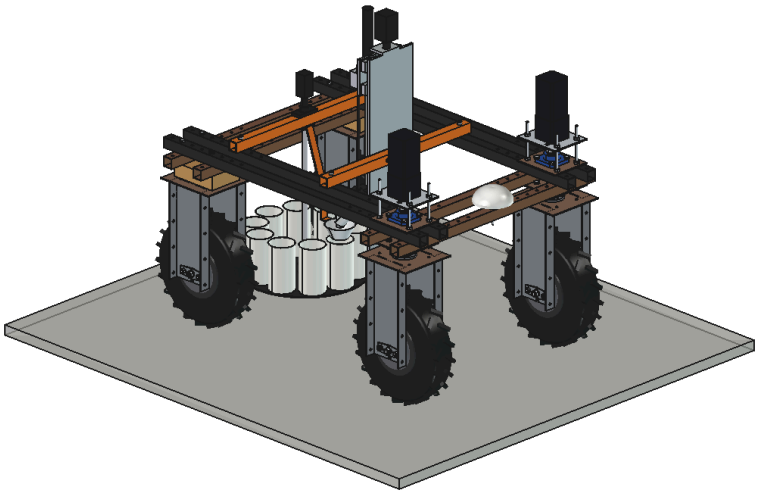
- Pull 629 N on grass, grip margin 1.44
- 0.56 ha/h at 1.5 m/s
- 85 kg

Folder: `grassland harrow/simple/`

IMPLEMENT · CROP & SOIL

Soil sampler

Takes soil cores at GPS points with a gouge on a ball-screw Z axis in the middle of the robot and collects 40 cores per plot in a cup of a 10-cup carousel.

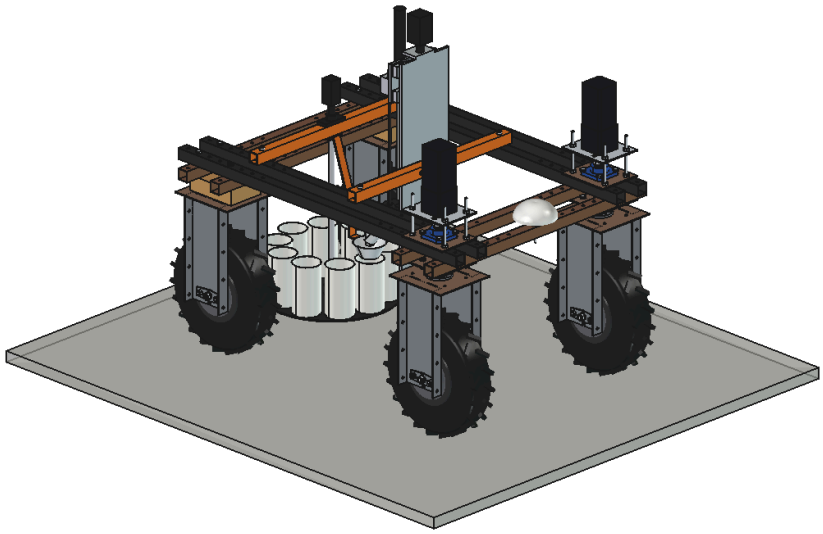


Simple

1 variant: Simple
Folder: `soil sampler/`

SOIL SAMPLER · VARIANT 1 OF 1

Simple



SBR20 rails and an SFU1605 ball screw with a NEMA23 on a mast plate between two bridges on the upper beams, O25 gouge with ejector, sprung foot, swing chute and funnel, carousel of 10 cups of 1.3 l.

> 0-10 cm grass needs 0.5..0.8 kN, 1.3 kN available

> 38 s per core, 25 min per plot

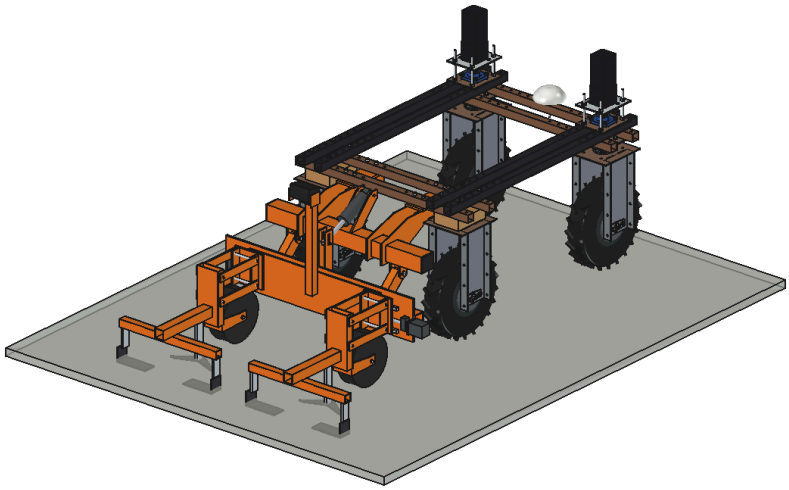
> 33 kg

Folder: `soil sampler/simple/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Inter-row hoe

Single-axis guided hoe behind the robot for row crops such as maize: two hoe units in the inter-rows behind the wheels, kept on the rows by a camera-steered ball-screw side shift.

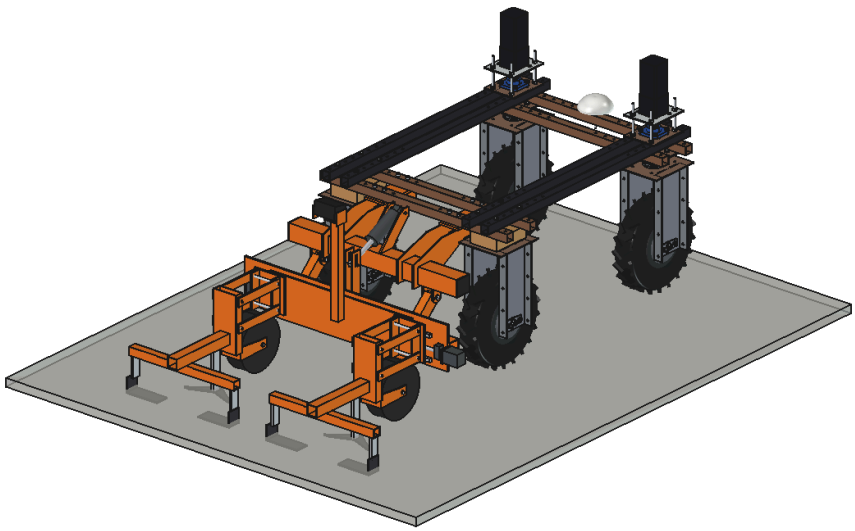


Simple

1 variant: Simple
Folder: `inter row hoe/`

INTER-ROW HOE · VARIANT 1 OF 1

Simple



Rear toolbar and lift arms of the grassland harrow (3000 N actuator), carrier with 2 HGR15 rails, 8 mm shift plate moved +-75 mm by an SFU1610 ball screw and a closed-loop NEMA23. Two parallelogram units with a gauge wheel O250, a goosefoot 250 and two L-blades (550 mm per inter-row, 100 mm band each side of the row). Camera on a mast above the middle row.

> 1.5 m per pass (2 x 750), ~0.43 ha/h at 1 m/s

> Pull 764 N at 20 mm, grip margin 1.23

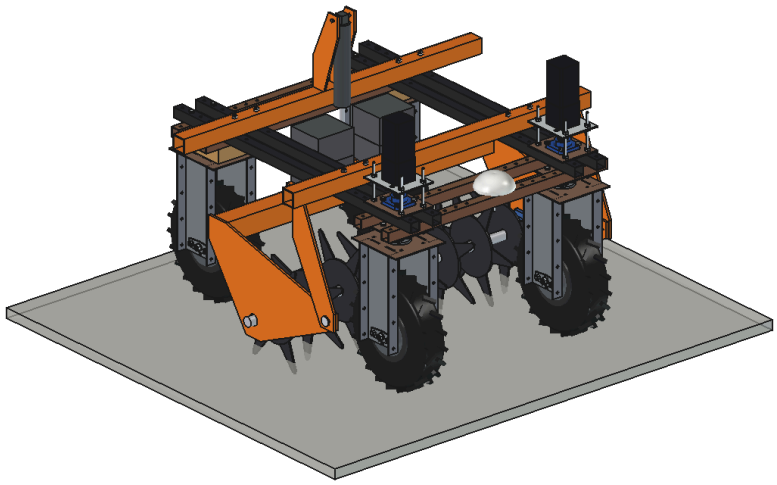
> Side shift 80 mm/s, ~119 kg, needs the 2 x 20 kg front counterweight

Folder: `inter row hoe/simple/`

IMPLEMENT · SOIL CULTIVATION & WEED CONTROL

Grassland aerator

Slitter aerator in the middle of the robot, between the axles: a passive roller with star discs cuts slits into grassland, pressed in by frame weight and ballast, both axles sharing the reaction.

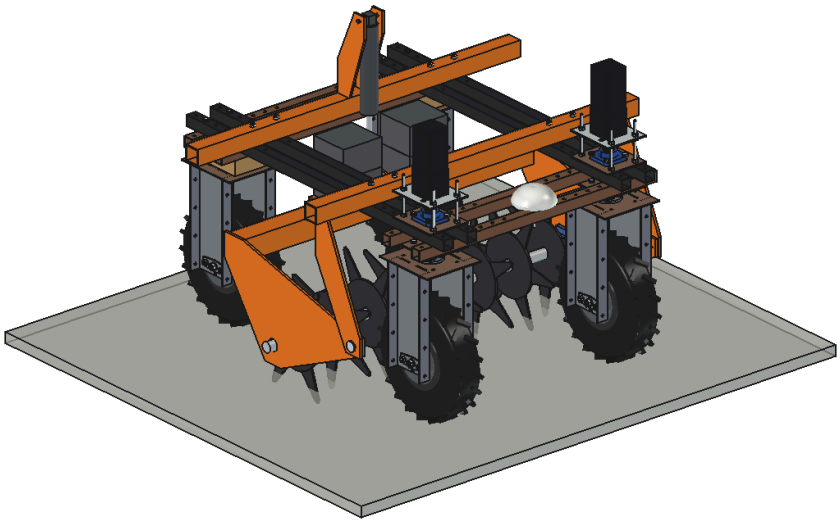


Simple

1 variant: Simple
Folder: `grassland aerator/`

GRASSLAND AERATOR · VARIANT 1 OF 1

Simple



2 bridges on the upper beams, side arms on pivots outside the steering sweep, roller with 8 stars of 6 slitting knives (O450) in 2 UCFL208 bearings, cross beam with 4 x 20 kg ballast, 24 V 2000 N actuator from a mast on the rear bridge (lifts 14 degrees, can press).

> 1.2 m, slits 150 x 236 mm, 70 mm deep

> ~0.41 ha/h at 1.2 m/s, pull 325 N

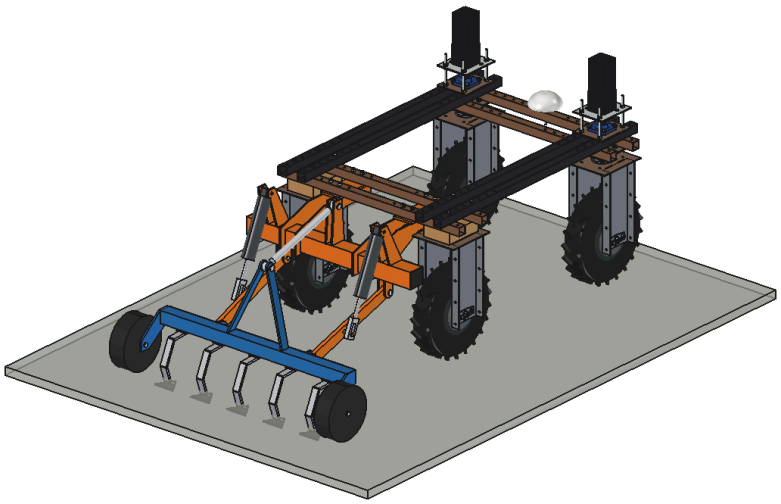
> 201 kg incl. 80 kg ballast, 47 % on the front axle

Folder: `grassland aerator/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Three-point hitch

Category 0 three-point linkage at the rear with two electric lift rods, so standard sub-compact implements can hang on the robot. Drawn with a small 5-tine cultivator, which covers the small cultivator idea.

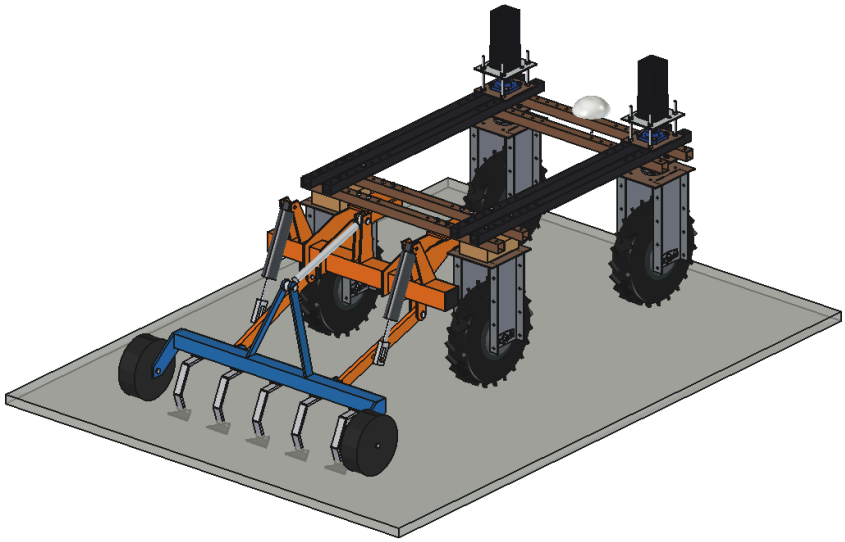


Simple

1 variant: Simple
Folder: `three point hitch/`

THREE-POINT HITCH · VARIANT 1 OF 1

Simple



Rear toolbar of the grassland harrow with lugs for 2 lower links (50 x 20, O16 pins, 500 spacing), a top link bracket and 2 x 24 V 3000 N actuators as lift rods with a slotted eye (the implement floats). Demo implement: 1.0 m cultivator, 5 S-tines with duckfoot shares, 2 gauge wheels.

> Lower pins 220 .. 425 mm, implement level within 2 degrees

> Carries ~56 kg lifted with the 2 x 20 kg counterweight (front axle >= 20 %)

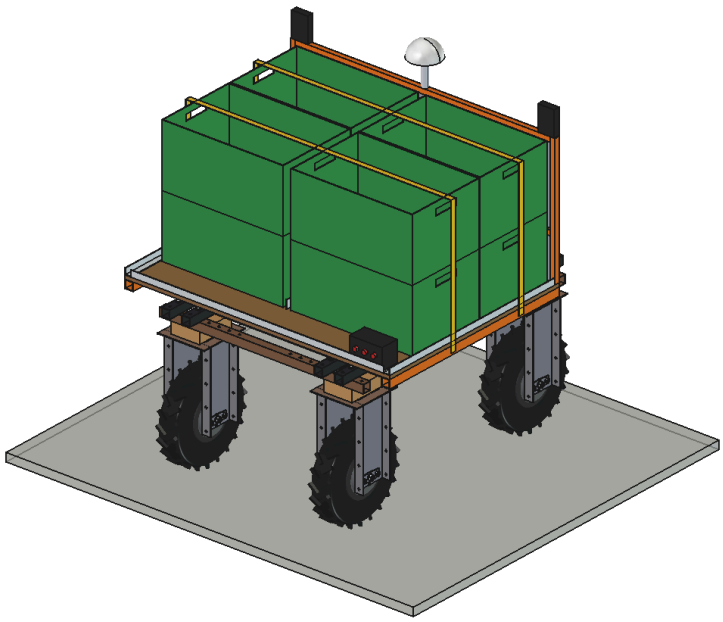
> Cultivator: 808 N draft at 40 mm, grip margin 1.29, 0.29 ha/h

Folder: `three point hitch/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Crate carrier

Flat deck on the upper beams for 8 harvest crates, carried from the rows to the field edge following a picker (UWB) or on an RTK route. The same deck carries tools and material (covers the tool cart idea).

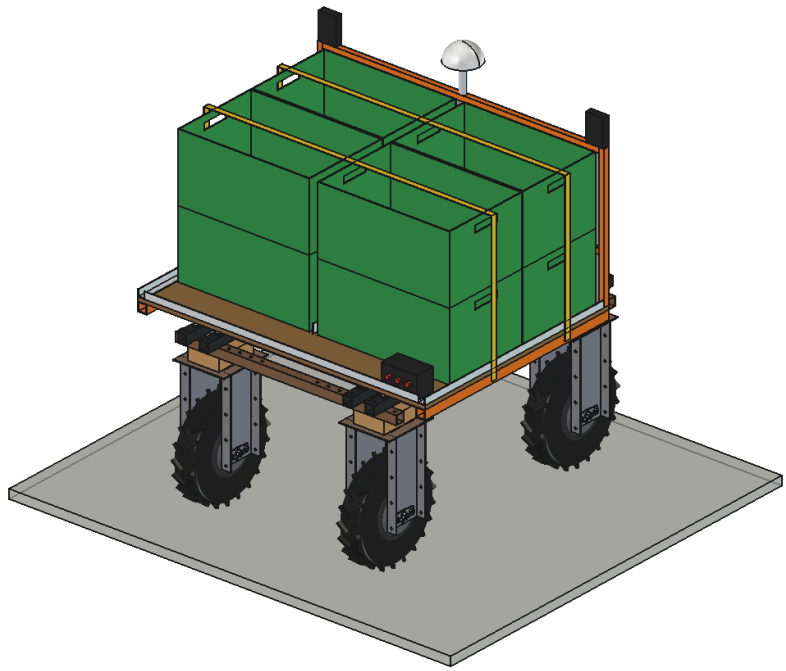


Simple

1 variant: Simple
Folder: `crate carrier/`

CRATE CARRIER · VARIANT 1 OF 1

Simple



3 bearers across the upper beams, side rails and 18 mm plywood (1300 x 1000, loading height 757), angle guides, divider, mesh headboard in front of the crates, 2 ratchet straps; 2 UWB anchors, a stop/go/follow box and the GNSS antenna moved to a mast above the stack.

> 8 crates 600 x 400 x 300 in 2 layers

> Apples: 141 kg, 48 % front axle, side tip angle 28 degrees

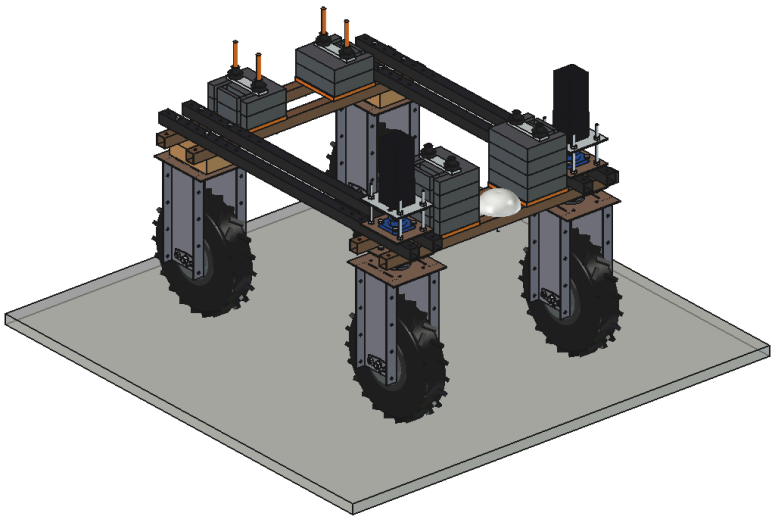
> ~107 crates per hour on a 150 m shuttle, 51 kg

Folder: `crate carrier/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Counterweights

Stackable 11 kg ballast plates on the lower cross beams, front and rear, to keep enough load on the front axle with rear implements (and on the rear axle with the push blade).

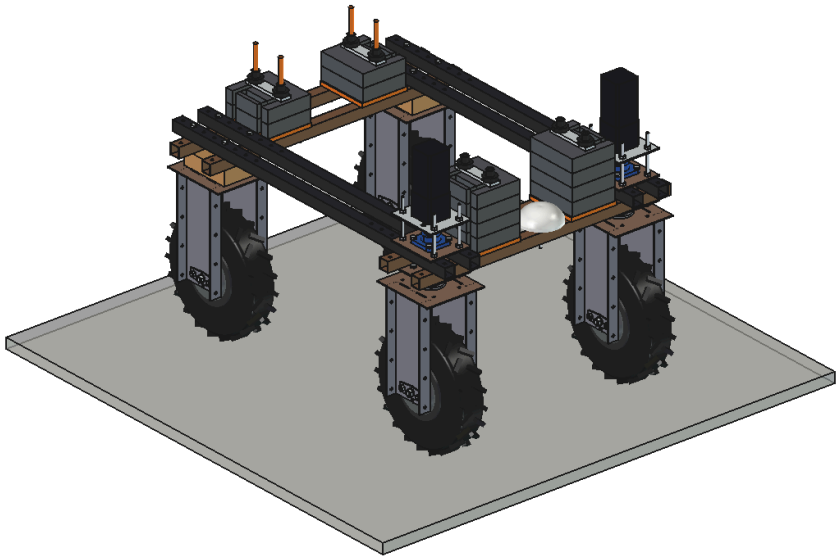


Simple

1 variant: Simple
Folder: `counterweights/`

COUNTERWEIGHTS · VARIANT 1 OF 1

Simple



Per stack (4 stacks): a 15 mm base plate on the two lower beams with 2 M16 studs, up to 4 plates of 170 x 200 x 50 with open slots that slide on from the side, a clamp bar and 2 knob nuts. Fits together with the push blade and the rear toolbar.

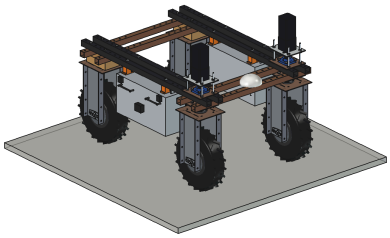
- 11 kg per plate, 0..99 kg per axle end
- Harrow, hitch, spreader: 1 plate per stack front
- Inter-row hoe: 2 plates per stack front, 5 % -> 21 % front axle

Folder: `counterweights/simple/`

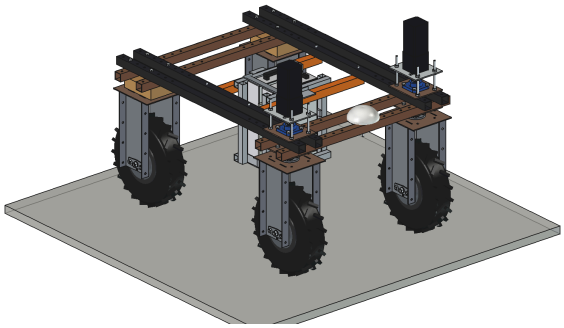
IMPLEMENT · POWER & DRIVE

Battery pack

Two exchangeable 48 V LiFePO4 pods (2.56 kWh each) hanging under the upper beams between the wheels: low, centred, and free of all toolbars and decks.



Simple

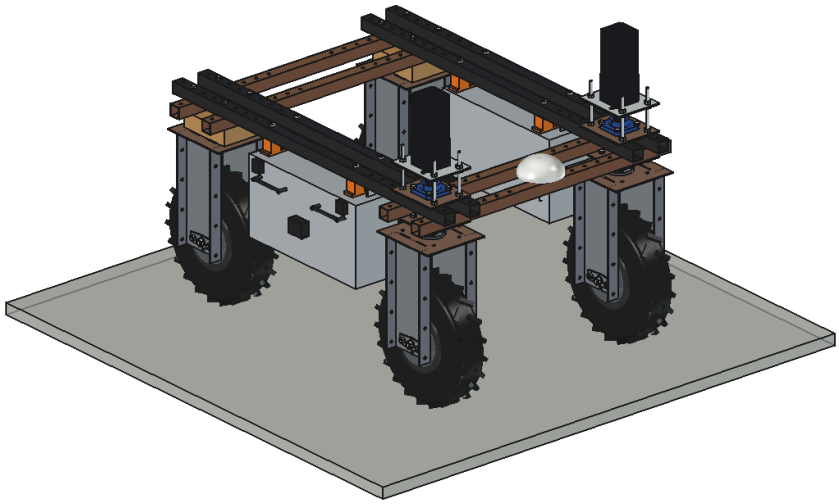


MBH3+ box

2 variants: Simple · MBH3+ box
Folder: `battery pack/`

BATTERY PACK · VARIANT 1 OF 2

Simple



Per side: 2 hangers (flat bar 40 x 8, M10 through the upper beams) and an aluminium pod with 16 LiFePO4 cells 50 Ah, a 16S BMS, hook strips, latches, handles and an Anderson SB50 connector.

> 51.2 V, 2 x 2.56 kWh, 26 kg per pod

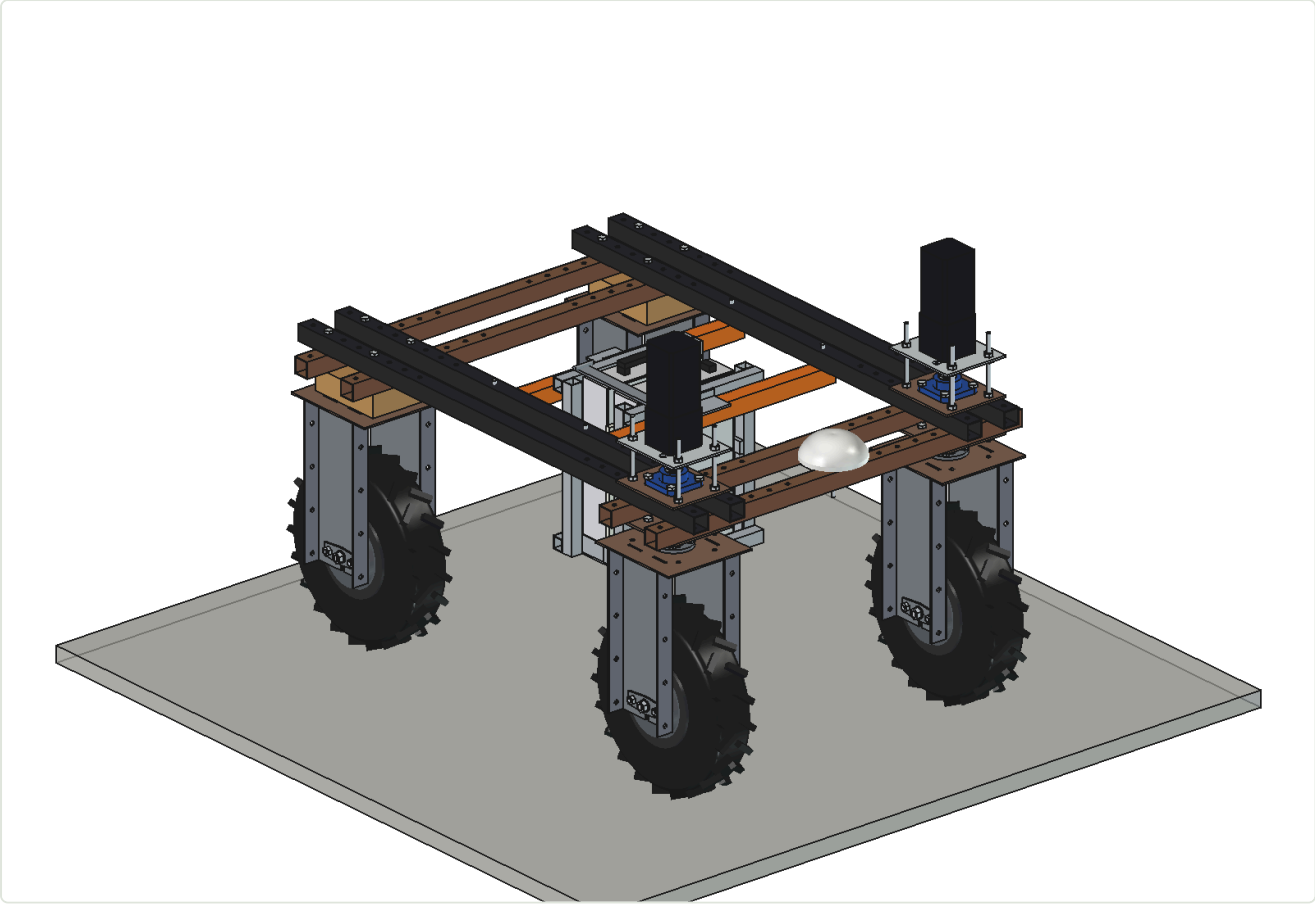
> ~230 N extra grip, 52 mm from the steered tyres

> Inter-row hoe 1.8 -> 6.0 h, harrow 1.4 -> 4.9 h

Folder: `battery pack/simple/`

BATTERY PACK · VARIANT 2 OF 2

MBH3+ box



Open rack for BatteryLabs MBH3+ drop-in batteries (14S 48 V 50 Ah, 2.56 kWh each) in tray segments of 100 mm, so 2 batteries grow to 3 or 4; on top, sunk in the centre well (recommended) or in side cradles. Also a closed alu cabinet and a wooden box version, with placeholders for MBH3+ and MU3+.

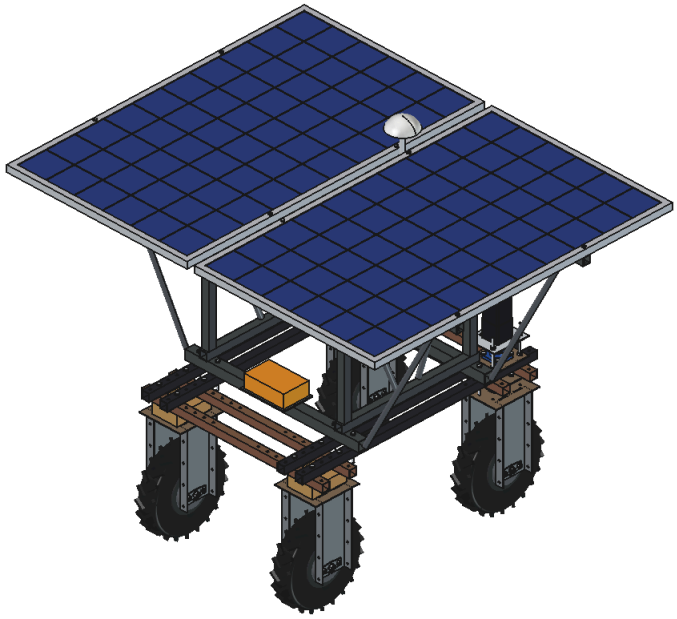
- 2 / 4 batteries: 4.6 / 9.2 kWh usable
- Centre well: cg 502 mm
- Inter-row hoe 4.3 / 8.5 h

Folder: `battery pack/mbh3_box/`

IMPLEMENT · POWER & DRIVE

Solar canopy

Roof of solar panels on a tube frame above the robot that charges the battery through an MPPT controller and carries the GNSS antenna.

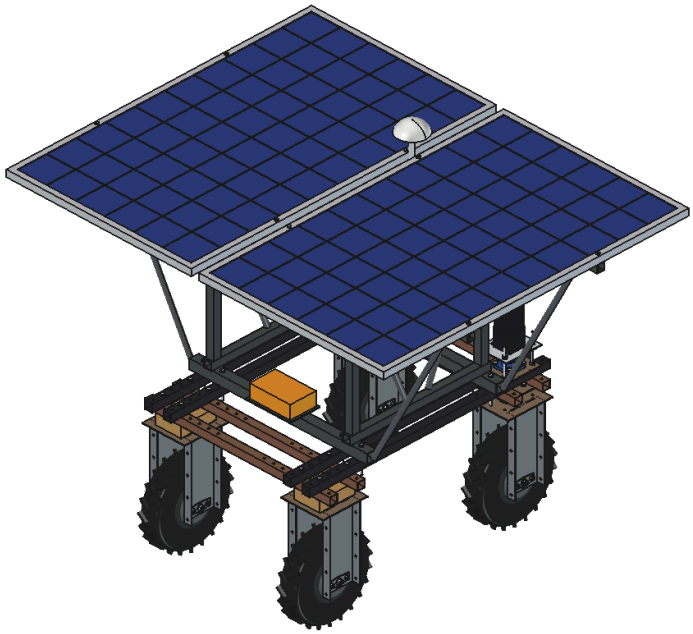


Canopy

1 variant: Canopy
Folder: `solar/`

SOLAR CANOPY · VARIANT 1 OF 1

Canopy



2 panels 1000 x 1650 mm (320 Wp each, 3 possible) on a steel tube frame: base tubes on the upper beams, 4 posts, top rails and cross members, diagonal struts to the overhangs; MPPT box, GNSS antenna on a mast between the panels.

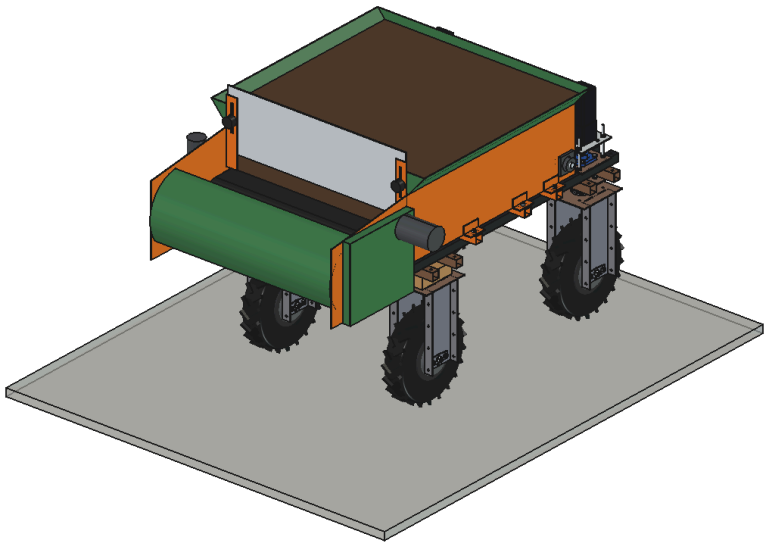
- 640 Wp, ~480 kWh per year (NL)
- 2.3 kWh per day in June, 0.2 in December
- 91 kg, tips over at ~29 degrees static

Folder: `solar/`

IMPLEMENT · SPREADING & BEDDING

Compost spreader

Bin on the upper beams with a moving floor belt and a beater behind the robot: top-dresses beds, orchard strips and grassland with compost in an even 1.1 m layer, dose from the belt speed.

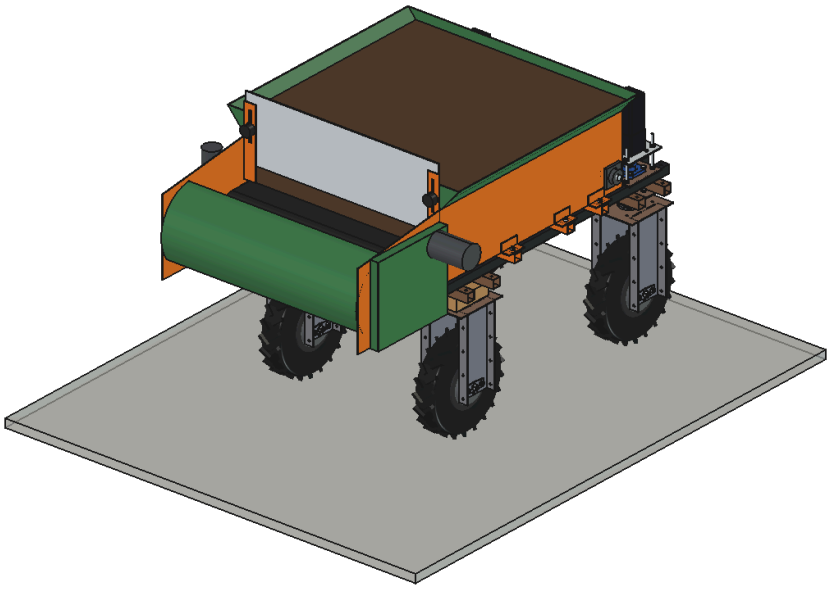


Simple

1 variant: Simple
Folder: `compost spreader/`

COMPOST SPREADER · VARIANT 1 OF 1

Simple



299 l bin (150 kg payload) on 3 bearers, floor belt 920 wide on a UHMW slide bed round 2 drums O100 driven by a 24 V worm gearmotor, slide gate 30..150 mm, paddle beater O280 at 300 rpm on a 48 V 1 kW BLDC with a poly-V belt and a deflector hood.

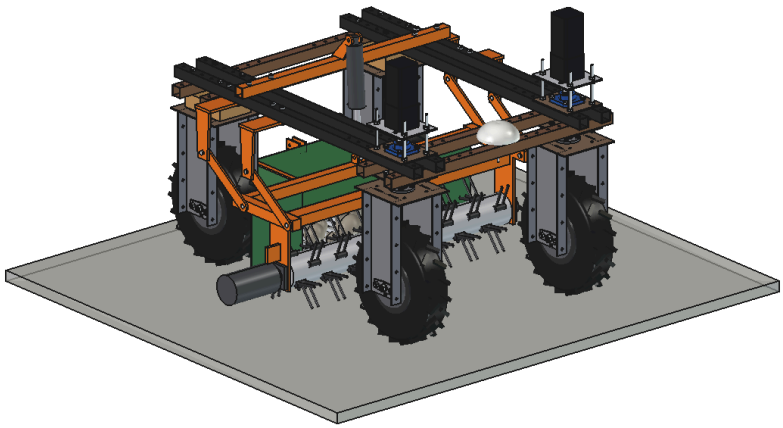
- 1..3 kg/m2, 50..150 m2 per bin, ~1.1 t/h with refilling
- Needs 1 counterweight plate per stack (front axle 33 %)
- 190 kg empty, climbs 8 degrees full

Folder: `compost spreader/simple/`

IMPLEMENT · CROP & SOIL

Stone picker

Mid-mounted under the robot between the axles: a 1 m tine rotor flicks stones over a comb into a sieve basket, on a parallelogram lift; the floor drops open at the heap.

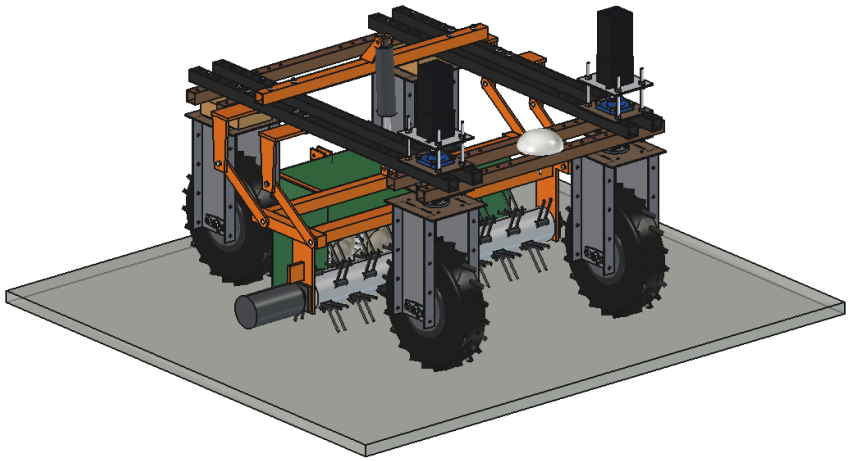


Simple

1 variant: Simple
Folder: `stone picker/`

STONE PICKER · VARIANT 1 OF 1

Simple



Rotor O114 with 48 double spring tines (tip radius 150, 20 mm deep) on a 48 V 750 W gearmotor at 250 rpm, comb of 12 mm bars, 40 l basket with a sieve floor (25 mm gaps) opened by a 24 V actuator, frame on 4 swing links outside the wheel line lifted 100 mm by one 1500 N actuator.

> 1 m wide, 1200..1700 m2/h, ~1 ha per day

> 44 kg of stones per basket, front axle 41..45 %, no ballast

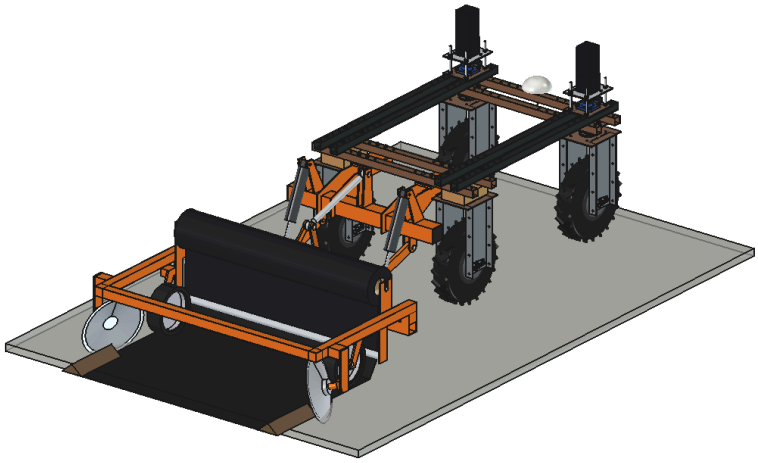
> 92 kg, 525 W rotor

Folder: `stone picker/simple/`

IMPLEMENT · CROP & SOIL

Mulch layer

Category 0 implement for the three-point hitch that lays 1.2 m mulch foil over the bed behind the robot: press wheels push the edges down, angled discs bury them.

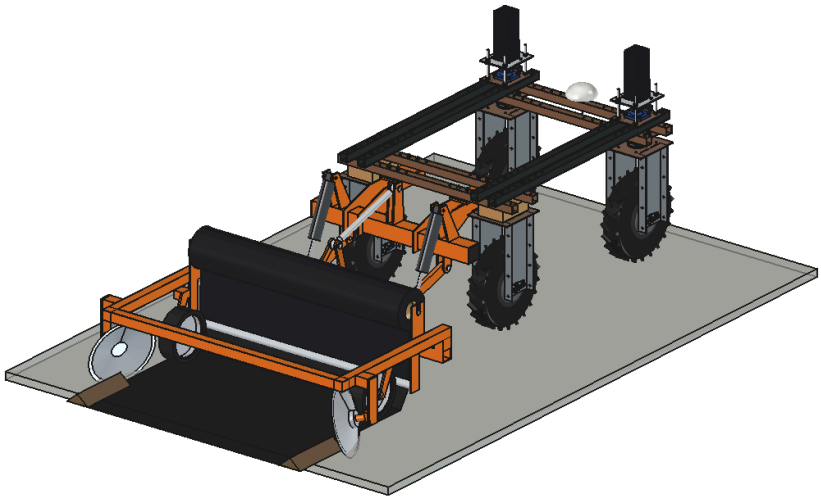


Simple

1 variant: Simple
Folder: `mulch layer/`

MULCH LAYER · VARIANT 1 OF 1

Simple



Frame with a 60 x 60 toolbar, A-frame mast and 50 x 50 side beams on the hitch of three point hitch/simple (imported), foil roll O200 x 1200 in open V-holders with a band brake, guide bar, 2 semi-pneumatic press wheels O250 on the edges and 2 concave covering discs O350 at 22 degrees, 50 mm deep.

> 12 beds of 100 m per hour, a 1075 m roll lasts ~50 min

> Pull 730 N, grip margin 1.51

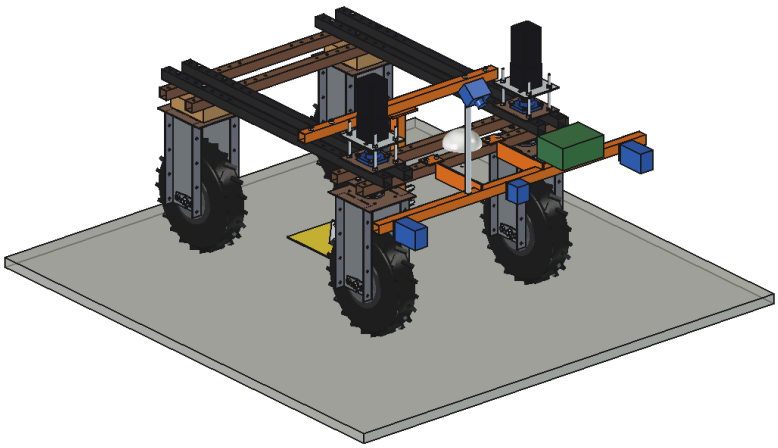
> 94 kg with a full roll: needs 3 counterweight plates per front stack

Folder: `mulch layer/simple/`

IMPLEMENT · MEASURING & SENSORS

Grass meter

Maps grass cover while driving: front boom with 2 active NDVI sensors, a lidar and a camera, plus a sledge plate meter between the axles for compressed height and dry matter. Covers the NIR sensor idea.

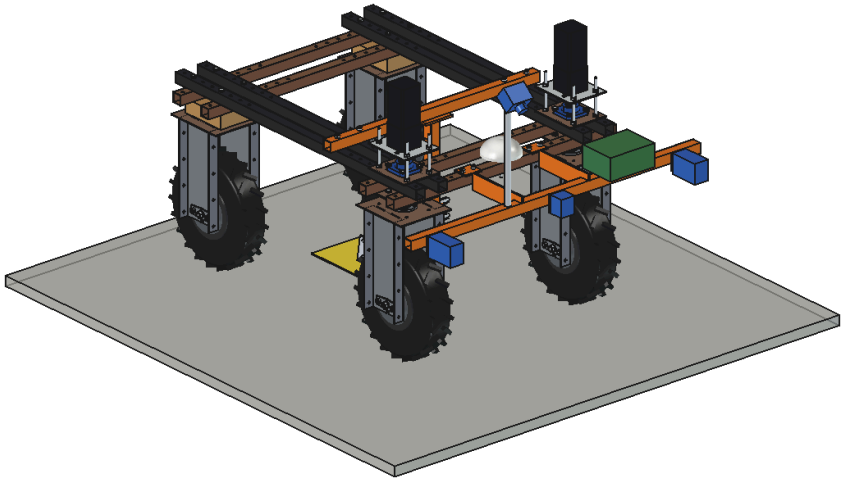


Simple

1 variant: Simple
Folder: `grass meter/`

GRASS METER · VARIANT 1 OF 1

Simple



Boom 1300 on two flat brackets on the front outer lower beam (above the tyres), 2 active NDVI sensors at x +-550, a lidar and an RGB camera; plate 0.1 m2 at 5 kg/m2 on a 350 mm parallelogram from a post under the upper beams, angle sensor on the pivot, 24 V actuator to lift it.

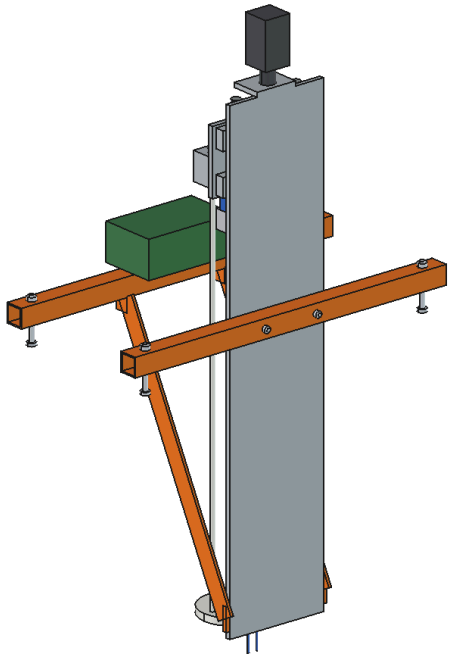
- 4.4 ha/h on 10 m transects, ~35 ha per battery
- Plate 0..200 mm, 0.3..0.5 mm resolution
- 22 kg, stays on with most implements

Folder: `grass meter/simple/`

IMPLEMENT · MEASURING & SENSORS

Soil compaction meter

Automatic cone penetrometer in the middle of the robot: a ball-screw Z axis pushes an ASABE cone at 30 mm/s to 600 mm, a load cell reads the force; compaction maps per depth layer.

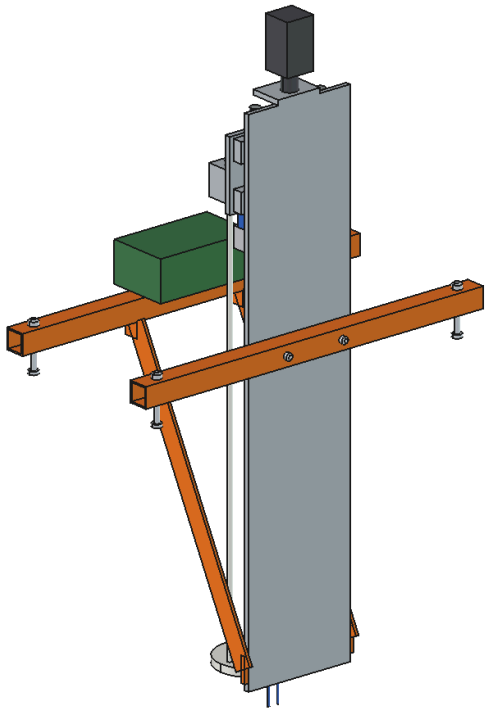


Simple

1 variant: Simple
Folder: `soil compaction meter/`

SOIL COMPACTION METER · VARIANT 1 OF 1

Simple



Bridges, mast, SBR20 rails and SFU1605 screw as the soil sampler (mast z 120..1150 with stays), S-type 2 kN load cell, rod O9.5 x 700 with a 30 degree cone O12.83, sprung foot with a PTFE rod bush and a capacitive moisture probe.

> Up to 10.2 MPa with 70 % of the robot weight

> 35 s per 600 mm profile, ~47 points per hour

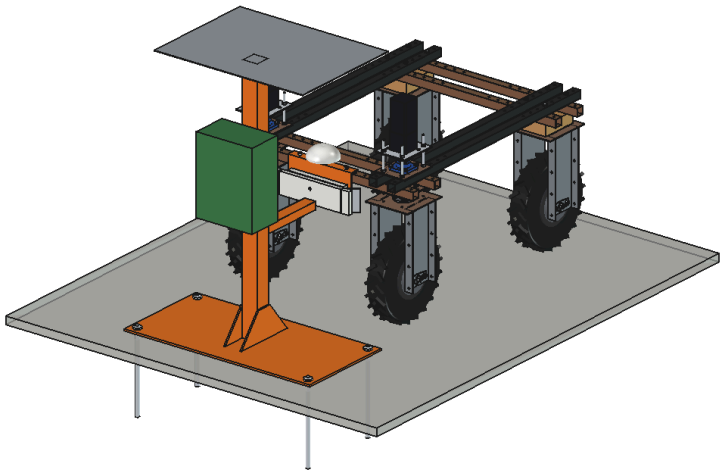
> 32 kg

Folder: `soil compaction meter/simple/`

IMPLEMENT · POWER & DRIVE

Charging dock

Fixed docking station at the field edge or in the shed: the robot drives in and charges through sprung copper contacts (the contact version of the wireless charging dock idea).

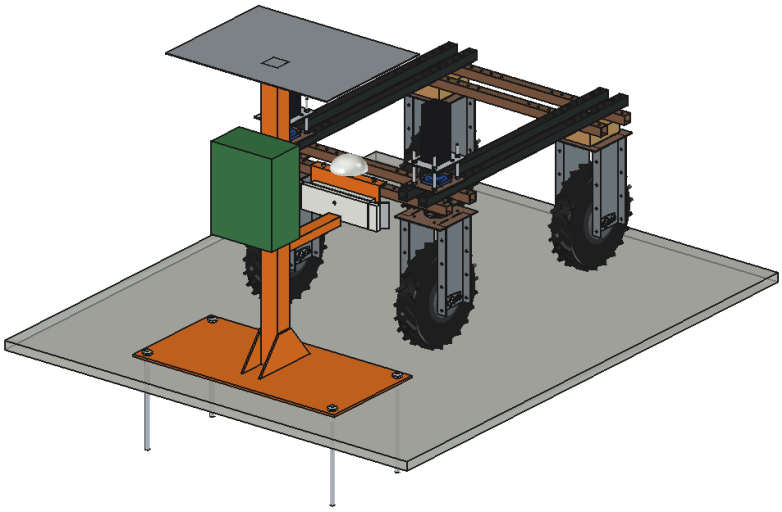


Simple

1 variant: Simple
Folder: `charging dock/`

CHARGING DOCK · VARIANT 1 OF 1

Simple



Robot side: a 3.9 kg contact bracket on the front outer lower beam with 2 sprung copper pads and a pilot pin. Station: base plate with ground anchors, post with a 48 V 1 kW IP65 charger and a roof, an arm back between the front wheels to a PE contact head with 2 wide copper bars and a V-funnel.

> 17 A at 54 V, 0.3 W contact loss

> Robot 20 -> 100 % in 2 h

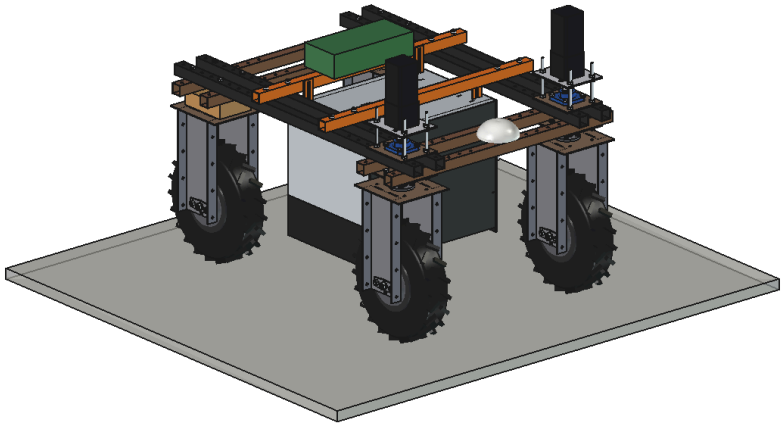
> Catches 75 mm sideways; 80 kg station

Folder: `charging dock/simple/`

IMPLEMENT · WEED & CROP PROTECTION

UV-C mildew light

Reflecting hood with 6 UV-C lamps under the robot between the axles: drives over the bed at night and gives a low UV-C dose against powdery mildew, without chemicals.

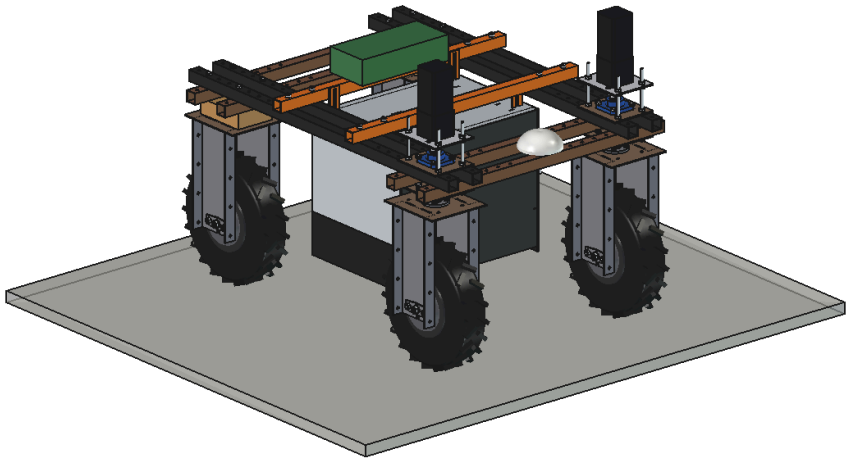


Simple

1 variant: Simple
Folder: `uv mildew light/`

UV-C MILDEW LIGHT · VARIANT 1 OF 1

Simple



Aluminium hood 580 x 440 on 4 slotted hangers from 2 bearers on the upper beams, rubber side skirts and brush curtains, 6 compact UV-C tubes (PL-L 55 W class) behind guard wires, ballasts and interlock in a box on the rear bearer.

> 200 J/m2 at 0.58 m/s, ~1020 m2/h

> 489 W, ~4000 m2 per battery

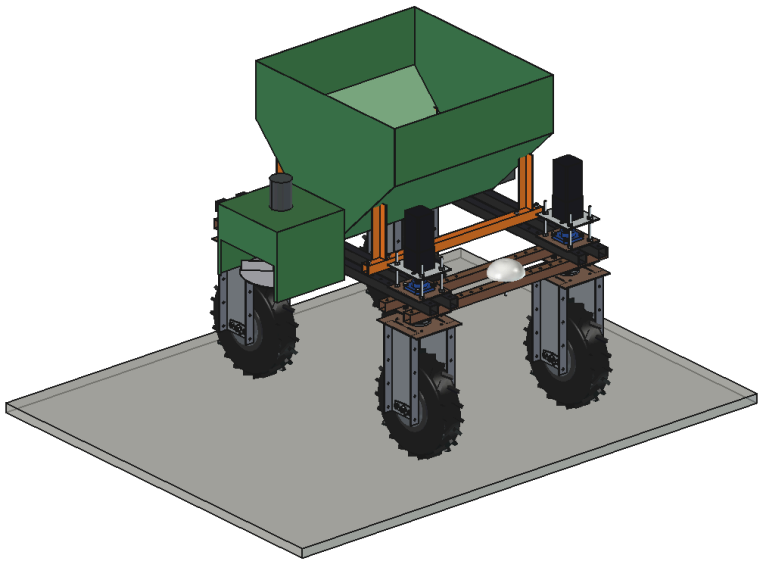
> 21 kg, 25 mm from the steered tyres

Folder: `uv mildew light/simple/`

IMPLEMENT · SPREADING & BEDDING

Sawdust spreader

Bedding spreader for cubicles: a 308 l bin with an auger across the robot feeds a spinner on the right side that throws sawdust, chopped straw or a lime mix into the cubicles from the alley.

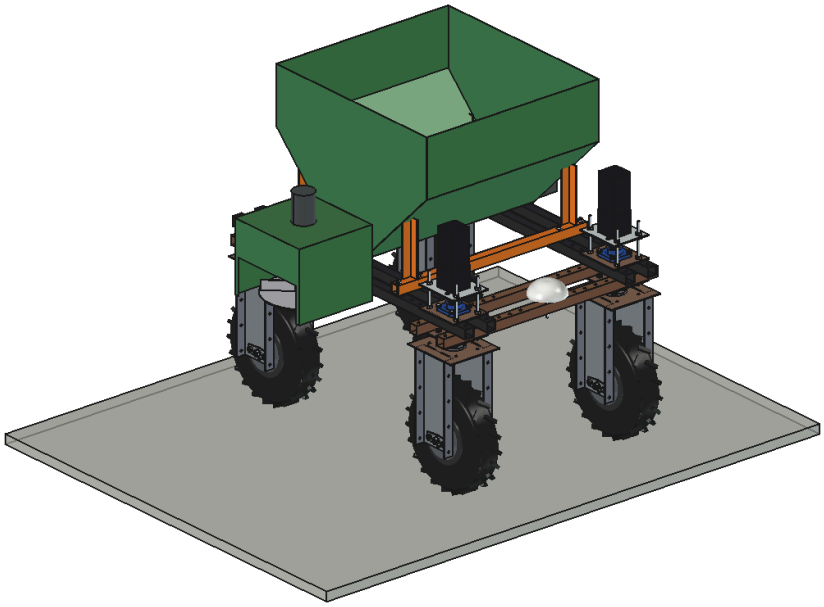


Simple

1 variant: Simple
Folder: `sawdust spreader/`

SAWDUST SPREADER · VARIANT 1 OF 1

Simple



Bin 2 mm with 45 degree walls to a trough on 2 bearers on the upper beams, auger O150 across the robot on a 24 V worm gearmotor, spinner housing outside the right outer upper beam with a O320 disc (4 vanes, 350 rpm) open to the right.

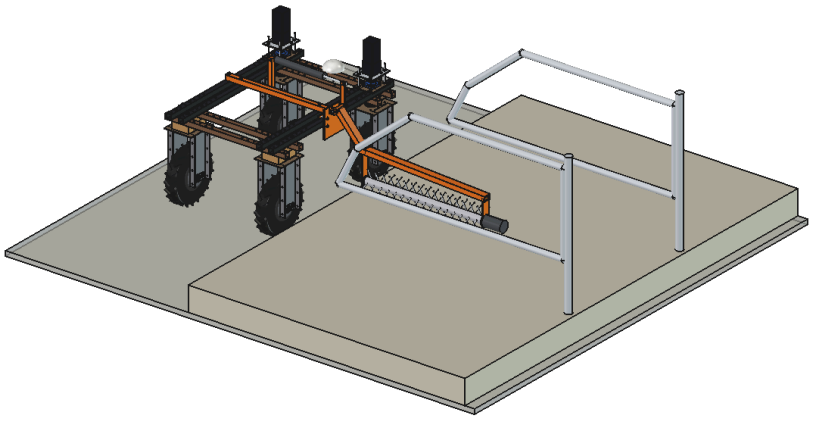
- 61 cubicles per fill of sawdust, ~570 per hour
- Throws ~1.3 m beyond the wheel line
- 75 kg empty

Folder: `sawdust spreader/simple/`

IMPLEMENT · LIVESTOCK & FARM

Bedding rake

Cubicle groomer on a side arm: reaches from the alley over the curb, under the divider loops, with a tine rotor along the bed that loosens and levels the bedding in the rear metre of each cubicle.



Simple

1 variant: Simple
Folder: `bedding rake/`

BEDDING RAKE · VARIANT 1 OF 1

Simple

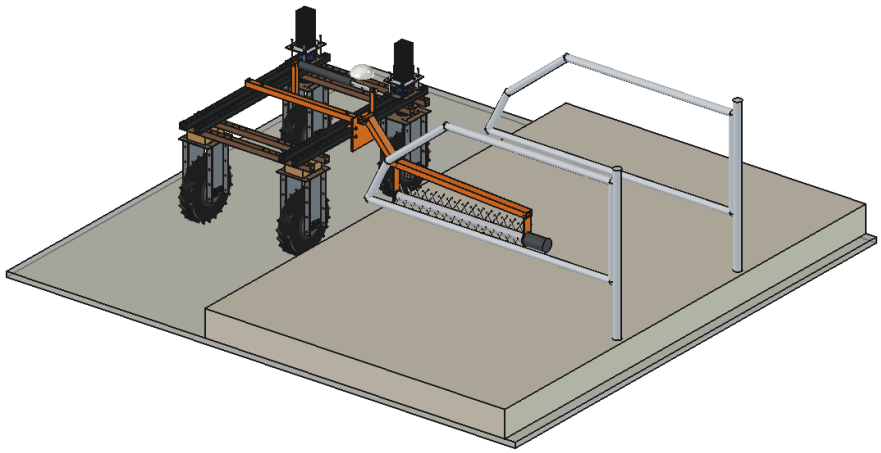


Plate bracket with pivot lugs on the right outer upper beam, arm 60 x 40 from a pivot above the beams down past the curb and out over the bed, tine rotor O240 x 970 along the bed on a 48 V 400 W gearmotor, 24 V actuator across the robot top to a crank lifts the arm 40 degrees. A row of cubicles is modelled for the checks.

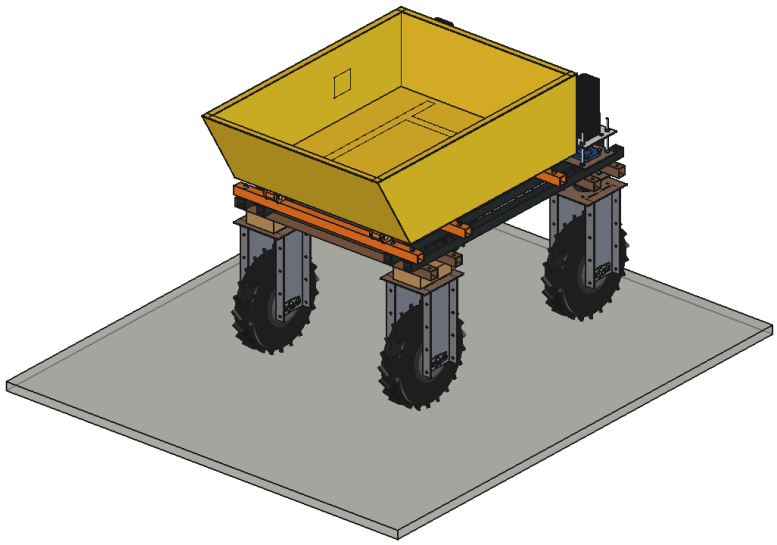
- Grooms 100..1070 mm behind the curb, 90 mm under the loops
- ~655 cubicles per hour, ~300 W
- 37 kg, side tip angle 27 degrees

Folder: `bedding rake/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Dumper barrow

Tipping skip on the upper beams: a 353 l steel bin hinged at the rear end of the robot, tipped backwards to 48 degrees by two actuators so the load slides out behind the rear wheels.

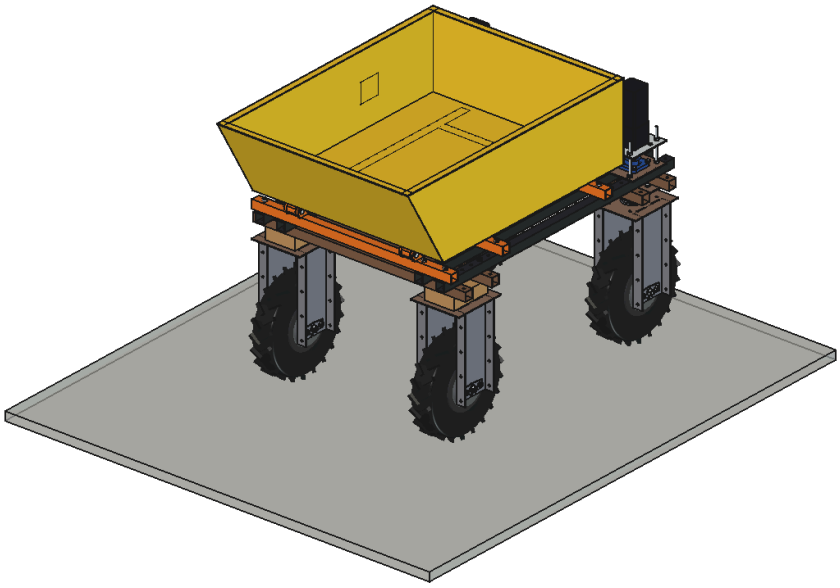


Simple

1 variant: Simple
Folder: `dumper barrow/`

DUMPER BARROW · VARIANT 1 OF 1

Simple



3 bearers on the upper beams (rest pads, actuator lugs, hinge lugs), subframe of 2 rails 40 x 40 with hinge bosses and a cross tube, 2 mm bin 900 x 1020 x 350 with a 30 degree rear lip, 2 x 24 V 3000 N actuators (stroke 300).

> 150 kg of soil (100 l) or 353 l of chips / green waste

> Tips 48 degrees in ~36 s, cg stays in front of the rear axle

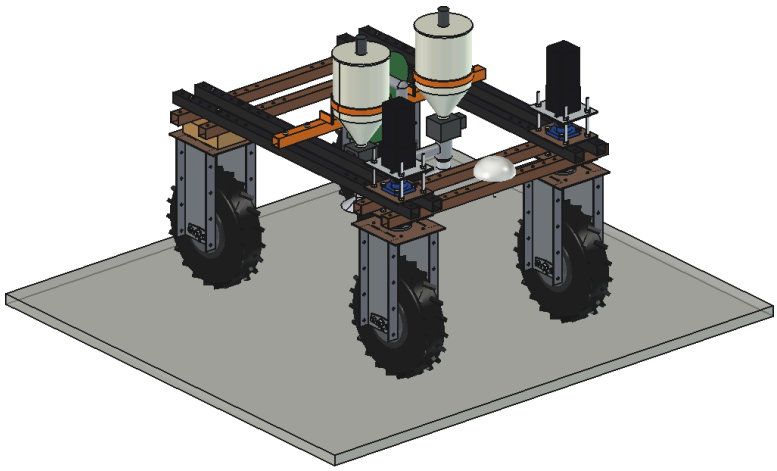
> 72 kg, ~950 kg of soil per hour on 100 m

Folder: `dumper barrow/simple/`

IMPLEMENT · WEED & CROP PROTECTION

Biocontrol spreader

Releases predatory mites and parasitic wasps (in bran, vermiculite or sawdust) over two rows: 2 hoppers with metering wheels and a blower that carries the product through hoses to deflector nozzles.

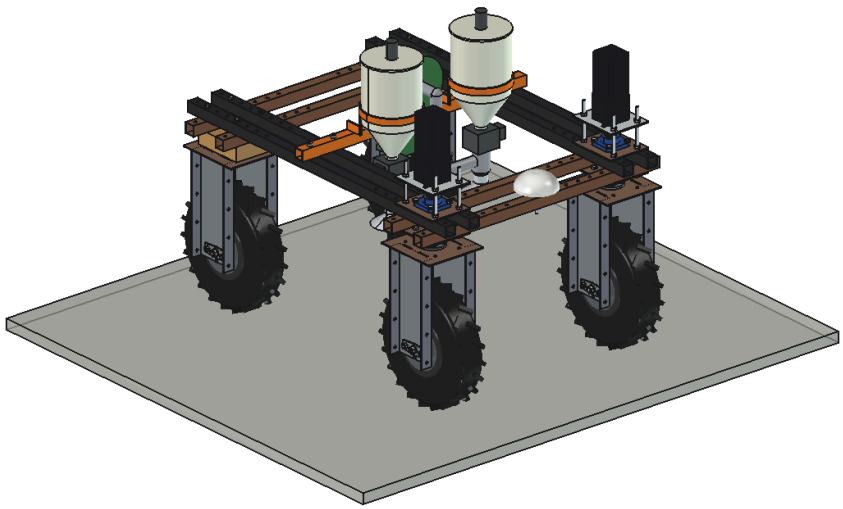


Simple

1 variant: Simple
Folder: `biocontrol spreader/`

BIOCONTROL SPREADER · VARIANT 1 OF 1

Simple



Bearer across the upper beams with 2 PE hoppers of 7.9 l (agitator, NEMA17 metering wheel) over the rows at x +-187.5, a 24 V 60 W centrifugal fan, venturis, hoses O40 and deflector nozzles 450 mm above the ground.

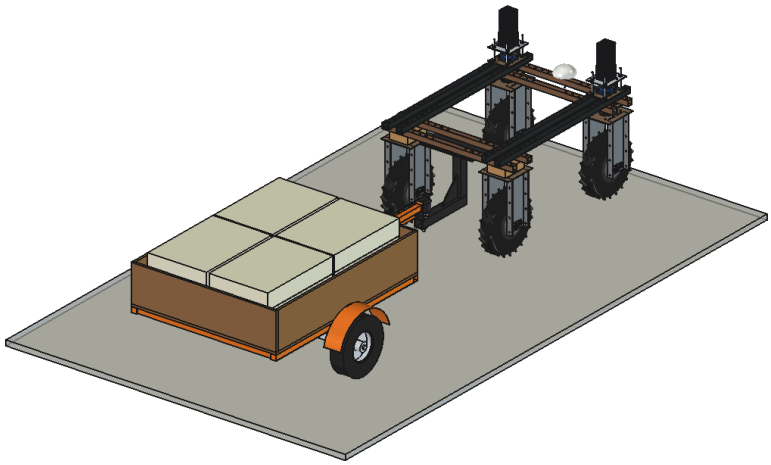
- Predatory mites 2 ml/m2: 0.8 ha per fill
- ~1350 m2/h, ~12 h per battery
- 18 kg

Folder: `biocontrol spreader/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Bag trailer

Two-wheeled trailer on the 50 mm tow ball of the cambridge roll (hitch, coupling and drawbar imported): a plywood deck with boards and a drop tailgate for 12 bags of 25 kg.

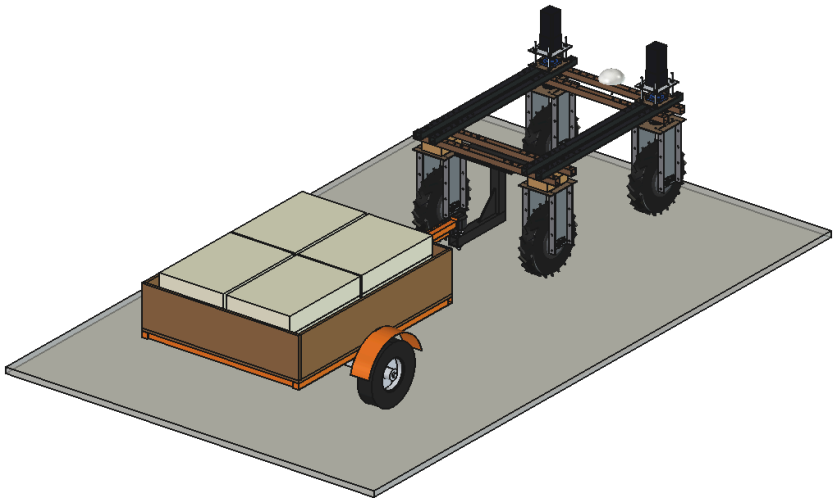


Simple

1 variant: Simple
Folder: `bag trailer/`

BAG TRAILER · VARIANT 1 OF 1

Simple



Frame of 40 x 40 rails and cross members behind the cbr cross beam and drawbar, plywood deck 1110 x 1410 at 358 mm with side and front boards and a drop tailgate, axle tube on hangers, 2 x 4.00-8 wheels with mudguards.

> 12 bags of 25 kg, 24 kg on the ball

> Steepest slope loaded: 3.5 deg wet grass, 6.5 dry

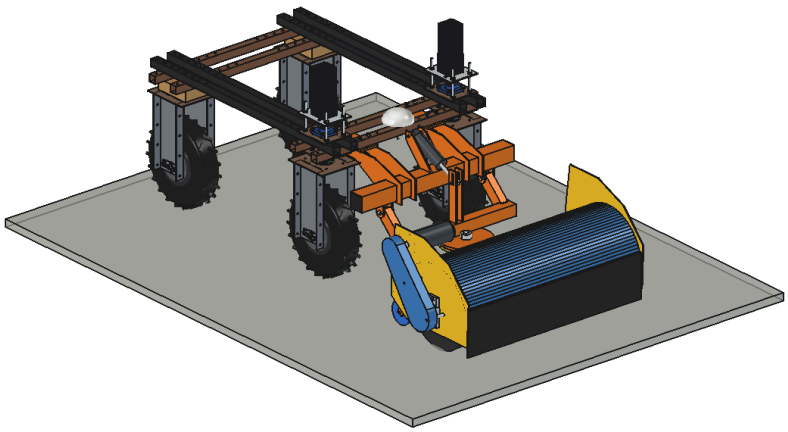
> 110 kg empty

Folder: `bag trailer/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Broom / sweeper

Front angle broom on the push blade's toolbar, lift arms and king pin (imported): a roller brush O400 x 1000 under a hood, 48 V belt drive, on 2 castors, angled +-25 degrees to throw the material aside.

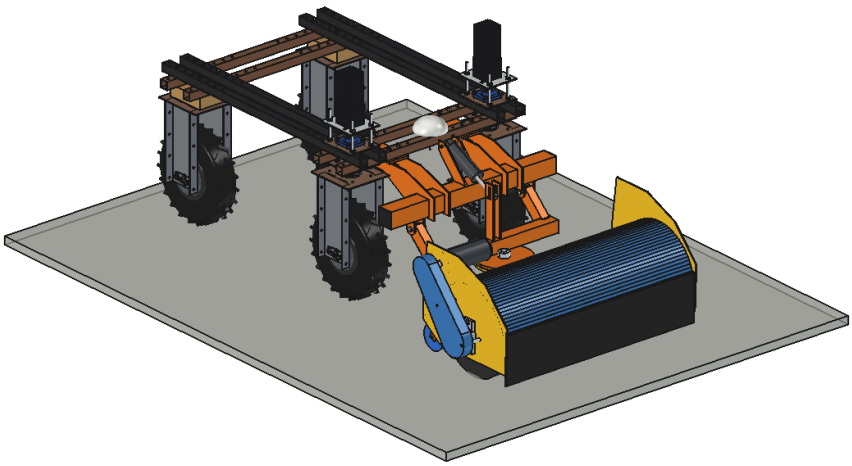


Simple

1 variant: Simple
Folder: `broom sweeper/`

BROOM / SWEEPER · VARIANT 1 OF 1

Simple



Head frame with the swivel plate, cross beam 80 x 60 and 5 mm end plates, PP brush O400 x 1000 in UCFL 208 bearings at 400 rpm, 48 V 1000 W gearmotor with an HTD belt, aluminium hood and rubber flap, 2 swivel castors O125; 3000 N version of the blade actuator.

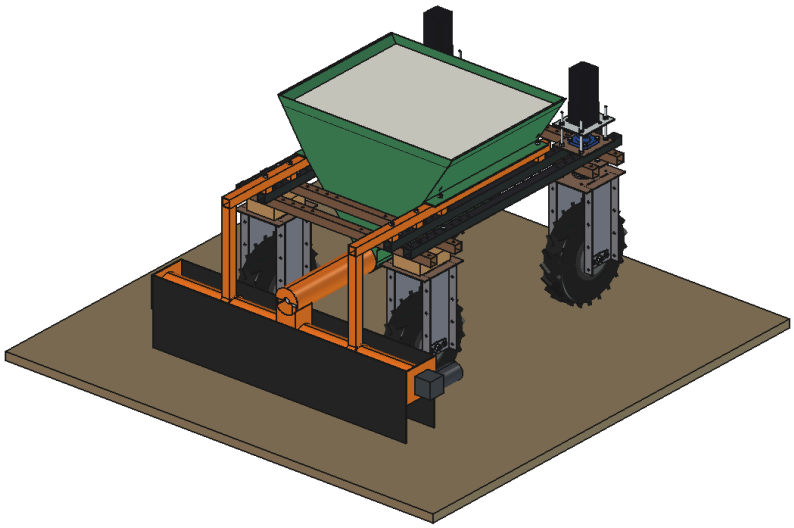
- ~2900 m2/h angled, ~4900 m2 per charge
- ~910 W brush, rear axle 54 % working
- Lifting needs 4..9 counterweight plates; 124 kg

Folder: `broom sweeper/simple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Lime spreader

Drop spreader for granulated or ground lime: a 178 l V-hopper centred over the robot, a trough auger to a 1.3 m cross auger behind the rear wheels that lets the lime fall through a row of holes inside a rubber dust skirt.

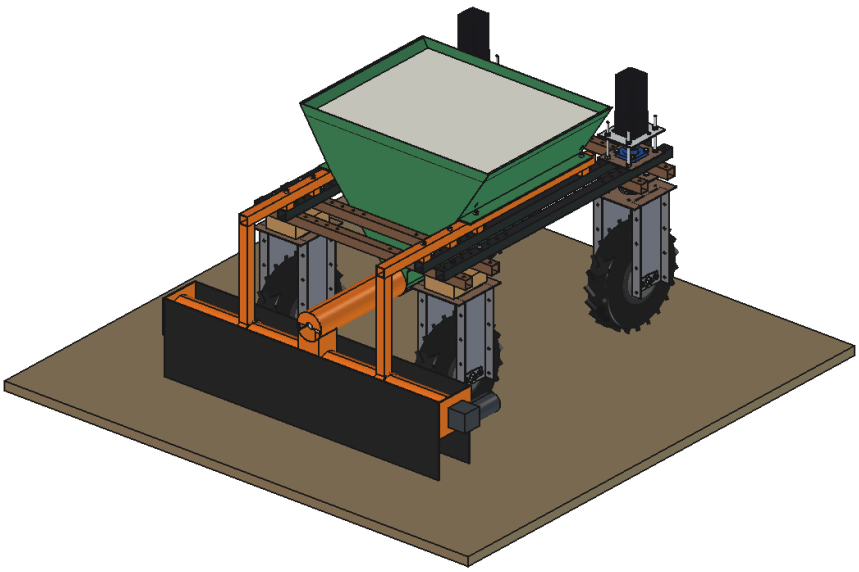


Simple

1 variant: Simple
Folder: `lime spreader/`

LIME SPREADER · VARIANT 1 OF 1

Simple



Rails on the inner upper beams, steel V-hopper down between the beams to a U-trough, trough auger O100 in a tube under the rear lower beams, cross auger O100 x 1300 with 25 outlet holes O20 and flights from the middle outward, 2 x 24 V worm gearmotors, rubber skirt 20 mm above the ground.

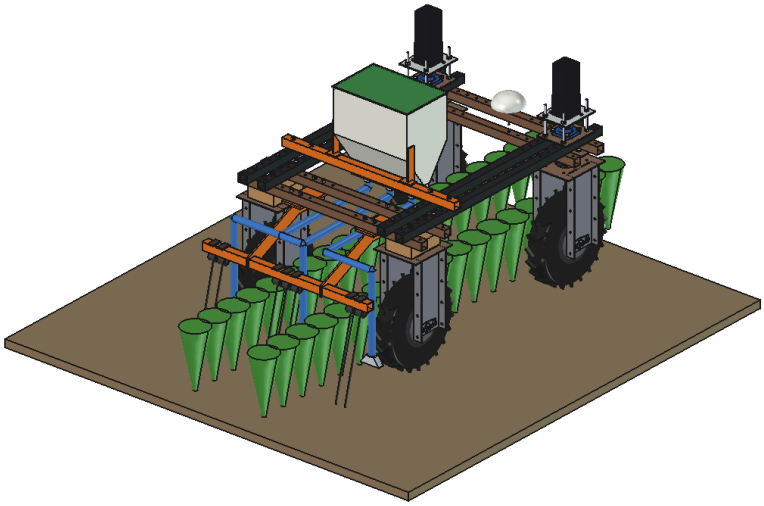
- 205 kg granulated lime per fill: 0.21 ha at 1000 kg/ha
- Up to 3.5 t/ha, 0.30 ha/h, no dust cloud
- 97 kg empty, rear axle 60 % full

Folder: `lime spreader/simple/`

IMPLEMENT · CROP & SOIL

Cover crop undersower

Sows a cover crop into a standing 2-row crop: a 28 l hopper with 3 fluted metering cells feeds hoses to fishtail deflectors in the free middle inter-row and behind both rear tyres; spring tines on a bar high over the crop cover the seed.

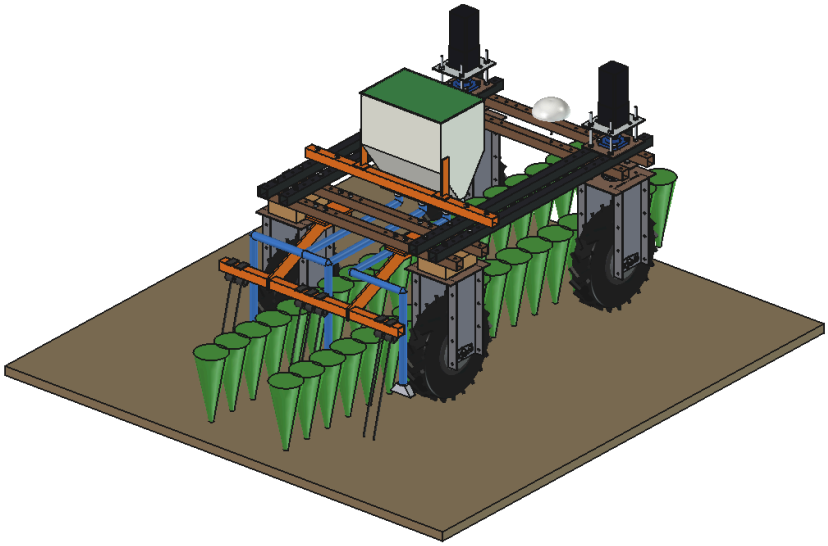


Simple

1 variant: Simple
Folder: `cover crop undersower/`

COVER CROP UNDERSOWER · VARIANT 1 OF 1

Simple



Bearer across the upper beams with a PE V-hopper, aluminium metering housing with 3 fluted rollers on one 12 V speed-following shaft, hoses O32 routed over the crop rows, fishtail deflectors 40 mm above the ground, tine bar at 430 mm on 2 hangers under the rear lower beams with 7 spring tines O8.

> Ryegrass 0.43 ha, clover 3.6 ha per fill

> 0.22 ha/h, ~1.1 ha per charge

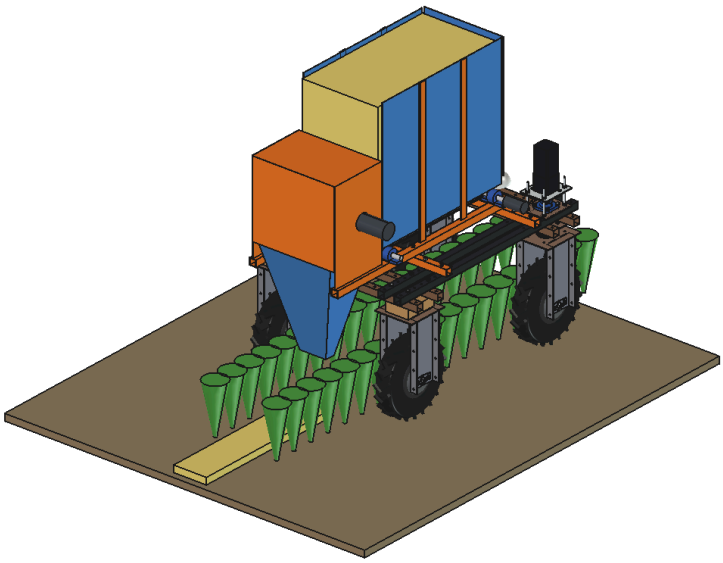
> 26 kg; crop rows untouched (checked)

Folder: `cover crop undersower/simple/`

IMPLEMENT · SPREADING & BEDDING

Straw spreader

Lays a straw mulch strip in the free middle inter-row: 2 small bales on a slatted floor chain are fed against a spiked shredder drum, the loose straw falls through a chute to a 160 mm outlet between the crop rows.

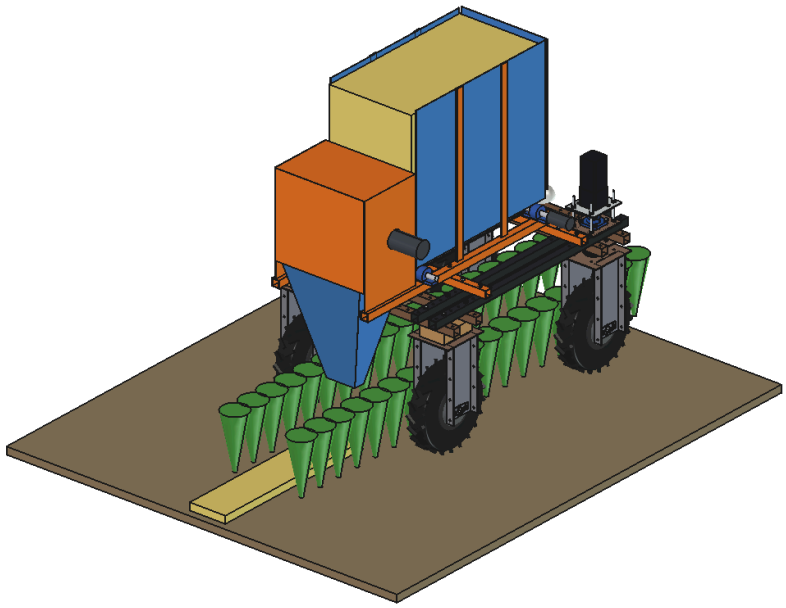


Simple

1 variant: Simple
Folder: `straw spreader/`

STRAW SPREADER · VARIANT 1 OF 1

Simple



2 bearers and 2 rails on the upper beams, slatted floor of 2 chains 08B in UCP 204 shafts with a 24 V gearmotor, aluminium bale chamber, spiked drum O320 at 250 rpm on a 48 V 750 W motor in a 3 mm housing, aluminium chute to z 300.

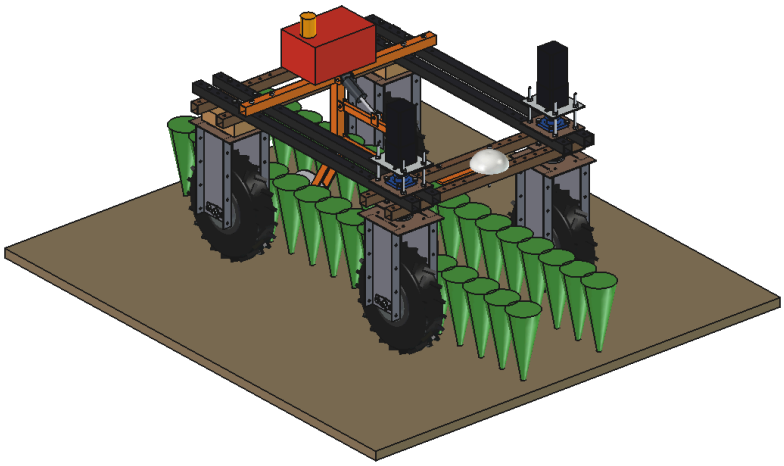
- 260 m of strip per fill at 0.5 kg/m2
- ~440 W drum, ~3.5 km of strip per charge
- 102 kg empty; crop rows untouched (checked)

Folder: `straw spreader/simple/`

IMPLEMENT · WEED & CROP PROTECTION

Electric weeder

Spot electric weed control in the free middle inter-row: a camera finds weeds, a 5 kV generator switches on a comb of spring-steel fingers only while a weed is under it, the current returns through a coulter disc.

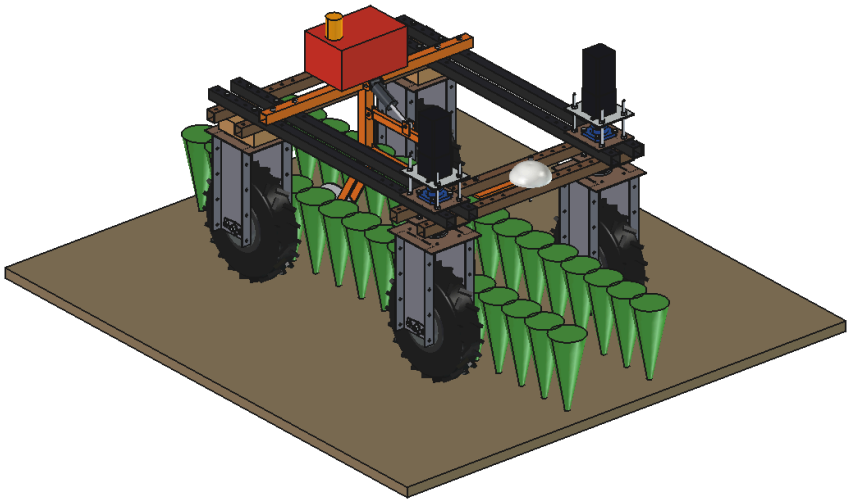


Simple

1 variant: Simple
Folder: `electric weeder/`

ELECTRIC WEEDER · VARIANT 1 OF 1

Simple



Bearer on the upper beams with the HV box and beacon, post with a parallel-arm lift (24 V actuator), PE holder with 9 spring-steel fingers and rubber curtains, coulter disc O250 as earth return, camera under the inner front lower beam.

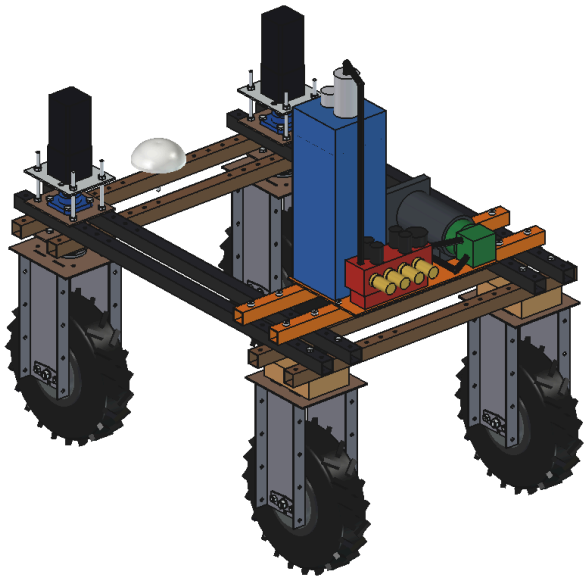
- 4-leaf weeds at 0.2 m/s: 1.8 kJ per weed
- ~3000-3800 weeds per charge
- 27 kg; crop rows untouched (checked)

Folder: `electric weeder/simple/`

IMPLEMENT · TRANSPORT & GENERAL

Hydraulic power pack

48 V hydraulic power unit on the rear half: BLDC motor with a 4 cc gear pump (11 l/min, 150 bar), 16 l tank, 2-section solenoid valve block and 2 pairs of ISO-A quick couplers for hydraulic implements.

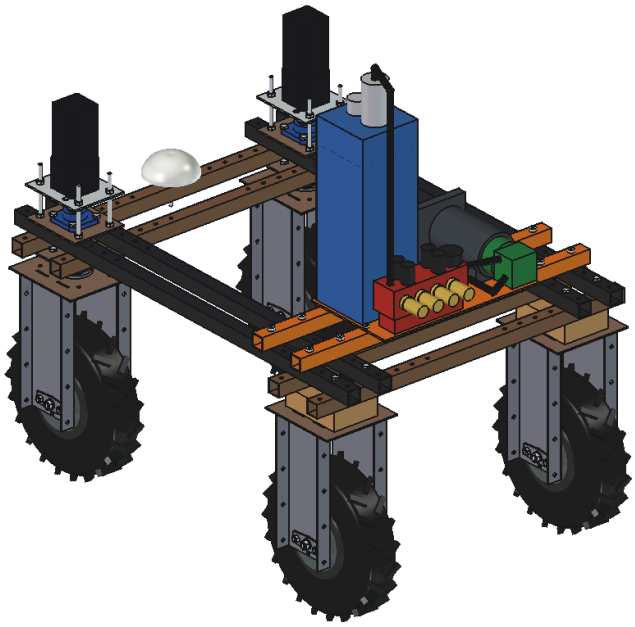


Simple

1 variant: Simple
Folder: `hydraulic pump/`

HYDRAULIC POWER PACK · VARIANT 1 OF 1

Simple



2 bearers and a 5 mm base plate on the upper beams, 16 l steel tank with return filter, BLDC 48 V 3.5 kW with bell housing and group-1 gear pump, motor controller, valve block with relief valve, gauge and 2 double-acting 24 V sections, couplers screwed into its rear face.

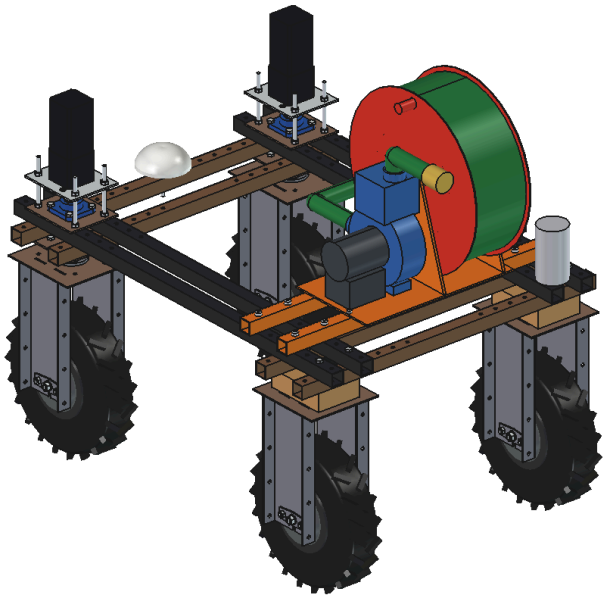
- 11 l/min, 150 bar: 19-30 kN cylinders
- 44 min of full flow at 100 bar per charge
- 49 kg + 12 kg oil, rear axle 62 %

Folder: `hydraulic pump/simple/`

IMPLEMENT · SPRAYING, FERTILIZING & IRRIGATION

Irrigation pump

Direct pump from ditches and ponds: a self-priming 1.5" pump on a 48 V 1.5 kW motor with an 8 m suction hose on a reel, delivering to a Camlock coupling at the rear; mount of the hydraulic power pack.

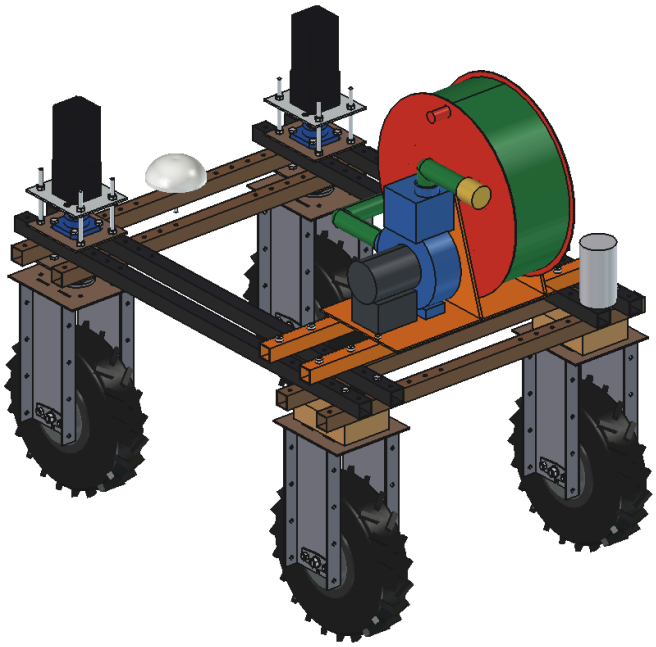


Simple

1 variant: Simple
Folder: `irrigation pump/`

IRRIGATION PUMP · VARIANT 1 OF 1

Simple



Bearers and base plate from the hydraulic pack, self-priming centrifugal pump with priming chamber and BLDC 1.5 kW, reel O520 with 8 m of spiral suction hose, rotary union and strainer with foot valve, discharge to a 1.5" Camlock at the rear edge.

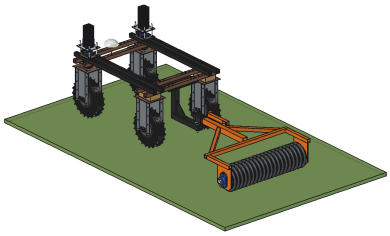
- 8.5 m3/h at 3 m lift into a drip line
- 10-17 m3 per charge
- 62 kg

Folder: `irrigation pump/simple/`

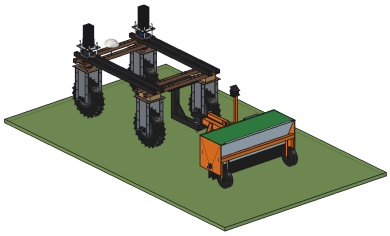
IMPLEMENT · SOWING & PLANTING

Cambridge roll and overseeder

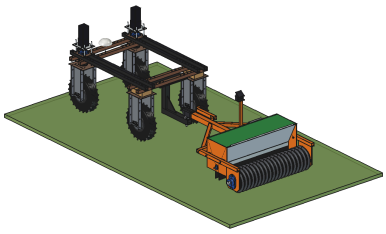
Simple trailed lawn-repair implements on a 50 mm tow ball under the rear beams: a Cambridge roll, a drop seeder that a camera switches on over bare soil, and the two combined.



Cambridge roll



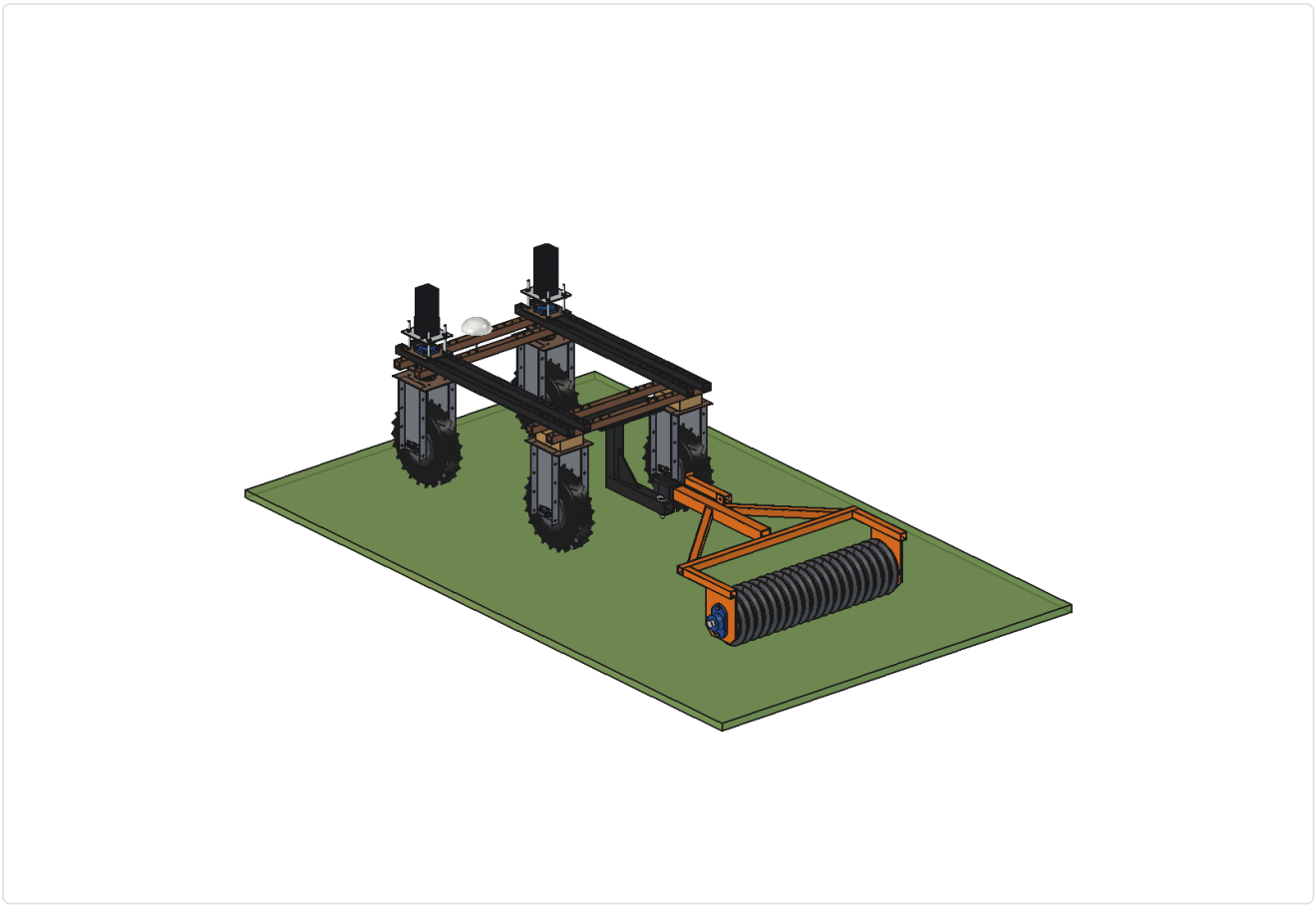
Overseeder (drop seeder)



Overseeder with Cambridge roll

3 variants: Cambridge roll · Overseeder (drop seeder) · Overseeder with Cambridge roll
Folder: `overseeding/`

Cambridge roll

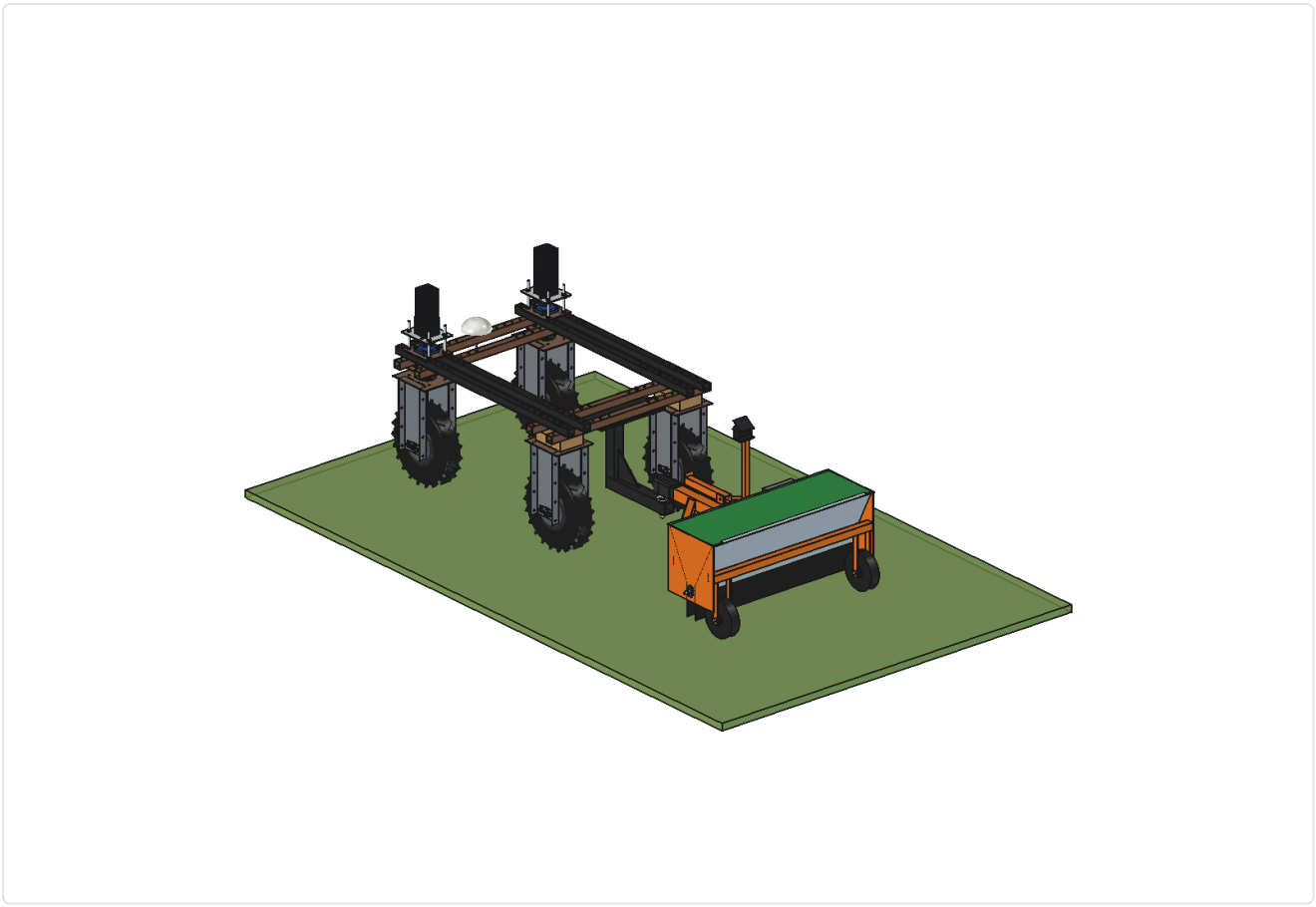


1 m ring roller with 20 cast-iron rings Ø280 on a Ø40 shaft, on the tow hitch with coupling head and drawbar.

- 178 kg
- 100 ... 250 N draft
- About €850 with the hitch

Folder: `overseeding/cambridge roll/`

Overseeder (drop seeder)

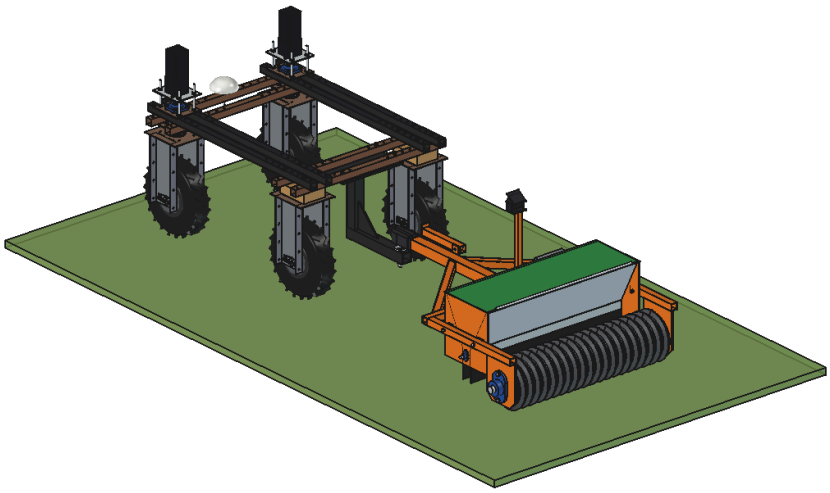


1 m drop seeder with a 49 l hopper and a fluted roller on a 24 V worm gearmotor that a camera switches on only over bare soil, on two small wheels.

- 61 kg
- About 60 N draft
- About €500

Folder: `overseeding/overrseeder/`

Overseeder with Cambridge roll



The seeder unit hangs between the arms of the roll: seed falls 300 mm in front of the rings and is rolled in. Includes the turn kinematics of the trailer and an animation with bare patches.

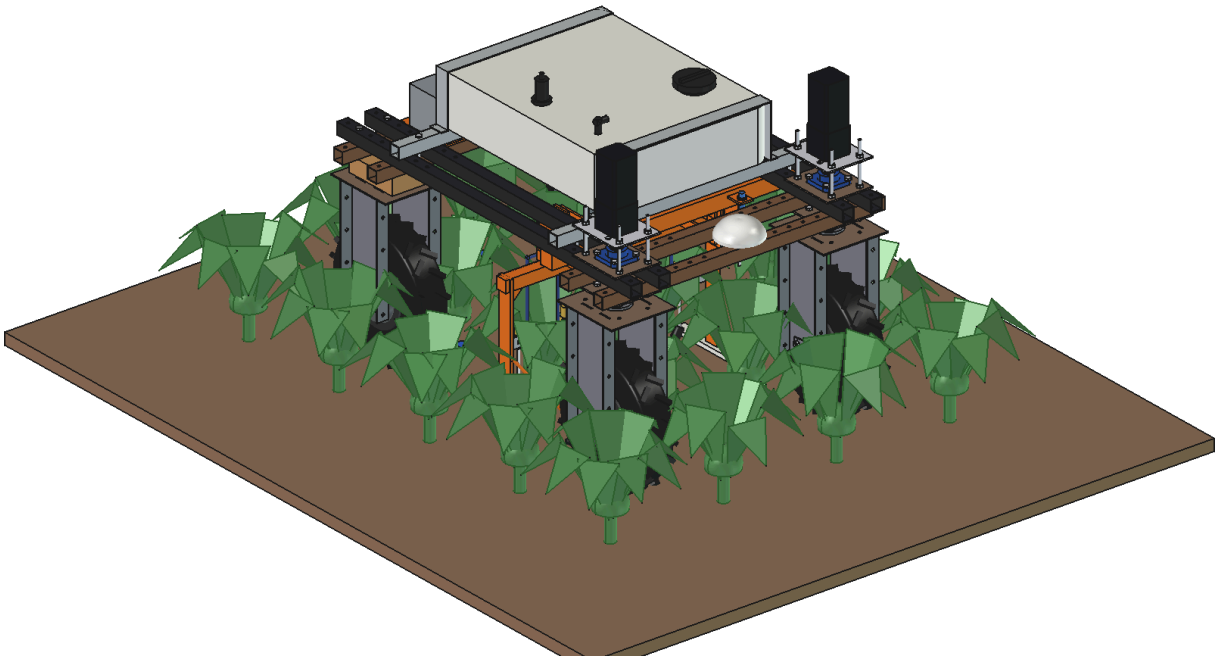
- 216 kg (234 kg full)
- 185 ... 310 N draft
- About €1,200

Folder: `overseeding/overseeder with cambridge roll/`

IMPLEMENT · WEED & CROP PROTECTION

Cauliflower root-crown applicator

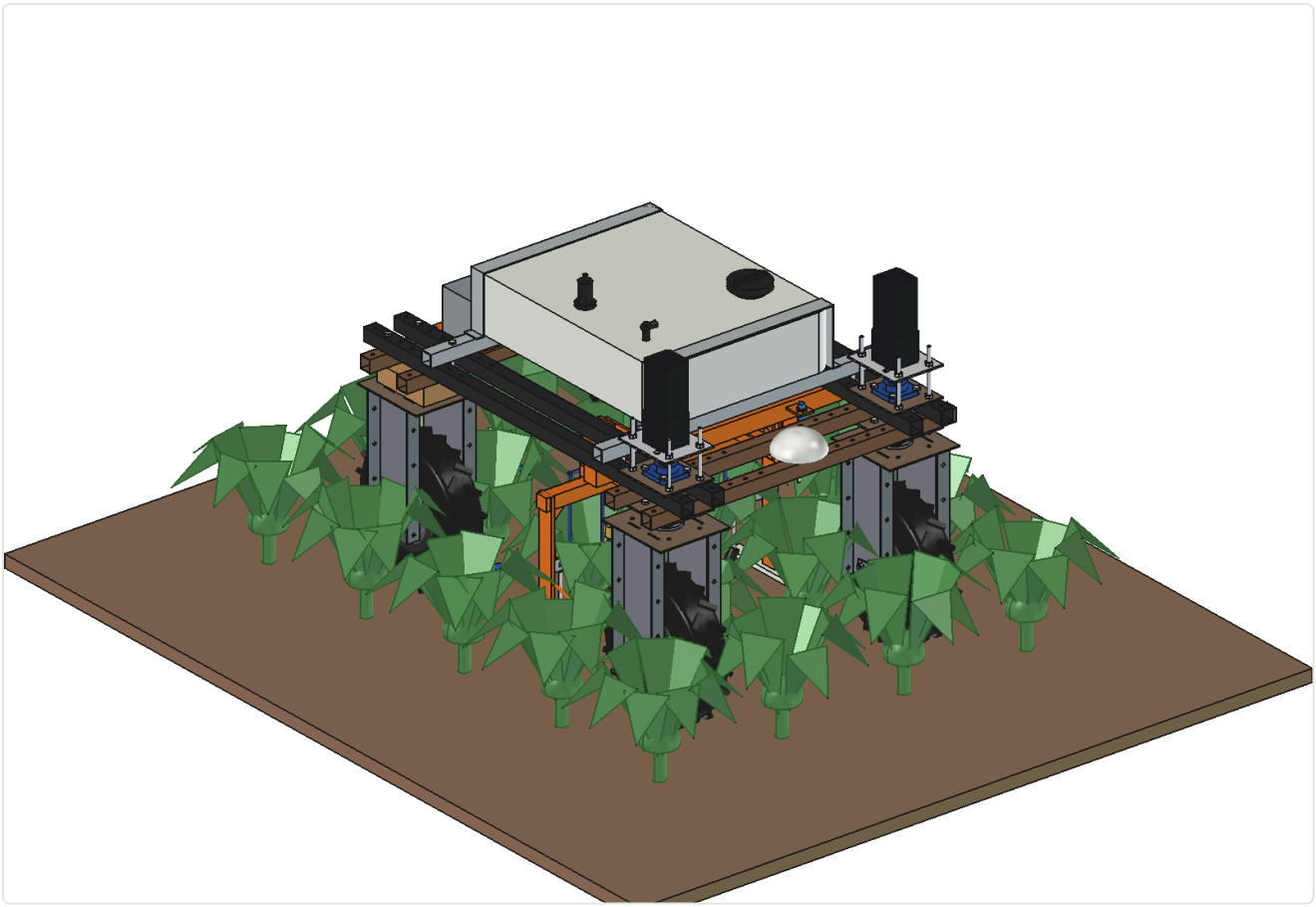
Doses insecticide in pulses onto the soil 40 mm beside the stem of every cauliflower plant (mid-season, 40 x 40 cm), not over the crop.



Simple

1 variant: Simple
Folder: `cauliflower applicator/`

Simple



Two rows pass under the robot; per row a parallelogram unit on UHMW skids follows the soil, leaf lifter rods clear the stem base and two 24 V solenoid valves fire 30° drop nozzles inside splash shrouds. A distance sensor finds the crown, wheel odometry times the pulse. 60 l tank, diaphragm pump, regulator and accumulator on the rear beams.

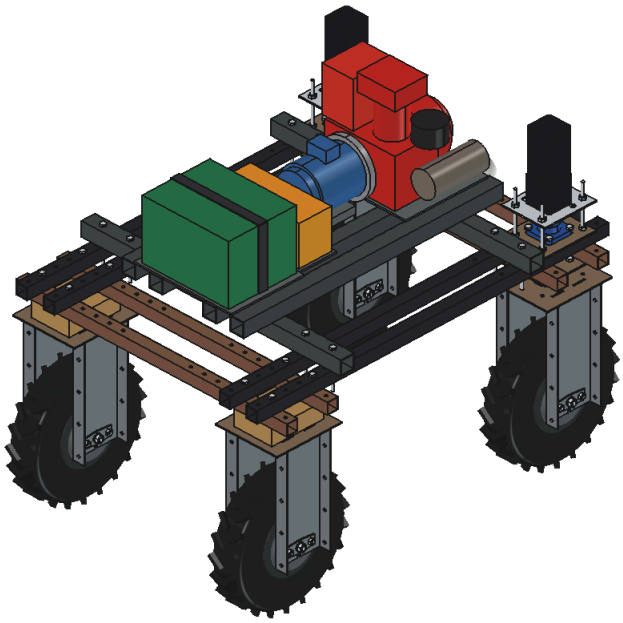
- 4 ml/plant = 250 l/ha
- 0.11 ha/h, 0.24 ha per tank
- 56.5 kg

Folder: `cauliflower applicator/simple/`

IMPLEMENT · POWER & DRIVE

Hybrid range extender

Petrol generator set on a steel tube base on top of the robot that charges the battery while driving; the battery delivers the peak power for heavy pulling (the worked-out generator / engine idea).

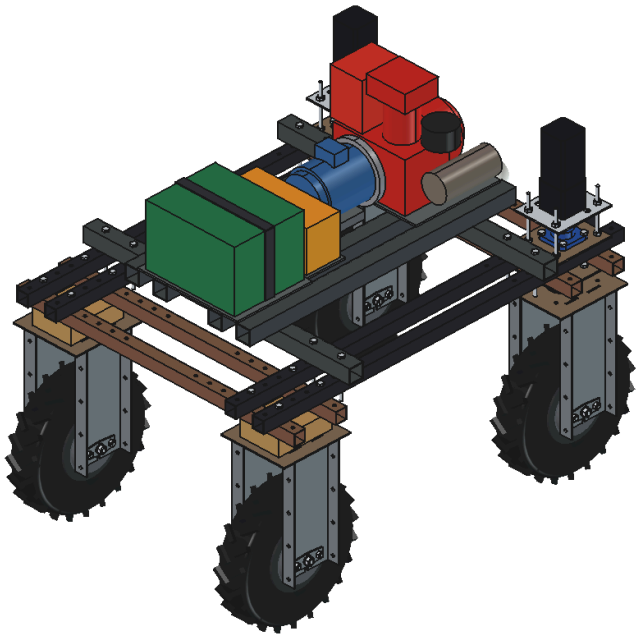


Range extender

1 variant: Range extender
Folder: `Hybrid/`

HYBRID RANGE EXTENDER · VARIANT 1 OF 1

Range extender



In-line train on 50 x 50 tubes bolted to the upper beams: 212 cc petrol engine (4.8 kW), jaw coupling, 3 kW permanent-magnet generator, controller box and a 16S 59 V 40 Ah battery (2.4 kWh). Clear of the steering stacks.

- About 76 kg, cg z 868 mm
- About 1.5 l/h at full output
- Light task: battery +2.1 kW

Folder: `Hybrid/`

IMPLEMENT IDEAS

Ideas not worked out yet

24 ideas (from the website's list and own brainstorming) that have no model yet. Each has a folder in `implement ideas/` with a `README.md` describing what it does, so it can be worked out later.

Soil Cultivation & Weed Control

- **Drop-on-demand boom (coarse droplets)**

Sowing & Planting

- **Tray setter**

Power & Drive

- **Heavy-duty electric motors**
- **Charging robot (mobile charger)**

Transport & General

- **Seat, cabin & frame extension**
- **Speech speaker**
- **Magnet sweeper**
- **Debris vacuum**

Cooperation & Swarm Operation

- **Drone combination**
- **Swarm operation (multiple robots per field)**

Crop & Soil

- **Mole and tunnel mapper**

Weed & Crop Protection

- **Laser or UV weed treatment**
- **Hot steam / hot foam weeder**
- **Pheromone trap service**

Livestock & Farm

- **Fence inspector**
- **Drinking trough check**
- **Chicken house robot**

Measuring & Sensors

- **Yield estimator**
- **Disease and pest scout**
- **Soil moisture sensor placer**
- **Windbreak and tree-row inspector**
- **Security camera patrol**
- **Metal detector**

Spraying, Fertilizing & Irrigation

- **Biostimulant sprayer (full field)**

IDEA · SOIL CULTIVATION & WEED CONTROL

Drop-on-demand boom (coarse droplets)

Not worked out yet

WHAT IT DOES

Boom with individually controlled injectors that fire single coarse droplets on demand at low pressure. Every droplet is placed straight onto the weed, which practically eliminates drift and cuts chemical usage to a fraction of a conventional spray pass.

CONCEPT (FIRST THOUGHTS)

Same carrier as the spot sprayer (camera mast in front, boom behind), but with drop-on-demand valves instead of PWM flat-fan nozzles. Droplets are fired at the detected plant position, so the boom needs a finer position resolution along the boom than 250 mm.

REUSE FROM EXISTING IMPLEMENTS

Camera mast, tank, pump, folding boom and control box of the spot sprayer (spotspray unit/boom); pulsed solenoid drop nozzles and timing by odometry of the cauliflower applicator.

OPEN QUESTIONS

- Injector type, droplet size and firing rate
- Injector pitch along the boom
- Aiming accuracy at 0.5 m/s: time from camera to droplet
- Product types and viscosity the injectors can handle

Folder: `implement ideas/drop_on_demand_boom/`

IDEA · SOWING & PLANTING

Tray setter

Not worked out yet

WHAT IT DOES

Front-mounted unit that takes seedlings out of the tray and sets them out at the right spacing.

CONCEPT (FIRST THOUGHTS)

A magazine of trays above the front beams, a gripper or pin-picker takes one seedling per cycle and plants it in a dibbled hole. Spacing by the robot's odometry; the robot stops or crawls per plant.

REUSE FROM EXISTING IMPLEMENTS

XYZ stage idea of the field marker for positioning; camera and odometry of the in-row weeder.

OPEN QUESTIONS

- Tray type and plug size
- Pick-and-place mechanism and cycle time
- Planting depth and soil closing
- Tray reloading, again from a refiller?

Folder: `implement ideas/tray_setter/`

IDEA · POWER & DRIVE

Heavy-duty electric motors

Not worked out yet

WHAT IT DOES

Electric motors such as the Ego Power+ or Greenworks units to drive implements that need real power, replacing small petrol engines.

CONCEPT (FIRST THOUGHTS)

A standard interface plate for battery-powered garden-tool motors (their battery or the robot's bus via adapter), with a belt or shaft output to an implement. Gives a ready, cheap drive where a bought motor is easier than a custom 48 V BLDC.

REUSE FROM EXISTING IMPLEMENTS

Motor plates and V-belt drives of the around-pole mower and the weed picker.

OPEN QUESTIONS

- Which motors: brands, power, output shaft type
- Supply: own battery or the robot's 48 V
- Interface plate and quick-change mounting
- Protection against dust and water

Folder: `implement ideas/heavy_duty_motors/`

IDEA · TRANSPORT & GENERAL

Seat, cabin & frame extension

Not worked out yet

WHAT IT DOES

Front or rear-mounted seat, with the option to widen the frame tubes at the sides and a small cabin as a protective enclosure for the operator.

CONCEPT (FIRST THOUGHTS)

A seat on a tube frame so a person can ride on the robot, and side extensions that widen the frame for wider implements. Needs a manual-drive mode and clear operator-safety rules. A light cabin or canopy over the seat is the next step and only makes sense together with the seat.

REUSE FROM EXISTING IMPLEMENTS

Toolbar and hanger design of the disc brushes; tube sizes from the robot.

OPEN QUESTIONS

- Operator weight and the axle loads
- Controls and an emergency stop within reach
- Side-extension width versus stability
- Safety rules for a ridden robot
- Cabin weight, entry and exit, visibility and rollover protection

Folder: `implement ideas/seat_frame_extension/`

IDEA · COOPERATION & SWARM OPERATION

Drone combination

Not worked out yet

WHAT IT DOES

The robot works together with a drone as a mobile base station. The drone flies out from the platform, maps the plot from above and passes its imagery straight to the robot, which converts it into a driving and treatment plan. The drone charges and swaps batteries on the robot, so the pair can keep scouting and working alternately for a full day without anyone coming out to the field.

CONCEPT (FIRST THOUGHTS)

A landing platform on the robot's top plate with a charging contact or battery-swap mechanism, a data link to the robot computer and a way to align the drone on landing (RTK or visual markers). The plan generation reuses the task-map pipeline of the crop scanner.

REUSE FROM EXISTING IMPLEMENTS

Top-plate pattern; solar panel roof as an alternative platform; task-map pipeline of the grass meter and NIR sensor.

OPEN QUESTIONS

- Platform size and landing aid
- Charging versus battery swap
- Weather limits for flying and landing
- Regulations for autonomous flights

Folder: `implement ideas/drone_combination/`

IDEA · COOPERATION & SWARM OPERATION

Swarm operation (multiple robots per field)

Not worked out yet

WHAT IT DOES

Several Bruut units work the same field as one swarm. The job is split into sections, the robots divide the passes among themselves and keep each other informed of position, progress and battery state, so they never overlap or cross. Capacity scales simply with the number of units instead of with machine size, work continues when one unit drops out, and the light platforms keep soil compaction low.

CONCEPT (FIRST THOUGHTS)

Mostly software: a job planner that splits the field into passes, shared position and progress messages over a radio link, collision avoidance by reserved corridors, and hand-over of a pass when one unit drops out. The mobile refiller already works with a working robot, which is a two-unit special case.

REUSE FROM EXISTING IMPLEMENTS

The pass pattern of the mobile refiller (simple) and the headland-turn animations of the agbot slim.

OPEN QUESTIONS

- Communication link and message set
- Pass allocation and re-allocation
- Collision and crossing rules
- Charging and refilling logistics for many units

Folder: `implement ideas/swarm_operation/`

IDEA · CROP & SOIL

Mole and tunnel mapper

Not worked out yet

WHAT IT DOES

Finds and maps molehills (and with a probe, mole tunnels) so they can be treated or spread flat later.

CONCEPT (FIRST THOUGHTS)

A camera looking down finds fresh hills; each gets a GPS point. A probe or an acoustic sensor could later detect tunnels. The map feeds the molehill spreader or a trap placing action.

REUSE FROM EXISTING IMPLEMENTS

Camera and detection pipeline of the spot sprayer; GPS logging of the field marker.

OPEN QUESTIONS

- Detection of fresh versus old hills
- Tunnel detection: is a probe or a ground-radar feasible?
- Map format and who uses it

Folder: `implement ideas/mole_mapper/`

IDEA · TRANSPORT & GENERAL

Speech speaker

Not worked out yet

WHAT IT DOES

A speaker on the robot that can speak: warnings when the robot approaches, status messages for the operator, instructions in the field, and a way to speak to people near the robot.

CONCEPT (FIRST THOUGHTS)

A weatherproof loudspeaker (or a pair, front and rear) on the top plate or mast with a small amplifier, fed by the robot computer. Text-to-speech for messages such as 'robot is turning', 'tank empty' or 'battery low'; optionally a microphone and a radio link so an operator can speak through the robot. Volume follows the distance to detected people.

REUSE FROM EXISTING IMPLEMENTS

Top-plate pattern and mast mounting of the spot sprayer and the field marker; computer and radio link of the mobile refiller.

OPEN QUESTIONS

- Speaker type, power and IP rating
- Languages and voice (text-to-speech engine, offline)
- Which events trigger which messages
- Noise limits near animals (cows, horses) and neighbours
- Microphone and two-way speech

Folder: `implement ideas/speaker/`

IDEA · WEED & CROP PROTECTION

Laser or UV weed treatment

Not worked out yet

WHAT IT DOES

Kills individual weeds with a laser or a focused UV beam, without chemicals and without moving the soil.

CONCEPT (FIRST THOUGHTS)

Camera finds the weeds (as in the spot sprayer), a galvo mirror aims the beam at the growing point. Needs a shroud against stray light and an interlock, and mostly works on small seedlings.

REUSE FROM EXISTING IMPLEMENTS

Camera mast and detection of the spot sprayer; shroud and safety interlock ideas of the weeds burner.

OPEN QUESTIONS

- Laser class and the safety rules that follow
- Power needed per weed and dwell time
- Working width versus galvo scan area
- Cost compared with a spot sprayer

Folder: `implement ideas/laser_weeder/`

IDEA · WEED & CROP PROTECTION

Hot steam / hot foam weeder

Not worked out yet

WHAT IT DOES

Kills weeds with hot steam or hot foam under a light hood, as an alternative to the flame weeder.

CONCEPT (FIRST THOUGHTS)

A small boiler or heat exchanger with a hood. Hot foam holds the heat on the leaf longer than steam alone. Electric heating is possible but needs a lot of power, so it may stay a small-area tool.

REUSE FROM EXISTING IMPLEMENTS

Hood, heat shield and castors of the weeds burner; water tank from the mobile refiller concept.

OPEN QUESTIONS

- Power and water use per hectare
- Boiler type: propane, electric or both
- Effect on perennial weeds with deep roots
- Weight and heat shielding

Folder: `implement ideas/steam_foam_weeder/`

IDEA · WEED & CROP PROTECTION

Pheromone trap service

Not worked out yet

WHAT IT DOES

Visits insect traps and pheromone dispensers at the field edge: checks them with a camera, counts catches and swaps dispensers on schedule.

CONCEPT (FIRST THOUGHTS)

A small arm or an XYZ stage that opens a trap, photographs the catch and replaces the lure. The robot stops at stored trap positions.

REUSE FROM EXISTING IMPLEMENTS

XYZ stage and camera ideas of the field marker; trap positions kept like the marker's point list.

OPEN QUESTIONS

- Trap types and how they open
- Catch counting by image
- Lure storage and replacement by the robot
- Frequency of visits

Folder: `implement ideas/pheromone_trap_service/`

IDEA · LIVESTOCK & FARM

Fence inspector

Not worked out yet

WHAT IT DOES

Drives along the fence, measures the voltage of the electric fence per metre, reports breaks, shorts and where the voltage drops, and straightens leaning posts.

CONCEPT (FIRST THOUGHTS)

A contactless (capacitive or inductive) or contact sensor on an arm along the wire to measure voltage and current with a GPS log per metre, a camera for posts and a simple push or pull tool for leaning posts. Reports the position of every fault. The sensor can also ride on the fence mower arm.

REUSE FROM EXISTING IMPLEMENTS

Around-pole fence mower arm and post-following geometry.

OPEN QUESTIONS

- Contactless versus contact sensing, calibration to a known voltage
- How to straighten a post without damage
- Fault classification: break, short, vegetation
- Reporting to the farmer: map, push message, alarm threshold

Folder: `implement ideas/fence_inspector/`

IDEA · LIVESTOCK & FARM

Drinking trough check

Not worked out yet

WHAT IT DOES

Visits drinking troughs, measures the water level, tops it up and reports dirt or damage with a camera.

CONCEPT (FIRST THOUGHTS)

A mast with a level sensor, a camera and a small refill nozzle on a hose; the robot parks at stored trough positions. Water from a tank on the robot or a nearby tap.

REUSE FROM EXISTING IMPLEMENTS

Nozzle and docking approach of the mobile refiller (simple); camera mast.

OPEN QUESTIONS

- Level sensing: ultrasonic or float
- Water supply and tank size
- Frost and winter use
- Cleaning: only reporting or also scrubbing

Folder: `implement ideas/drinker_check/`

IDEA · LIVESTOCK & FARM

Chicken house robot

Not worked out yet

WHAT IT DOES

Drives through a poultry house to turn the litter, collect floor eggs, and check birds and climate with sensors.

CONCEPT (FIRST THOUGHTS)

A small low variant with a litter-turning tool, a soft-brush egg collector and sensors for ammonia, temperature and humidity. Must be quiet and slow around the birds.

REUSE FROM EXISTING IMPLEMENTS

Disc brush and drive from the brushes; camera and logging from the crop scanner idea.

OPEN QUESTIONS

- Robot height and width for a poultry house
- Egg pick-up mechanism that does not break eggs
- Speed and noise near birds
- Cleaning and disinfection

Folder: `implement ideas/chicken_house_robot/`

IDEA · MEASURING & SENSORS

Yield estimator

Not worked out yet

WHAT IT DOES

Counts fruits, heads or plants with a camera and logs them per GPS position, so the harvest can be estimated and planned.

CONCEPT (FIRST THOUGHTS)

Cameras on a mast or in the row, an image model that counts and sizes the crop, and a yield map per row.

REUSE FROM EXISTING IMPLEMENTS

Global-shutter camera and detection pipeline of the in-row weeder and the spot sprayer.

OPEN QUESTIONS

- Crop to count (strawberry, broccoli, apple ...)
- Lighting at night versus day
- Count accuracy and counting error
- Output: map, report or plucking plan

Folder: `implement ideas/yield_estimator/`

IDEA · MEASURING & SENSORS

Disease and pest scout

Not worked out yet

WHAT IT DOES

Looks at the leaves close up, finds disease and pest damage early and stores a photo per plant with its position.

CONCEPT (FIRST THOUGHTS)

A camera (RGB or multispectral) on a low mast or sideways arm, with its own light so it works at night. The images feed a classification model; alerts go to the farmer. Combines well with the UV light and the biocontrol spreader.

REUSE FROM EXISTING IMPLEMENTS

Camera and lighting of the in-row weeder; NIR sensor idea.

OPEN QUESTIONS

- Which diseases and crops to detect
- Training data and model
- Lighting and shading of the camera
- Spot treatment links (UV, spray)

Folder: `implement ideas/disease_scout/`

IDEA · MEASURING & SENSORS

Soil moisture sensor placer

Not worked out yet

WHAT IT DOES

Places wireless soil moisture sensors at set GPS points, reads them out and picks them up again.

CONCEPT (FIRST THOUGHTS)

A drop tube or probe that pushes a sensor into the soil, with a small magazine of sensors. A reader on the robot collects values by radio on each visit.

REUSE FROM EXISTING IMPLEMENTS

XYZ stage and ground sensing of the field marker; irrigation hose reel for timing of the water.

OPEN QUESTIONS

- Sensor type, battery life and radio protocol
- Pick-up of sensors: magnet or tether
- Number of points per field

Folder: `implement ideas/soil_moisture_placer/`

IDEA · MEASURING & SENSORS

Windbreak and tree-row inspector

Not worked out yet

WHAT IT DOES

Drives along hedgerows, windbreaks and orchard rows and logs gaps, damage, dead trees and growth with a camera.

CONCEPT (FIRST THOUGHTS)

A sideways-looking camera pair and a logging unit, producing a tree-by-tree status list with photos.

REUSE FROM EXISTING IMPLEMENTS

Camera masts of the spot sprayer and yield estimator.

OPEN QUESTIONS

- Image quality in shade and at row ends
- Per-tree identification
- Status classes and reports

Folder: `implement ideas/windbreak_inspector/`

IDEA · POWER & DRIVE

Charging robot (mobile charger)

Not worked out yet

WHAT IT DOES

A robot on wheels that carries a large battery and charges working robots in the field, so they do not have to return to the shed.

CONCEPT (FIRST THOUGHTS)

A support robot like the mobile refiller, with a battery pack and a charging plug on a boom or at a fixed side. The working robot docks at a stop point, or charging happens in the headland turn.

REUSE FROM EXISTING IMPLEMENTS

Docking approach, RTK antenna above the connection and procedure of the mobile refiller; battery pack idea.

OPEN QUESTIONS

- Charging connector and automatic plug-in
- Charging power and time
- Battery capacity on the charger robot
- Wireless versus conductive contact

Folder: `implement ideas/charging_robot/`

IDEA · MEASURING & SENSORS

Security camera patrol

Not worked out yet

WHAT IT DOES

A mast with camera and thermal imaging that patrols the farm or field at night, reports intruders and keeps an eye on animals and equipment.

CONCEPT (FIRST THOUGHTS)

Mast with pan/tilt camera, thermal sensor and microphone; the robot drives a set route and alerts on movement. Combines with the speaker for warnings.

REUSE FROM EXISTING IMPLEMENTS

Camera mast of the spot sprayer; speaker idea; solar roof for power.

OPEN QUESTIONS

- Camera and thermal sensor choice
- Alert route to the farmer
- Privacy and rules for filming
- Night-time driving and obstacles

Folder: `implement ideas/security_patrol/`

IDEA · TRANSPORT & GENERAL

Magnet sweeper

Not worked out yet

WHAT IT DOES

Picks up nails, screws, wire and other loose iron from fields, paths and yards after building work or a festival, before it ends up in the feed or in tyres.

CONCEPT (FIRST THOUGHTS)

A wide permanent or electro magnet a few cm above the ground across the robot width, on a height-adjustable bar. With an electromagnet the load is dropped in a bin at the field edge; with permanent magnets a release plate strips it off. The robot drives the field in overlapping passes and logs the covered area.

REUSE FROM EXISTING IMPLEMENTS

Toolbar and lift of the push blade or the three-point hitch; pass pattern and logging of the field marker.

OPEN QUESTIONS

- Permanent magnet or electromagnet (48 V power)
- Working height versus pick-up of small parts in grass
- Release and collecting of the load
- Mass of the magnet bar and the axle loads

Folder: `implement ideas/magnet_sweeper/`

IDEA · MEASURING & SENSORS

Metal detector

Not worked out yet

WHAT IT DOES

Scans a field for buried or hidden metal and stores a GPS point at every hit, so the spots can be checked or cleaned later.

CONCEPT (FIRST THOUGHTS)

A pulse-induction search coil on a non-metallic boom away from the robot's steel frame and hub motors, low above the ground. Every pulse above a threshold gives a GPS point with its signal strength; the map can feed the field marker to stake out the spots, or the magnet sweeper.

REUSE FROM EXISTING IMPLEMENTS

Front boom of the grass meter; GPS point list and staking out of the field marker.

OPEN QUESTIONS

- Interference from the robot's steel and motors: boom length
- Coil size and detection depth
- Threshold and false hits
- Map format

Folder: `implement ideas/metal_detector/`

IDEA · TRANSPORT & GENERAL

Debris vacuum

Not worked out yet

WHAT IT DOES

Sucks up litter and debris from grassland: trash after a festival, or shards of solar panels that a fire on a nearby roof spread over the fields. Several passes clean up the mess.

CONCEPT (FIRST THOUGHTS)

A wide suction nozzle close to the grass with a fan, a cyclone or bag and a collecting bin on the robot. Glass and panel shards are heavy and sharp, so the nozzle and bin need wear protection; a rotating brush in the nozzle lifts pieces out of the sward.

REUSE FROM EXISTING IMPLEMENTS

Bin on the upper beams of the dumper barrow or compost spreader; brush and lift of the broom sweeper.

OPEN QUESTIONS

- Fan power and air flow on 48 V
- Nozzle width and height above the grass
- Separation of debris and grass, dust filtering
- Wear protection against glass shards

Folder: `implement ideas/debris_vacuum/`

IDEA · SPRAYING, FERTILIZING & IRRIGATION

Biostimulant sprayer (full field)

Not worked out yet

WHAT IT DOES

Sprays biostimulants over the whole field at the right moment: compost tea, seaweed extract, or nitrogen-fixing products such as Corteva's Utrisha N (Blue N). It needs no plant-by-plant precision, but every pass is logged, so it is clear what was applied where and when.

CONCEPT (FIRST THOUGHTS)

100 to 200 l/ha is little to carry: a tank on the robot with the boom of the spot sprayer with all nozzles open. The main part is the record keeping: a log per pass with product, rate, time, weather and GPS track, to compare treated and untreated strips afterwards.

REUSE FROM EXISTING IMPLEMENTS

Folding boom, tank and pump of the spot sprayer; tank and pump of the liquid fertilizer applicator.

OPEN QUESTIONS

- Nozzles and filtering for compost tea (particles, living organisms)
- Tank size versus field size
- Log format and registration
- Timing: weather and crop stage

Folder: `implement ideas/biostimulant_sprayer/`